Amaro

de Matos, João Amaro - Theoretical Foundations of Corporate

- MM propositions with and without taxes: Ch. 2, §2.1 (p. 40–47)
- Miller (1977): personal and corporate taxation equilibrium: Ch. 2, §2.2.1–2.2.4 (p. 48–54)

ratio is irrelevant. The introduction of corporate taxes in this context, discussed by Modigliani and Miller (1958, 1963), allows firms to deduct the interest on debt in computing taxable profits. This generates tax advantages for debt that would induce firms to be completely financed through debt. The fact that this is not observed in practice led several authors, starting with Modigliani and Miller themselves, to argue for the relevance of bankruptcy costs. This factor, together with other possible costs associated with debt, could therefore explain the debt-equity ratios observed in the market. The work of Modigliani and Miller will be described in some detail in Section 2.1. However, some years later, Miller argued with convincing market figures that bankruptcy costs are in general of too small a magnitude to affect the equilibrium to be sustained. He suggested a different equilibrium, based on demand and supply of bonds induced by differentiated taxes for individuals and corporations. In Section 2.2 this equilibrium is described and compared with that of MM.

2.1 The MM Propositions

Modigliani and Miller (1958, 1963) have worked out a set of rather strict conditions under which the structure of capital does not affect the actual value of the firms. Of course, it is never claimed that the real world follows these assumptions. What is hoped is that by modeling the universe of the firms in such a way, and by relaxing the assumptions one by one, it is possible to identify and discuss the weight or relevance of each of the different factors in the search for an optimal capital structure.

2.1.1 The Irrelevancy Statement

Assume that a firm starts at time $t = 0$ and has payoffs $\Gamma_j(t)$ in state j at any point in time t, with $j = 1, 2, \ldots k(t)$ denoting the possible states of nature at time t.

In the setting to be considered here, it is assumed that markets do not allow for arbitrage opportunities and also that markets are perfect, in the sense that there are no frictions of any type, taxes included. Also, there are two ways to finance the firm: either through equity only or through equity together with debt. When debt is issued, the firm is said to be *levered*; otherwise it is *unlevered*.

Equityholders and debtholders have different claims on the final payoff of the firm. Debtholders, also known as *bondholders* or *creditors*, buy a debt contract issued by the firm giving them the claim to a face value F at time T. If at that point in time the state of nature is such that $\Gamma_j(T)$ is less

than F, the firm is said to go *bankrupt*, in which case the creditholder will receive $\Gamma_j(T)$, if positive. Let $V_j(T)$ denote the positive part of $\Gamma_j(T)$, or $V_j(T) \equiv [\Gamma_j(T)]^+$. *Bankruptcy* is then defined by the set of states of nature where the firm is unable to pay its debt. The payoff to a debt instrument in the j-th state of nature is thus

$$B_j(T) = \min\{F, V_j(T)\}.$$

If K is the number of possible states of nature at time T, then, under the absence of arbitrage opportunities, there exists a strictly positive vector $\psi \in R^K$ such that the present value of the debt is

$$B = \sum_{j=1}^{K} \psi_j B_j(T).$$

Notice that, if there is no state leading to bankruptcy, $B_j(T) = F$ and the preceding expression implies that

$$B = F \sum_{j=1}^{K} \psi_j \Rightarrow B = \frac{F}{R}. \tag{2.1}$$

In other words, in the absence of bankruptcy, debt is contracted at the risk-free rate. On the other hand, given the positiveness of the vector ψ, the existence of bankruptcy in some states of nature leads to

$$B < \frac{F}{R}.$$

This means that F should be discounted at a higher rate than the risk-free rate in order to provide the no-arbitrage value of the debt.

The equityholders, on the other hand, are the true owners of the firm, having the right to receive what is left from the payoff of the firm once the creditholders are paid, provided that the remaining value is positive. They may issue *shares* in order to transact this residual claim, in which case the equityholders are also known as *shareholders*. The positiveness of this residual claim constitutes a protection to shareholders and is termed *limited liability*. In other words, the payoff to equityholders in state j is

$$\begin{aligned} S_j(T) &= [\Gamma_j(T) - \min\{F, V_j(T)\}]^+ \\ &= V_j(T) - \min\{F, V_j(T)\} \\ &= \max[0, V_j(T) - F], \end{aligned} \tag{2.2}$$

implying that

$$V_j(T) = S_j(T) + B_j(T).$$

The present value of the equity is thus

$$S = \sum_{j=1}^{K} \psi_j S_j(T).$$

In this setting, it is clear that the value of the unlevered firm V_u is equal to the value V_l of a levered firm since

$$V_l = S + B = \sum_{j=1}^{K} \psi_j [B_j(T) + S_j(T)] = \sum_{j=1}^{K} \psi_j V_j(T) = V_u$$

This result is known as the first proposition of Modigliani and Miller and may be stated as

Proposition 2.1.1 (MM1). *The market value of any firm is independent of its capital structure*

$$V_l = V_u. \tag{2.3}$$

This statement implies that if equity is not sufficient to finance a firm, debt can be contracted without affecting the value of the project. Also, the consumption of equityholders is not bounded by the fact that they are investing in the project. In fact, equity may be used for consumption and replaced in the capital structure by debt without any side effect on the total value of the firm.

2.1.2 Cost of Capital

Consider now the situation of an unlevered firm. Its cost of capital is simply the expected return on equity. By construction

$$R_s^u = \frac{EV_j(T)}{V}$$

where $V \equiv V_l = V_u$. From Chapter 1 it is known that under complete markets there is a portfolio θ^* such that its expected return is $E(R^*)$, allowing us to write

$$R_s^u = R + \beta_s^u [E(R^*) - R],$$

where β_s^u is the systematic risk of equity in the unlevered firm, as defined in (1.18).

In a levered firm one must consider the *cost of debt*, that is, the expected return to debt

$$R_b = \frac{EB_j(T)}{B}.$$

Under the absence of bankruptcy possibilities, $EB_j(T) = F$ and expression (2.1) implies that $R_b = R$. The presence of bankruptcy, however, leads to $R_b > R$, implying the existence of a systematic risk β_d that is different from zero, as defined by equation (1.18).

Also, there is the *cost of equity*, given by the expected return to equity

$$R_s = \frac{ES_j(T)}{S}.$$

Rewriting this last term,

$$R_s = \frac{E[V_j(T) - B_j(T)]}{S} = \frac{R_s^u V}{S} - \frac{R_b B}{S} = R_s^u + (R_s^u - R_b)\frac{B}{S}.$$

This result is known as the second proposition of Modigliani and Miller:

Proposition 2.1.2 (MM2). *The expected return on equity is equal to the expected return R_s^u for a pure equity stream, plus a premium related to the financial risk equal to the debt-to-equity ratio times the spread between R_s^u and R_b . That is to say,*

$$R_s = R_s^u + (R_s^u - R_b)\frac{B}{S}. \tag{2.4}$$

In other words, the cost of equity changes linearly with the debt-equity ratio B/S. From this expression, the systematic risk of equity in the unlevered firm can be obtained from the linearity of the covariance operator as

$$\beta_s = \beta_s^u\left(1 + \frac{B}{S}\right) - \beta_d\frac{B}{S}.$$

Finally, the *average cost of capital* is defined as

$$\bar{R} = \frac{E[S_j(T) + B_j(T)]}{S + B} = \frac{S}{S + B}R_s + \frac{B}{S + B}R_b$$

and is expressed as the weighted average of the cost of equity and the cost of debt. The weights turn out to be the relative values of equity and debt with respect to the total value. Clearly, this average cost of capital has a correspondent systematic risk given by

$$\bar{\beta} = \frac{S}{S + B}\beta_s + \frac{B}{S + B}\beta_d.$$

2.1.3 The MM Propositions with Taxes

The presence of frictions and, in particular, the presence of taxes may drastically change the preceding conclusions. In this section it is assumed that firms pay taxes at a rate $\tau \in [0, 1]$.

Consider the case of a levered firm. In that case, let T be the point in time where both taxes and the creditholders are paid. In case of bankruptcy, no taxes are paid; otherwise, the firm pays $\tau[V_j(T) - F]$. In other words, in any case the taxes are calculated in state j as

$$\theta_j^\tau(T) = \tau[V_j(T) - F]^+.$$

Under the absence of arbitrage opportunities, the present value of taxes is then

$$\theta^\tau = \tau \sum_{j=1}^{K} \psi_j [V_j(T) - F]^+.$$

The debtholders will receive in state j

$$B_j(T) = \min\left\{F, \left[V_j(T) - \theta_j^\tau(T)\right]^+\right\} = \min\{F, V_j(T)\},$$

exactly the same payoff as without taxes. The equityholders are left in state j with

$$S_j^\tau(T) = V_j(T) - B_j(T) - \theta_j^\tau(T). \tag{2.5}$$

It follows that

$$V = S^\tau + B + \theta^\tau \tag{2.6}$$

where

$$S^\tau = \sum_{j=1}^{K} \psi_j S_j^\tau(T).$$

The Main Result

Given the payoffs in the presence of taxes, the value of the unlevered firm is

$$V_u^\tau = (1 - \tau) \sum_{j=1}^{K} \psi_j V_j(T).$$

In the case of the levered firm, the firm generates for equity and debtholders the value

$$V_l^\tau = \sum_{j=1}^{K} \psi_j [S_j^\tau(T) + B_j(T)].$$

Notice from (2.5) that

$$\begin{aligned} S_j^\tau(T) + B_j(T) &= (1-\tau)V_j(T) + \tau V_j(T) - \theta_j^\tau(T) \\ &= (1-\tau)V_j(T) + \tau\{V_j(T) - [V_j(T) - F]^+\} \\ &= (1-\tau)V_j(T) + \tau \min\{V_j(T), F\}. \end{aligned}$$

In other words,

$$\begin{aligned} V_l^\tau &= (1-\tau)\sum_{j=1}^{K} \psi_j V_j(T) + \tau \sum_{j=1}^{K} \psi_j B_j(T) \\ &= V_u^\tau + \tau B. \end{aligned}$$

This result is known as the proposition of Modigliani and Miller under taxes and is summarized as

Proposition 2.1.3. *Under the absence of arbitrage opportunities and in the presence of taxes, the value of the firm increases linearly with the level of debt as*

$$V_l^\tau = V_u^\tau + \tau B. \tag{2.7}$$

This decomposition of the value of the firm affects the different costs of capital.

Cost of Capital under Taxation

The fact that the payoff to the debtholders is exactly the same as in the case with no taxes implies that the cost of debt is the same, as is its systematic risk. The cost of equity, however, is affected by the payment of taxes. By construction

$$\begin{aligned} R_s^\tau &= \frac{ES_j^\tau(T)}{S^\tau} = \frac{E\big[V_j(T) - B_j(T) - \theta_j^\tau(T)\big]}{S^\tau} \\ &= \frac{R_s^u V}{S^\tau} - \frac{R_b B}{S^\tau} - \frac{E\theta_j^\tau(T)}{S^\tau}. \end{aligned}$$

Use of equality (2.6) simplifies this expression to

$$R_s^\tau = R_s^u + (R_s^u - R_b)\frac{B}{S^\tau} + \frac{R_s^u \theta^\tau - E\theta_j^\tau(T)}{S^\tau},$$

from which expression (2.4) is recovered by making $\tau = 0$. This is a general expression, allowing for the possibility of bankruptcy. A particular case can be worked out considering that there is no bankruptcy in any state. Under this assumption,

$$\theta_j^\tau = \tau[V_j(T) - B_j(T)], \forall j$$

leading to

$$\theta^\tau = \tau[V - B]$$

and

$$E\theta_j^\tau(T) = \tau[R_s^u V - RB],$$

where $R_b = R$, given the absence of default. Substituting in the expression for R_s^τ gives the equivalent to Proposition MM2 with taxes:

$$R_s^\tau = R_s^u + (R_s^u - R)\frac{B}{S^\tau}(1 - \tau).$$

The systematic risk of equity in this case is then

$$\beta_s^\tau = \beta_s^u\left[1 + \frac{B}{S^\tau}(1 - \tau)\right].$$

Finally, the average after-tax cost of capital is defined as

$$\bar{R}^\tau = \frac{E[V_j(T)(1 - \tau)]}{S^\tau + B} = R_s^\tau\frac{S^\tau}{S^\tau + B} + R_b\frac{B}{S^\tau + B}.$$

2.1.4 Empirical Evidence

According to equation (2.7), this theory suggests that there is a big incentive for firms to be financed only through debt. The existence of tax advantages for debt financing is clear from equation (2.7). The value of levered firms is always greater than the value of unlevered equivalent firms. This increase in value is usually termed *debt tax shield*.

The evidence for the existence of a relation between debt-equity ratios and tax shields is mixed. In fact, there are other variables (depreciations, net operating losses, carryforwards, and investment tax credits, for example) that provide similar sorts of tax advantages as the debt tax shield.

Models predict that firms with more taxable income and fewer non-debt advantages from taxes should present higher debt-equity ratios.

In contrast to such predictions, some studies[1] show that the more non-debt tax shields firms have, the more debt they seem to have.

As Barclay and Smith (1999) stress, however, these results cannot lead to conclusions about the irrelevancy of taxes in the capital structure decisions. In fact, the results are difficult to interpret because they are ambiguous. The tax variables may be seen as crude proxies for the firms' effective marginal rate and, therefore, possibly highly correlated with other variables that influence the choice of capital structure.

As an example, consider a firm with a high level of depreciation. Typically, it will be a firm with mainly tangible fixed assets and, thus, with low contracting costs associated with debt financing. This is so, since tangible fixed assets provide good collateral. Evidence reported in the study just cited above is perfectly compatible with this type of interpretation.

Additional recent studies have dealt carefully with this issue. Mackie-Mason (1990) found more solid evidence that firms with a high marginal tax rate are more likely to issue debt than firms in with a low marginal tax rate. In another attempt to avoid the problem of variables that are crude proxies for marginal tax rates, Graham (1996) developed a specific simulation method to provide an extremely sophisticated measure of such rates. Using those firms' marginal tax rates, evidence is found of a positive relation between changes in debt-equity ratios and those rates.

Therefore, evidence tends to support the idea that the existence of taxes does somehow affect the capital structure decisions. However, the impression that one might come away with—namely, that firms have the incentive to raise external capital only in the form of debt—is not correct. In reality, firms are never fully financed with debt. The reason is that there are typical costs associated with debt, such as the flotation costs and the costs of bankruptcy. As the level of debt increases, the higher the probability is that firms will not be able to pay their debt obligations. Aware of that, the debtholders increase the required premium, increasing the cost of capital. These costs increasing with debt may well offset the tax advantage here described, leading to the equilibrium structure observed in the market.

Modigliani and Miller's framework does not take these effects into account. Therefore, the correctness of their results is to be understood in the strict context of their assumptions.

2.1.5 Exercises

1. Let a firm start at time $t = 0$ and have projects with payoffs $\Gamma_j(T) =$ [10 8 6] for the three different states of nature at time $t = T$. The firm is partially financed with debt with face value $F = 5$ and pays taxes at rate $\tau = 0.4$. Assume that the three different states of nature have

equal assigned probabilities. There are no arbitrage opportunities and the vector of state prices is $\psi = [0.3\ 0.25\ 0.44]$.

(a) Check propositions MM1 and MM2. Compute the weighted average of the costs of debt and equity.

(b) Suppose now that $F = 7$, that is, the firm will be bankrupt if state of nature $j = 3$ occurs. How do the values of debt, equity, and the firm change? Compute the weighted average of the costs of debt and equity and compare them with the previous case.

2. Show how to explore arbitrage opportunities when $V_l \neq V_u$, if individuals can borrow capital with the same characteristics as firms.
3. Modigliani and Miller (1958) developed their theory under some different assumptions. They assumed that debt had an infinite life and that each period the firm had to pay interest on debt equal to rD. Moreover, firms could deduct the interest on debt in computing taxable profits. The return to the stockholders is defined as the time average of an infinite stream of payoffs and given by X. How would you obtain proposition MM1 with and without taxes?
4. In the context of the previous exercise, suppose that an individual in the economy faces a borrowing interest rate $\bar{r}$ and a lending interest rate $\underline{r}$ such that $\bar{r} > r > \underline{r}$, where r is the corporate interest rate. What happens to MM1?
5. Suppose that any time an individual wants to sell/purchase assets (bonds and shares), he must pay a fixed amount T of transaction costs. In the context of Modigliani and Miller (1958), what is the effect of these transaction costs on MM1?

2.2 Personal and Corporate Taxation

As discussed in this chapter's introduction, Miller (1977) argued that bankruptcy costs are not relevant enough to justify the debt-equity ratios observed in the market.

The U.S. corporate tax savings were around 48 cents per dollar at the time of Miller's paper—34 cents now, after the tax reforms in 1986—and the basis for comparison with bankruptcy costs was a familiar number around 20 cents per dollar. The point that Miller raises is that this 20% is applicable to *individuals*, not to firms. Actually, for large, publicly held corporations, that figure does not hold. An empirical study by Warner (1977) using a sample of 11 railroads reports an average bankruptcy cost of around 5.3 cents per dollar invested, presenting a strong inverse size effect: The largest corporation had a cost of 1.7%.

The fact that the debt-equity rate has a stable behavior through time and does not explode to infinity, as the Modigliani and Miller theory suggests, remains to be explained.

The idea of Miller in (1977) to explain this fact is quite simple. Taxes on interest income and dividends are higher than taxes on capital gains (most of which can be deferred). Hence, companies would issue debt to investors who either are tax-exempted or pay little tax, and would issue equity to high–tax-bracket investors. The equilibrium value for the return on the bonds would be defined by the condition that firms would not increase returns above the level at which the saving in taxes to the investors exceeded the firms' corporate tax liability.

On the basis of Miller's argument, there is the underlying assumption that taxes are differentiated. Consider here that besides the tax on corporate profits under which interest payments are deductible, hereby denoted by τ_c, there are also personal income taxes denoted by τ_b and τ_s that are applicable, respectively, to bonds and to shares.

Under this assumption it is possible to describe the equilibrium suggested by Miller. In what follows, rates of return will be used instead of total returns, as before.

2.2.1 Demand for Bonds

To study the demand for bonds, consider the existence of two types of bonds: fully tax-exempt bonds, with equilibrium rate of return $r_\circ$, and the other bonds, which are not tax-exempt and have a required rate of return r_b. The after-tax rate of return on these latter bonds is $(1 - \tau_b)r_b$. There is demand for the nonexempt bonds provided that this after-tax rate of return is greater than $r_\circ$. This means that

$$r_b \geq \frac{r_\circ}{1 - \tau_b}.$$

Consider a firm that raises equity at a cost of capital before taxes denoted by ρ. If this equity is within the same class of risk as the tax-exempt bonds, with rate of return $r_\circ$ as just described, one would require the same rate of return for the equity after paying the taxes due. Thus

$$r_\circ = (1 - \tau_s)(1 - \tau_c)\rho.$$

If a small amount of *aggregate* debt is issued, it can be sold for relatively low rates of return, that is, the bonds will be owned by *tax-exempt* investors. In this case, $\tau_b = 0$ and the demanded rate of return to the bonds is at its lowest possible level, $r_b^d = r_\circ$.

Increasing the amount of issued debt will require bondholders in higher tax brackets. Thus, the average τ_b paid by the investors is an increasing function of the aggregate issued debt,

$$\tau_b \equiv \tau_b(B) \text{ such that } \tau_b'(B) \geq 0.$$

As a consequence, the *demanded rate of return* on the bonds becomes

$$r_b^d \equiv r_b^d(B) = \frac{r_\circ}{1 - \tau_b(B)}, \tag{2.8}$$

which is an increasing function of B since

$$\frac{dr_b^d(B)}{dB} = \frac{dr_b^d}{d\tau_b(B)}\tau_b'(B) = \frac{r_\circ}{(1-\tau_b)^2}\tau_b'(B) \geq 0.$$

2.2.2 Supply of Bonds

From the supply side, corporate taxable firms prefer to issue debt in comparison to equity as long as the bonds are cheaper than equity. In this case firms require a before-tax rate of return r_b that is smaller than the minimum before-tax rate of return $r_\circ$ required by investors on equity.

In other words, corporations prefer to raise capital through bonds if

$$r_b \leq \rho. \tag{2.9}$$

Hence, firms are willing to supply bonds whenever the rates of return r_b satisfy

$$r_b \leq \frac{r_\circ}{(1-\tau_s)(1-\tau_c)}.$$

The maximum value of the available rates of return r_b is taken as the *supplied rate of return* r_b^s for bonds,

$$r_b^s = \frac{r_\circ}{(1-\tau_s)(1-\tau_c)}. \tag{2.10}$$

The aggregate supply of bonds as a function of the aggregate debt is therefore described by a straight line defining a constant level of rate of return, independent of the aggregate level of debt.

2.2.3 The Equilibrium

The equilibrium that follows from the preceding supply and demand curves for rates of return on bonds is achieved at a debt level B^* and a rate of return on bonds r_b^* satisfying

$$r_b^* \equiv r_b^s = r_b^d(B^*).$$

The solution of this equation is illustrated graphically in Figure 2.1.

Given that r_b^s is constant, equal to ρ, it follows that

$$r_b^* = \rho \quad \text{and} \quad B^* = \left[r_b^d\right]^{-1}(\rho),$$

where $[r_b^d]^{-1}$ denotes the *inverse function* of $r_b^d(x)$. In terms of the tax rates, since

$$\rho = \frac{r_\circ}{(1-\tau_s)(1-\tau_c)} \quad \text{and} \quad \rho = r_b^d(B^*) = \frac{r_\circ}{[1-\tau_b(B^*)]},$$

the equilibrium relation may be rewritten as

$$(1-\tau_s)(1-\tau_c) = 1-\tau_b(B^*). \tag{2.11}$$

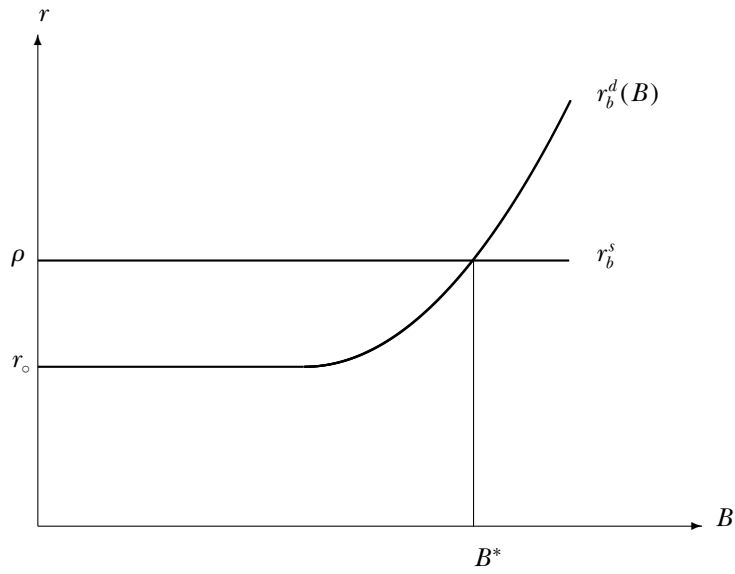


Figure 2.1. *The Equilibrium in Miller's Model*

But perhaps the most striking aspect of Miller's equilibrium is that, although there is an optimal level of aggregate debt in the economy, there will be *no tax advantage for any particular firm to issue either debt or equity.* To see why, notice that to raise a certain amount K of capital, one can issue either debt or equity. In the first case, the stockholders pay back $r_b(1-\tau_c)K$, whereas in the second, they pay $\rho(1-\tau_c)K$. But since in equilibrium $r_b^* = \rho$, there is no relative advantage in one form of raising capital with respect to the other.

Alternatively, the relative advantage of leverage for stockholders, G_L, is written as

$$\begin{aligned} G_L &= [\rho(1-\tau_c) - r_b^*(1-\tau_c)]K \\ &= \left[1 - \frac{(1-\tau_s)(1-\tau_c)}{1-\tau_b(B^*)}\right]\frac{r_\circ K}{(1-\tau_s)}. \end{aligned}$$

The equilibrium relation between the tax rates expressed in (2.11) implies that $G_L = 0$, which is to say, there is no advantage of leverage for the stockholders.

The nature of this equilibrium is strongly driven by the existence of personal taxations. Following Hamada and Scholes (1985), this will be called the *after-tax* equilibrium. However, as stressed by some authors,[2] the U.S. federal tax statutes allow for many different ways of saving and investing at *before-tax* rates of return. The result is that the *effective* personal tax rate applicable may be zero. In this case, one would expect to retrieve the results of Miller and Modigliani that firms will have incentive to be totally financed through debt.

In fact, making personal tax rates equal to zero (i.e., all tax rates except the corporate tax rate τ_c), it follows that the riskless rate under this condition will be greater than $r_\circ$ in equation (2.9) and will be given by $r'_\circ$ satisfying

$$r'_\circ = \rho(1-\tau_c) > r_\circ.$$

Also, the supply curve for bonds defined in (2.10) becomes a straight line at the constant level ρ, independent of the aggregate debt level,

$$r_b^s = \rho = \frac{r'_\circ}{1-\tau_c}, \tag{2.12}$$

and the demand curve for bonds in equation (2.8) becomes the parallel straight line at the constant value

$$r_b^d = r'_\circ = \rho(1-\tau_c). \tag{2.13}$$

Thus, in the before-tax model of Modigliani and Miller, equilibrium is reached only at an infinite amount of debt issued, since demand and supply are constants independent of the level of aggregate debt. This leads to a corner solution of 100% debt.

Notice moreover that $r_b^s \geq r_b^d$. This inequality will be strict as long as the constant values of r_b^d and r_b^s are different. If they are equal, both investors and firms will be indifferent to the amount of bonds issued. Comparing equations (2.13) and (2.12), one notices that this may happen in two cases: if $\tau_c = 0$, in which case the result of Proposition MM1 is retrieved, or if $r'_\circ = 0$.

For a noncorner solution of the before-tax equilibrium (i.e., to avoid the corner solution of 100% of debt financing), an additional assumption introducing some market imperfection would be required, such as leverage-related costs that increase with the *aggregate* debt issued. As noted by Hamada and Scholes (1985), and as will be extensively discussed, the nature of these costs can be multiple (e.g., bankruptcy costs, contracting or monitoring costs, information and signaling costs, differential flotation costs, incomplete markets costs). Which of these costs are relevant is an empirical issue. Whatever the true nature of the costs of debt is, they will lead firms to be willing to reduce the rate of return on bonds as B increases.

In such a case, the supply curve given in expression (2.12) will change into a decreasing function of the aggregate debt level B, $r_b^s(B)$, that will eventually cross the demand level $r_b^d = r'_\circ$ at some point, defining in that way a *before-tax equilibrium debt*

$$B^* = [r_b^s]^{-1}(r'_\circ)$$

and an *equilibrium rate of return on bonds*

$$r_b^* = r'_\circ.$$

2.2.4 Comparing with MM

The existence of these two competing theories to explain the equilibrium rate of return on bonds is compatible with the existence of possibly different types of marginal investors defining equilibrium. If these marginal investors can wash out the effect of personal taxation as suggested by Miller and Scholes (1978), the before-tax theory will dominate. Otherwise, in the presence of effective personal taxes, Miller's after-tax equilibrium will be the relevant one.

Notice that the two equilibria are quite different. As stressed before, in the *after-tax* equilibrium, the riskless rate $r_\circ$ is smaller than the risk-free

rate $r'_\circ$ in the *before-tax* equilibrium. This will be reflected in quite different costs of capital, as described in Hamada and Scholes (1985).

The cost of capital is given by the expected rate of return on a project at which a marginal investor would be indifferent between investing or not. From equation (1.17) in Chapter 1, the risk premium on a given project is proportional to the *systematic risk* β_p of the project. If R^* is the expected return on the portfolio with respect to which β is defined, for the after-tax theory this means that

$$E(R_p^{at}) - r_\circ = \beta_p[E(R^*) - r_\circ],$$

whereas for the before-tax theory this result reads

$$E(R_p^{bt}) - r'_\circ = \beta_p\big[E(R^*) - r'_\circ\big],$$

where $r_\circ = (1-\tau_s)r'_\circ$. Notice that the market portfolio may not be the same when taxes are considered. From these two relationships it is evident that the costs of capital associated with the two equilibria discussed here may be quite different, depending on the values of the various parameters.

2.2.5 Changes in Taxations: An Alternative Equilibrium

Once Modigliani and Miller established their result describing the tax advantages of debt, most of the research in the field of corporate finance involved the relevant costs that could define an optimal debt-equity ratio.

However, the arguments raised in the initial analysis of Miller's bond market equilibrium point to the possibility that there might be no real tax advantages. Hence, the costs previously discussed are apparently balancing a nonexistent advantage.

In order to solve this incompatibility, Miller (1977, p. 271) hinted at the origin of an alternative source of debt advantage. The point is that companies in default will not yield the issuing stockholders the whole interest deduction on debt, their tax shield. If that is the case, the rate of return on bonds supplied by the stockholders will tend to be lower than the rate of return given in (2.10) in order to account for this cost, and the relation between the tax rates will no longer be satisfied. Finally, there seems to exist a natural tax advantage of debt.

This idea has been worked out especially by DeAngelo and Masulis (1980). Their theory is essentially an analysis of the supply side of a generalized after-tax equilibrium. There are two main forces lowering the supplied rate of return on bonds as the aggregate debt increases; the first is the underutilization of tax shields and the second is the leverage-related deadweight costs driven from the before-tax equilibrium.

observed leverage and historical failure rates is reported. Also, Bradley, Jarrell, and Kim (1984) show that long-run average debt ratios are significantly negatively related to earnings volatility. This is compatible with the balancing theory discussed here, since the probability of financial distress for a given degree of financial leverage should increase with the variability of the earnings. Hence, there seems to be some evidence providing support for the generalized Miller's equilibrium.

2.2.7 Exercises

1. Show that in the presence of personal income tax and corporate income tax, the gain from leverage to stockholders is given by the expression

$$G_L = \left[1 - \frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}\right]B$$

   where B denotes the market value of the debt in the leveraged firm.
2. In the presence of underutilization of a tax shield or leverage-related deadweight costs, the after-tax equilibrium model discussed in Kim (1989) says that the equilibrium is reached at a value of t_c lower than the top tax rate. In this case, what is the best strategy for companies at the top corporate tax rate?

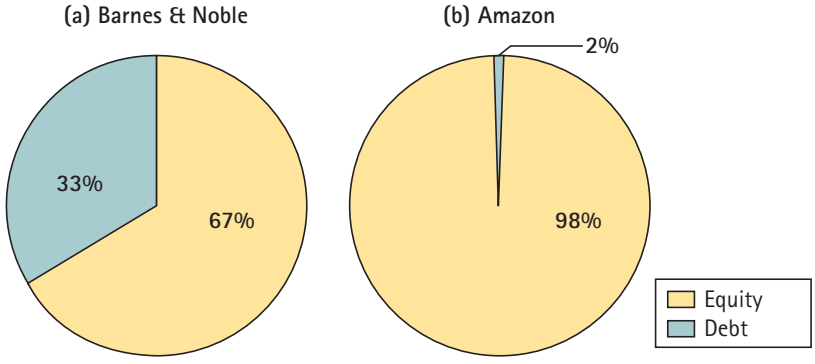
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Fundamentals of Corporate Finance, 2nd Edition {Jonathan Ber

- Capital structure in perfect markets: MM I and MM II: Ch. 16, §16.2 (p. 464–473)
- Debt and taxes: interest tax shield, recapitalisation: Ch. 16, §16.3 (p. 474–479)
- Distress costs and the trade-off theory: Ch. 16, §16.4–16.5 (p. 480–482)
- Dividends vs repurchases in a perfect market: Ch. 17, §17.2 (p. 503–509)

FIGURE 16.2
Capital Structures of Amazon.com and Barnes & Noble

The pie charts in panels (a) and (b) show the split between debt and equity in two competing firms, Amazon.com and Barnes & Noble. Both are part of the book retailing industry. Even within an industry, two companies can choose a different mix of debt and equity.



Source: Authors' calculations from http://finance.google.com (July 2010).

service, but it is a small part of its overall business. In this chapter, we will learn why these differences in assets and business plans would naturally lead to the different capital structures we observe today.

In the next section, we start by developing the critical theoretical foundation of any capital structure analysis. Then we move on to important real-world factors that managers must trade off when considering capital structure decisions.

Concept Check

1. What constitutes a firm's capital structure?
2. What are some factors a manager must consider when making a financing decision?

16.2 Capital Structure in Perfect Capital Markets

When a firm issues debt, equity, or other securities to fund a new investment, there are many potential consequences of its decision. By far the most important question for a financial manager is whether different choices will affect the value of the firm and thus the amount of capital it can raise. We begin by considering this question in a simple environment—a perfect capital market. A **perfect capital market** is a market in which:

perfect capital markets A set of conditions in which investors and firms can trade the same set of securities at competitive market prices; there are no frictions and the firm's financing decisions do not change the cash flows generated by its investments.

1. *Securities are fairly priced.* Investors and firms can trade the same set of securities at competitive market prices equal to the present value of their future cash flows.
2. *There are no tax consequences or transactions costs.* There are no tax consequences, transactions costs, or other issuance costs associated with financing decisions or security trading.
3. *Investment cash flows are independent of financing choices.* A firm's financing decisions do not change the cash flows generated by its investments, nor do they reveal new information about those cash flows.

The assumption of perfect capital markets may seem narrow and unrealistic. We will see, however, that by starting with them we will gain important insight into the true benefits and costs of leverage.

Application: Financing a New Business

Let's begin with an example of a possible financing decision in a perfect capital market. Imagine you have one year left in college and you want to earn some extra money before graduation. You have been offered the opportunity to run the coffee shop in the lobby of a nearby office building. The owner of the building is willing to grant you this right on a one-year basis prior to a major remodeling of the building.

Your research indicates that you will need to make an upfront investment of $24,000 to start the business. After covering your operating costs, including paying yourself a nice wage, you expect to generate a cash flow of $34,500 at the end of the year. The current risk-free interest rate is 5%. You believe, however, that your profits will be somewhat risky and sensitive to the overall market (which will affect the level of activity in the building and demand for your business), so that a 10% risk premium is appropriate, for a total discount rate of 15% (5% + 10%). You therefore calculate the NPV of this investment in the coffee shop as:

$$NPV = -\$24{,}000 + \frac{\$34{,}500}{1.15} = -\$24{,}000 + \$30{,}000 = \$6000$$

Thus, the investment has a positive NPV.

Even though the investment looks attractive, you still need to raise the money for the upfront investment. How should you raise the funds, and what is the amount you will be able to raise?

Equity Financing. First, you consider raising money solely by selling equity in the business to your friends and family. Given the cash flow estimates, how much would they be willing to pay for those shares? Recall that the value of a security equals the present value of its future cash flows. In this case, equity holders in your firm will expect to receive the payoff of $34,500 at the end of the year, with the same risk as the cash flows generated by the coffee shop. Therefore, your firm's equity cost of capital will be 15% and the value of its equity today will be:

$$PV(\text{equity cash flows}) = \frac{\$34{,}500}{1.15} = \$30{,}000$$

unlevered equity Equity in a firm with no debt.

Recall that the absence of debt means the absence of financial leverage. Equity in a firm with no debt is therefore called **unlevered equity**. Because the present value of the equity cash flows is $30,000, you can raise $30,000 by selling all the unlevered equity in your firm. Doing so allows you to keep the NPV of $6000 as profit after paying the investment cost of $24,000. In other words, the project's NPV represents the value to the initial owner of the firm (in this case, you, the entrepreneur) created by the project.

Levered Financing. As an alternative, you also consider borrowing some of the money you will need to invest. Suppose the business's cash flow is certain to be at least $16,000. Then you can borrow $15,000 at the current risk-free interest rate of 5%. You will be able to pay the debt of $15,000 × 1.05 = $15,750 at the end of the year without any risk of defaulting.

levered equity Equity in a firm with outstanding debt.

How much can you raise selling equity in your business now? Equity in a firm that also has outstanding debt is called **levered equity**. After the debt is repaid, equity holders can expect to receive $34,500 − $15,750 = $18,750. What discount rate should we use to value levered equity? What expected return will investors demand?

It's tempting to use the same 15% equity cost of capital as before. In that case, by selling levered equity, you could raise $18,750/1.15 = $16,304. If this result were correct, then using leverage would allow you to raise a total amount, including the debt, of $15,000 + $16,304 = $31,304, or $1304 more than in the case without leverage.

Thus, it would seem that simply financing a project with leverage can make it more valuable. *But if this sounds too good to be true, it is.* Our analysis assumed that your firm's equity cost of capital remained unchanged at 15% after adding leverage. But as we will see shortly, that will not be the case—leverage will increase the risk of the firm's equity and raise its equity cost of capital. To see why, and to understand what will actually happen, we turn to the hallmark work of researchers Franco Modigliani and Merton Miller.

Leverage and Firm Value

In an important paper, researchers Modigliani and Miller (or simply MM) considered whether leverage would increase the total value of the firm. Their answer to this question surprised researchers and practitioners at the time.[1] They argued that with perfect capital markets, the total value of a firm should *not* depend on its capital structure. Their reasoning: Your firm's total cash flows—those paid to both debt and equity holders—still equal the cash flows of the coffee shop, with the same expected value of $34,500 and the same total risk as before, as shown in Figure 16.3. Because the total cash flows of the debt and equity equal the cash flows of the unlevered firm, the Valuation Principle tells us that their market values must be the same. Specifically, we calculated earlier that the value of the unlevered firm, V^U, is:

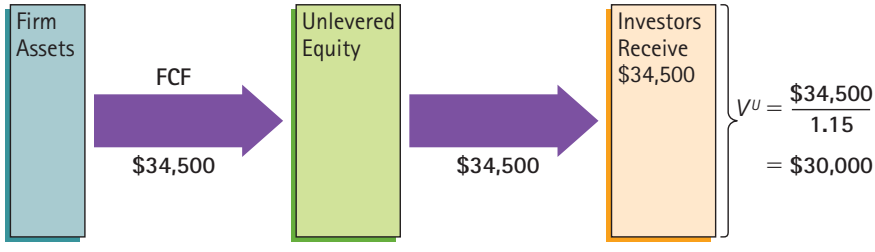
$$V^U = \$34{,}500/1.15 = \$30{,}000$$

FIGURE 16.3

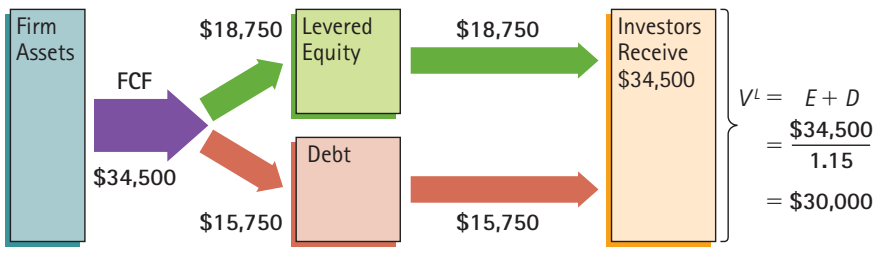
Unlevered Versus Levered Cash Flows with Perfect Capital Markets

When the firm has no debt, as shown in panel (a), the cash flows paid to equity holders correspond to the free cash flows generated by the firm's assets. When the firm has the debt, shown in panel (b), these cash flows are divided between debt and equity holders. However, with perfect capital markets, the total amount paid to all investors still corresponds to the free cash flows generated by the firm's assets. Therefore, the value of the unlevered firm, V^U, must equal the total value of the levered firm, V^L, which is the combined value of its debt D and levered equity E.

Panel (a), Unlevered Cash Flows



Panel (b), Levered Cash Flows



[1] F. Modigliani and M. Miller, "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review* 48 (3) (1958): 261–297.

Thus, the total value of the levered firm, V^L, which is the combined value of its debt, D, and levered equity, E, must be the same:

$$V^L = D + E = \$30{,}000$$

Therefore, if the initial market value of the debt is $D = \$15{,}000$ (the amount borrowed), the initial market value of the levered equity must be $E = \$30{,}000 - \$15{,}000 = \$15{,}000$.

Thus, Modigliani and Miller argued that leverage merely changes the allocation of cash flows between debt and equity without altering the total cash flows of the firm in a perfect capital market. As a result, they concluded that:

MM Proposition I: *In a perfect capital market, the total value of a firm is equal to the market value of the free cash flows generated by its assets and is not affected by its choice of capital structure.*

We can write this result in an equation as follows:

$$V^L = E + D = V^U \tag{16.1}$$

This equation states that the total value of the firm is the same with or without leverage.

Note in our example that because the cash flows of levered equity are smaller than those of unlevered equity, levered equity will sell for a lower price than unlevered equity ($15,000 versus $30,000). However, the fact that the equity is less valuable with leverage does not mean that you are worse off. You will still raise a total of $30,000 by issuing both debt and levered equity, just as you did with unlevered equity alone, and still keep the $6000 difference between the $30,000 you raise and your $24,000 cost as profit. As a consequence, you will be indifferent between these two choices for the firm's capital structure.

The Effect of Leverage on Risk and Return

Modigliani and Miller's conclusion went against the common view that even with perfect capital markets, leverage would affect a firm's value. In particular, it was thought that the value of the levered equity should exceed $15,000, because the present value of its expected cash flow at a 15% discount rate is $18,750/1.15 = $16,304, as we calculated earlier. The reason this is *not* correct is that leverage increases the risk of the equity of a firm. Therefore, it is inappropriate to discount the cash flows of levered equity at the same discount rate of 15% that we used for unlevered equity.

Let's take a closer look at the effect of leverage on the firm's equity cost of capital. If equity holders are only willing to pay $15,000 for the levered equity, then given its expected payoff of $18,750, their expected return is:

$$\text{Expected return of levered equity} = \$18{,}750/\$15{,}000 - 1 = 25\%$$

Although this return may seem like a good deal for investors, remember that the coffee shop cash flows are uncertain. Table 16.1 considers the different levels of demand and free

TABLE 16.1

Returns to Equity in Different Scenarios with and Without Leverage

Coffee Shop	Security Cash Flows			Security Returns		
Demand	Free Cash Flows	Unlevered Equity	Debt	Levered Equity	Unlevered Equity	Debt	Levered Equity
Weak	$27,000	$27,000	$15,750	$11,250	−10%	5%	−25%
Expected	**$34,500**	**$34,500**	**$15,750**	**$18,750**	**15%**	**5%**	**25%**
Strong	$42,000	$42,000	$15,750	$26,250	40%	5%	75%

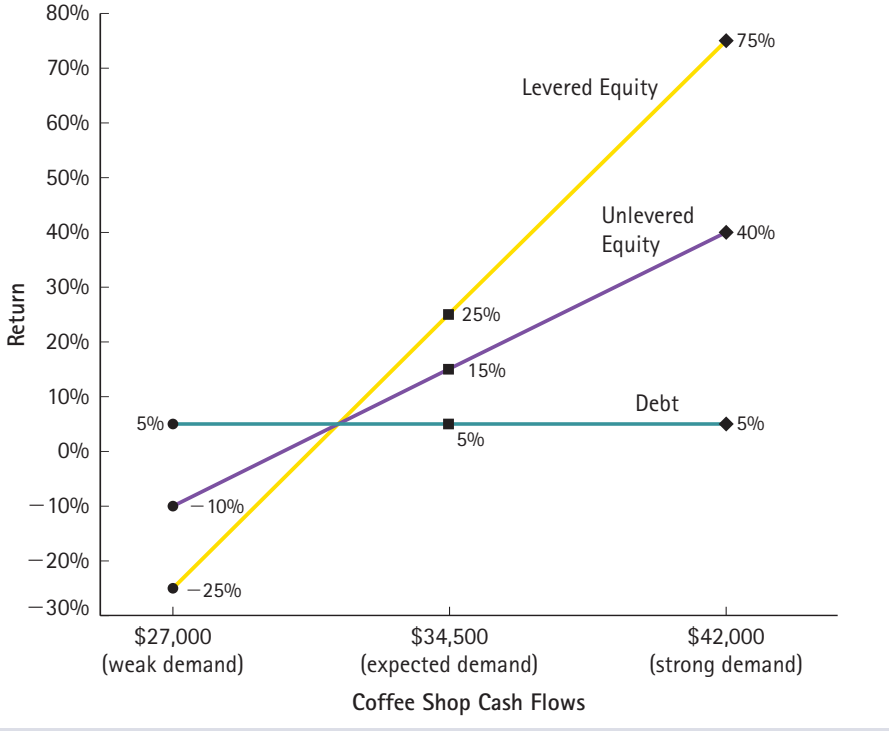
cash flows that the coffee shop may generate, and compares the security payoffs and returns with unlevered equity to the case in which you borrow $15,000 and raise an additional $15,000 using levered equity. Note that the returns are very different with and without leverage. With no debt, the returns of unlevered equity range from −10% to 40%, with an expected return of 15%. With leverage, the debt holders receive a risk-free return of 5%, whereas the returns of levered equity are much more volatile, with a range of −25% to 75%. To compensate for this higher risk, levered equity holders receive a higher expected return of 25%.

We further illustrate the effect of leverage on returns in Figure 16.4. By adding leverage, the returns of the unlevered firm are effectively "split" between low-risk debt and much higher-risk levered equity. Note that the returns of levered equity fall twice as fast as those of unlevered equity if the coffee shops cash flows decline. This doubling of the risk justifies a doubling of the risk premium, which is 15% − 5% = 10% for unlevered equity and 25% − 5% = 20% for levered equity. As this example shows, *leverage increases the risk of equity even when there is no risk that the firm will default.*

FIGURE 16.4

Unlevered Versus Levered Returns with Perfect Capital Markets

Leverage splits the firm's return between low-risk debt and high-risk levered equity compared to the equity of an unlevered firm. In this example, the returns of levered equity are twice as sensitive to the firm's cash flows as the returns of unlevered equity. This doubling of risk implies a doubling of the risk premium from 10% to 20%.



EXAMPLE 16.1

The Risk and Return of Levered Equity

Problem

Suppose you borrow only $6000 when financing your coffee shop. According to Modigliani and Miller, what should the value of the equity be? What is the expected return?

Solution

Plan

The value of the firm's total cash flows does not change: It is still \$30,000. Thus, if you borrow \$6000, your firm's equity will be worth \$24,000. To determine the equity's expected return, we will compute the cash flows to equity under the two scenarios. The cash flows to equity are the cash flows of the firm net of the cash flows to debt (repayment of principal plus interest).

Execute

The firm will owe debt holders \$6000 × 1.05 = \$6300 in one year. Thus, the expected payoff to equity holders is \$34,500 − \$6300 = \$28,200, for a return of \$28,200/\$24,000 − 1 = 17.5%.

Evaluate

While the total value of the firm is unchanged, the firm's equity in this case is more risky than it would be without debt, but less risky than if the firm borrowed \$15,000. To illustrate, note that if demand is weak, the equity holders will receive \$27,000 − \$6300 = \$20,700, for a return of \$20,700/\$24,000 − 1 = −13.75%. Compare this return to −10% without leverage and −25% if the firm borrowed \$15,000. As a result, the expected return of the levered equity is higher in this case than for unlevered equity (17.5% versus 15%), but not as high as in the previous example (17.5% versus 25%).

Homemade Leverage

MM showed that the firm's value is not affected by its choice of capital structure. But suppose investors would prefer an alternative capital structure to the one the firm has chosen. MM demonstrated that in this case, investors can borrow or lend on their own and achieve the same result. For example, an investor who would like more leverage than the firm has chosen can borrow and thus add leverage to his or her own portfolio. Adding leverage in this way will lower the out-of-pocket cost of the security, but increase the risk of the portfolio. When investors use leverage in their own portfolios to adjust the leverage choice made by the firm, we say that they are using **homemade leverage**. As long as investors can borrow or lend at the same interest rate as the firm, which is true in perfect capital markets, homemade leverage is a perfect substitute for the use of leverage by the firm. Thus, because different choices of capital structure offer no benefit to investors in perfect capital markets, those choices do not affect the value of the firm.

homemade leverage
When investors use leverage in their own portfolios to adjust the leverage choice made by a firm.

Leverage and the Cost of Capital

We can use the insight of Modigliani and Miller to understand the effect of leverage on the firm's cost of capital. Recall from Figure 16.3 and Eq. 16.1 that if we look at a portfolio of the equity and debt of a levered firm together, that portfolio has the same value and cash flows as the unlevered firm. Therefore, the expected return of the portfolio should equal the expected return of the unlevered firm. Recall from Chapter 12 that the expected return of the portfolio equity and debt is simply the weighted average of the expected returns of each security. Thus, with r_E representing the expected return of levered equity, r_D the expected return of debt, and r_U the expected return of the unlevered equity, we have:

Weighted Average Cost of Capital (Pretax)

$$\underbrace{r_E\frac{E}{E+D} + r_D\frac{D}{E+D}}_{\text{Pretax WACC}} = r_U \tag{16.2}$$

The amounts $\frac{E}{E+D}$ and $\frac{D}{E+D}$ represent the fraction of the firm's value financed by equity and debt, respectively. Thus, the left side of Eq. 16.2 is the weighted average cost

pretax WACC The weighted average cost of capital computed using the pretax cost of debt.

of capital (WACC) of the firm, which we defined in Chapter 13. Note that the cost of debt is not adjusted for taxes, because we are assuming "perfect capital markets" and thus are ignoring taxes. When we compute the weighted average cost of capital without taxes, we refer to it as the firm's **pretax WACC**. Equation 16.2 states that for any choice of capital structure, the firm's pretax WACC is unchanged and remains equal to the firm's unlevered cost of capital.

Let's check this result for our coffee shop example. With leverage, the firm's debt cost of capital was $r_D = 5\%$, and its equity cost of capital rose to $r_E = 25\%$. What about the portfolio weights? In this case, the firm borrowed $D = 15{,}000$, and issued equity worth $E = 15{,}000$, for a total value $V^L = E + D = 30{,}000$. Therefore, its pretax WACC is:

$$r_E \frac{E}{E+D} + r_D \frac{D}{E+D} = 25\%\left(\frac{15{,}000}{30{,}000}\right) + 5\%\left(\frac{15{,}000}{30{,}000}\right) = 15\%$$

Thus, the pretax WACC does indeed equal the firm's unlevered cost of capital, $r_U = 15\%$.

How is it that the firm's weighted average cost of capital stays the same even after adding leverage? There are two offsetting effects of leverage: we finance a larger fraction of the firm with debt, which has a lower cost of capital, but at the same time adding leverage raises the firm's equity cost of capital. Because the firm's total risk has not changed (it has just been split between these two securities), these two effects should exactly cancel out and leave the firm's WACC unchanged. In fact, we can use Eq. 16.2 to determine

COMMON MISTAKE

Capital Structure Fallacies

Here, we take a critical look at two incorrect arguments that are sometimes cited in favor of leverage.

Leverage and Earnings per Share

Fallacy 1: Leverage can increase a firm's expected earnings per share; by doing so, leverage should also increase the firm's stock price.

Consider the coffee shop example. In the all-equity case, if you issued 1000 shares, they would each be worth \$30 and the expected earnings per share (EPS) would be \$34,500/1000 = \$34.50. EPS would vary from \$27 per share to \$42 per share in the case of weak or strong demand. In the case with debt, as shown in Table 16.1, you only need to raise \$15,000 in equity and so could issue only 500 shares, each worth \$30. In this case, your expected EPS will be \$18,750/500 = \$37.50, with a range from \$22.50 to \$52.50. Thus, although the expected EPS is greater with leverage, the variation in EPS is also much greater, as shown in the blue and yellow bars. With leverage, EPS falls to \$22.50 when cash flows are low, which is much further than EPS would have dropped without leverage (\$27). Although EPS increases on average, this increase is necessary to compensate shareholders for the additional risk they are taking. As a result, the entrepreneur's share price does not increase as a result of issuing debt.



Equity Issuances and Dilution

Fallacy 2: Issuing equity will *dilute* existing shareholders' ownership, so debt financing should be used instead. *Dilution* means that if the firm issues new shares, the cash flows generated by the firm must be divided among a larger number of shares, thereby reducing the value of each individual share.

This line of reasoning ignores the fact that the cash raised by issuing new shares will increase the firm's assets. Consider Google's September 2005 seasoned equity offering of 14,159,265 Class A shares at \$295 apiece. Google priced the shares to match the market price of Class A shares on NASDAQ at the time of the offer. The amount raised was then \$4,176,983,175, so the total value of Google increased to \$60,560,157,355, which when divided by the new total number of shares (205,288,669), still results in a price of \$295 per share.

In general, as long as the firm sells the new shares of equity *at a fair price,* there will be no gain or loss to shareholders associated with the equity issue itself. The money taken in by the firm as a result of the share issue exactly offsets the dilution of the shares. *Any gain or loss associated with the transaction will result from the NPV of the investments the firm makes with the funds raised.*

the precise impact of leverage on the firm's equity cost of capital. Solving the equation for r_E, we have:

MM Proposition II: The Cost of Capital of Levered Equity

$$r_E = r_U + \frac{D}{E}(r_U - r_D) \tag{16.3}$$

Or, in words:

MM Proposition II: The cost of capital of levered equity is equal to the cost of capital of unlevered equity plus a premium that is proportional to the debt-equity ratio (measured using market values).

Let's check MM Proposition II for the coffee shop example. In this case:

$$r_E = r_U + \frac{D}{E}(r_U - r_D) = 15\% + \frac{15{,}000}{15{,}000}(15\% - 5\%) = 25\%$$

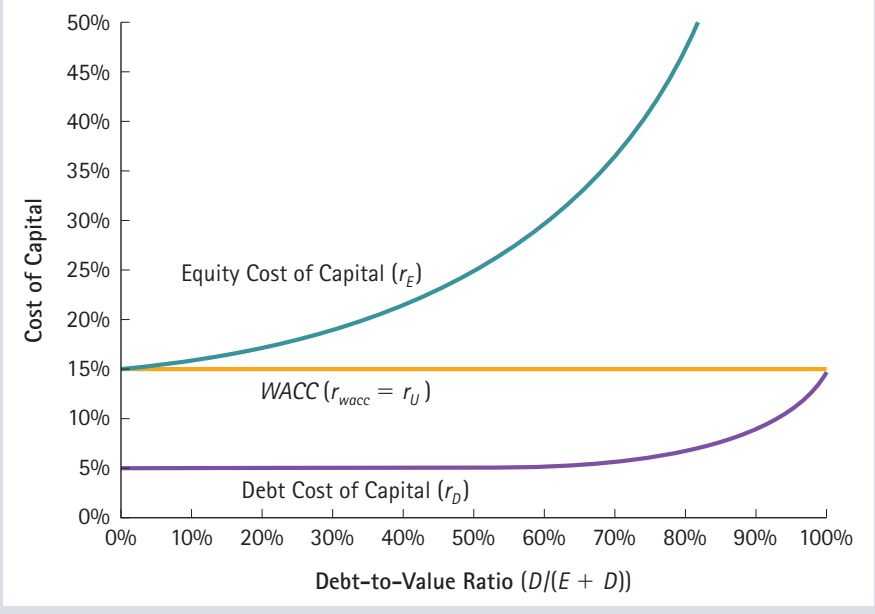
This result matches the expected return for levered equity we calculated in Table 16.1.

Figure 16.5 illustrates the effect of increasing the amount of leverage in a firm's capital structure on its equity cost of capital, its debt cost of capital, and its WACC. In the

FIGURE 16.5 WACC and Leverage with Perfect Capital Markets

Panel (a) represents the data in panel (b) for the coffee shop example. As the fraction of the firm financed with debt increases, both the equity and the debt become riskier and their cost of capital rises. Yet because more weight is put on the lower-cost debt, the weighted average cost of capital, r_{wacc}, remains constant.

Panel (a) Equity, Debt, and WACC for Different Amounts of Leverage



Panel (b) WACC Data for Alternative Capital Structures

E	D	r_E	r_D	$r_E \frac{E}{E+D} + r_D \frac{D}{E+D}$	$= r_{wacc}$
30,000	0	15.0%	5.0%	15.0% × 1.0 + 5.0% × 0.0	= 15%
24,000	6,000	17.5%	5.0%	17.5% × 0.8 + 5.0% × 0.2	= 15%
15,000	15,000	25.0%	5.0%	25.0% × 0.5 + 5.0% × 0.5	= 15%
3,000	27,000	75.0%	8.3%	75.0% × 0.1 + 8.3% × 0.9	= 15%

figure, we measure the firm's leverage in terms of its debt-to-value ratio, $D/(E + D)$. With no debt, the WACC is equal to the unlevered equity cost of capital. As the firm borrows at the low cost of capital for debt, its equity cost of capital rises, according to Eq. 16.3. The net effect is that the firm's WACC is unchanged. Of course, as the amount of debt increases, the debt becomes riskier because there is a chance the firm will default; as a result, the debt cost of capital also rises. With close to 100% debt, the debt would be almost as risky as the assets themselves (similar to unlevered equity). But even though the debt and equity costs of capital both rise when leverage is high, because more of the firm is financed with debt (which has lower cost), the WACC remains constant.

EXAMPLE 16.2
Computing the Equity Cost of Capital

Problem

Suppose you borrow only $6000 when financing your coffee shop. According to MM Proposition II, what will your firm's equity cost of capital be?

Solution

Plan

Because your firm's assets have a market value of $30,000, according to MM Proposition I the equity will have a market value of $\$24{,}000 = \$30{,}000 - \$6000$. We can use Eq. 16.3 to compute the cost of equity. We know the unlevered cost of equity is $r_u = 15\%$. We also know that r_D is 5%.

Execute

$$r_E = 15\% + \frac{6000}{24{,}000}(15\% - 5\%) = 17.5\%$$

Evaluate

This result matches the expected return calculated in Example 16.1 where we also assumed debt of $6000. The equity cost of capital should be the expected return of the equity holders.

MM and the Real World

Our conclusions so far may seem striking at first: In perfect capital markets, leverage does not affect either the cost of capital or a firm's value, and so the firm's choice of capital structure would be irrelevant! However, capital markets are not perfect in the real world. What then are we to make of Modigliani and Miller's results?

NOBEL PRIZE

Franco Modigliani and Merton Miller

Franco Modigliani and Merton Miller, authors of the Modigliani-Miller Propositions, each won the Nobel Prize in Economics for their work in financial economics, including their capital structure propositions. Modigliani won the Nobel Prize in 1985 for his work on personal savings and for his capital structure theorems with Miller. Miller earned his prize in 1990 for his analysis of portfolio theory and capital structure.

Miller once described the MM propositions in an interview this way:

> *People often ask, "Can you summarize your theory quickly?" Well, I say, you understand the MM theorem if you know why this is a joke: The pizza delivery man comes to Yogi Berra after the game and says, "Yogi, how do you want this pizza cut, into quarters or eighths?" And Yogi says, "Cut it in eight pieces. I'm feeling hungry tonight."*
>
> *Everyone recognizes that's a joke because obviously the number and shape of the pieces don't affect the size of the pizza. And similarly, the stocks, bonds, warrants, etc., issued don't affect the aggregate value of the firm. They just slice up the underlying earnings in different ways.**

Modigliani and Miller each won the Nobel Prize in large part for their observation that the value of a firm should be unaffected by its capital structure in perfect capital markets. Whereas the intuition underlying the MM propositions may be as simple as slicing pizza, their implications for corporate finance are far-reaching. The propositions imply that the true role of a firm's financial policy is to deal with (and potentially exploit) financial market imperfections such as taxes and transaction costs. Modigliani and Miller's work began a long line of research into these market imperfections, which we will look at in the rest of the chapter.

*Peter J. Tanous, *Investment Gurus* (New York: Institute of Finance, 1997).

As an analogy, consider Galileo's law of falling bodies. Galileo overturned the conventional wisdom by showing that without friction, free-falling bodies will fall at the same rate, independent of their mass. If you test this law, you will likely find it does not hold exactly. The reason, of course, is that unless we are in a vacuum, air friction tends to slow some objects more than others.

MM's results are similar. In practice, we will find that capital structure can have an effect on a firm's value. Galileo's law of falling bodies reveals that we must look to air friction, rather than any underlying property of gravity, to explain differences in the speeds of falling objects. MM's propositions reveal that any effects of capital structure must similarly be due to frictions that exist in capital markets. We explore the important sources of these frictions, and their consequences, in the remainder of this chapter.

Concept Check

3. How does leverage affect the risk and cost of equity for the firm?
4. In a perfect capital market, can you alter the firm's value or WACC by relying more on debt capital?

16.3 Debt and Taxes

So far, we have used the perfect capital markets setting to focus on the fundamental point that the firm's choice of projects and investments is the primary determinant of its value and risk, and hence its overall cost of capital. But in the real world, markets are imperfect and these imperfections can create a role for the firm's capital structure. In this section, we focus on one important market friction—corporate taxes—and show how the firm's choice of capital structure can affect the taxes that it must pay and therefore its value to investors.

The Interest Tax Deduction and Firm Value

As we discussed in Chapter 13, corporations can deduct interest expenses from their taxable income. The deduction reduces the taxes they pay and thereby increases the amount available to pay investors. In doing so, the interest tax deduction increases the value of the corporation.

To illustrate, let's consider the impact of interest expenses on the taxes paid by Safeway, Inc., a grocery store chain. In 2008, Safeway had earnings before interest and taxes of $1.85 billion, and interest expenses of $350 million. Given a corporate tax rate of 35%, we can compare Safeway's actual net income with what it would have been without debt, as shown in Table 16.2.

TABLE 16.2

Safeway's Income with and without Leverage, 2008 ($ millions)

	With Leverage	Without Leverage
EBIT	$1850	$1850
Interest expense	-350	0
Income before tax	1500	1850
Taxes (35%)	–525	–648
Net income	$975	$1202

As we can see from Table 16.2, Safeway's net income in 2008 was lower with leverage than it would have been without leverage. Thus, Safeway's debt obligations reduced the value of its equity. But more importantly, the *total* amount available to *all* investors was higher with leverage:

	With Leverage	Without Leverage
Interest paid to debt holders	350	0
Income available to equity holders	975	1202
Total available to all investors	**$1325**	**$1202**

With leverage, Safeway was able to pay out $1.325 billion in total to its investors, versus only $1.202 billion without leverage, representing an increase of $123 million.

It might seem odd that a firm can be better off with leverage even though its earnings are lower. But recall from Section 16.1 that the value of a firm is the total amount it can raise from all investors, not just equity holders who receive the earnings. Thus, if the firm can pay out more in total with leverage, it will be able to raise more total capital initially.

Where does the additional $123 million come from? Looking at Table 16.2 we can see that this gain is equal to the reduction in taxes with leverage: $648 million − $525 million = $123 million. Because Safeway does not owe taxes on the $350 million of pretax earnings it used to make interest payments, this $350 million is *shielded* from the corporate tax, providing the tax savings of 35% × $350 million = $123 million.

interest tax shield The reduction in taxes paid due to the tax deductibility of interest payments.

In general, the gain to investors from the tax deductibility of interest payments is referred to as the **interest tax shield.** The interest tax shield is the additional amount a firm can pay to investors by saving the taxes it would have paid if it did not have leverage. We can calculate the amount of the interest tax shield each year as follows:

$$\text{Interest Tax Shield} = \text{Corporate Tax Rate} \times \text{Interest Payments} \tag{16.4}$$

EXAMPLE 16.3
Computing the Interest Tax Shield

Problem

Shown below is the income statement for E. C. Builders (ECB). Given its marginal corporate tax rate of 35%, what is the amount of the interest tax shield for ECB in years 2007 through 2010?

	ECB Income Statement ($ million)	2007	2008	2009	2010
2	Total sales	$3,369	$3,706	$4,077	$4,432
3	Cost of sales	−2,359	−2,584	−2,867	−3,116
4	Selling, general, and administrative expense	−226	−248	−276	−299
5	Depreciation	−22	−25	−27	−29
6	**Operating income**	762	849	907	988
7	Other income	7	8	10	12
9	**EBIT**	769	857	917	1,000
10	Interest expense	−50	−80	−100	−100
11	**Income before tax**	719	777	817	900
12	Taxes (35%)	−252	−272	−286	−315
13	**Net income**	$467	$505	$531	$585

Solution

Plan

From Eq. 16.4, the interest tax shield is the tax rate of 35% multiplied by the interest payments in each year.

Execute

	($ million)	2007	2008	2009	2010
2	Interest expense	50	80	100	100
3	**Interest tax shield (35% × interest expense)**	**17.5**	**28**	**35**	**35**

Evaluate

By using debt, ECB is able to reduce its taxable income and therefore decrease its total tax payments by $115.5 million over the four-year period. Thus, the total amount of cash flows available to all investors (debt holders and equity holders) is $115.5 million higher over the four-year period.

Value of the Interest Tax Shield

When a firm uses debt, the interest tax shield provides a corporate tax benefit each year. To determine the benefit of leverage for the value of the firm, we must compute the present value of the stream of future interest tax shields the firm will receive.

As we saw with Safeway and in Example 16.3, each year a firm makes interest payments, the cash flows it pays to investors will be higher than they would be without leverage by the amount of the interest tax shield:

$$\begin{pmatrix}\text{Cash Flows to Investors}\\ \text{with Leverage}\end{pmatrix} = \begin{pmatrix}\text{Cash Flows to Investors}\\ \text{without Leverage}\end{pmatrix} + (\text{Interest Tax Shield})$$

Figure 16.6 illustrates this relationship. Notice how each dollar of pretax cash flows is divided. The firm uses some fraction to pay taxes, and it pays the rest to investors. By increasing the amount paid to debt holders through interest payments, the amount of the pretax cash flows that must be paid as taxes decreases. The gain in total cash flows to investors is the interest tax shield.[2]

Because the cash flows of the levered firm are equal to the sum of the cash flows from the unlevered firm plus the interest tax shield, according to the Valuation Principle the same must be true for the present values of these cash flows. Thus, letting V^L and V^U represent the value of the firm with and without leverage, respectively, we have the following change to MM Proposition I in the presence of taxes:

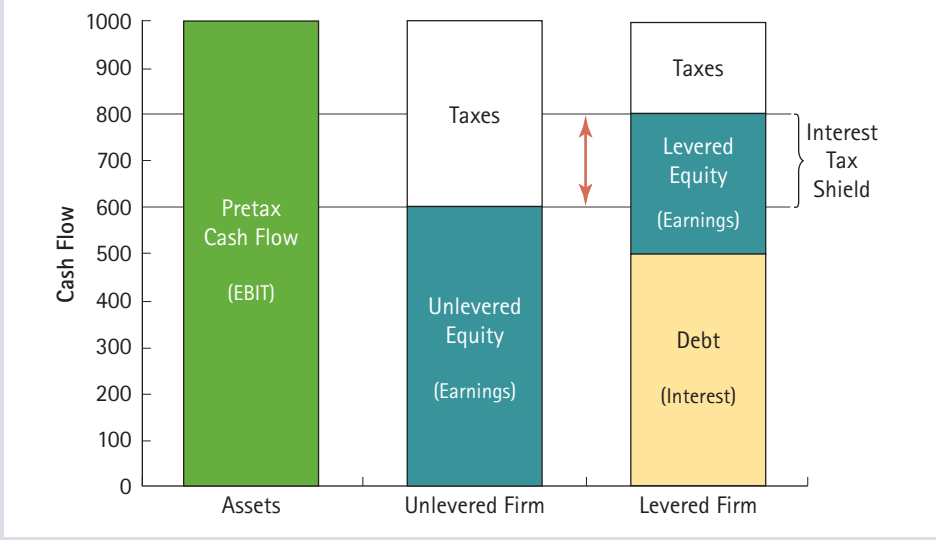
The total value of the levered firm exceeds the value of the firm without leverage due to the present value of the tax savings from debt:

$$V^L = V^U + PV\,(\text{Interest Tax Shield}) \tag{16.5}$$

Clearly, there is an important tax advantage to the use of debt financing. But how large is this tax benefit? To compute the increase in the firm's total value associated with the interest tax shield, we need to forecast how a firm's debt—and therefore its interest payments—will vary over time. Given a forecast of future interest payments, we can determine the interest tax shield and compute its present value by discounting it at a rate that corresponds to its risk.

FIGURE 16.6
The Cash Flows of the Unlevered and Levered Firm

By increasing the cash flows paid to debt holders through interest payments, a firm reduces the amount paid in taxes. The increase in total cash flows paid to investors is the interest tax shield. (The figure assumes a 40% marginal corporate tax rate.)



[2]If investors are taxed on interest income at a higher rate than they are on capital gains, this personal tax *disadvantage* of debt will partly offset the corporate tax advantage of debt.

EXAMPLE 16.4
Valuing the Interest Tax Shield

Problem

Suppose ECB from Example 16.3 borrows $2 billion by issuing ten-year bonds. ECB's cost of debt is 6%, so it will need to pay $120 million in interest each year for the next 10 years, and then repay the principal of $2 billion in year 10. ECB's marginal tax rate will remain 35% throughout this period. By how much does the interest tax shield increase the value of ECB?

Solution

Plan

In this case, the interest tax shield lasts for ten years, so we can value it as a ten-year annuity. Because the tax savings are as risky as the debt that creates them, we can discount them at ECB's cost of debt: 6%.

Execute

The interest tax shield each year is 35% × $120 million = $42 million. Valued as a ten-year annuity with a discount rate of 0.06, we have:

$$PV(\text{Interest Tax Shield}) = \$42 \text{ million} \times \frac{1}{0.06}\left(1 - \frac{1}{1.06^{10}}\right)$$
$$= \$309 \text{ million}$$

Because only interest is tax deductible, the final repayment of principal in year 10 is not deductible, so it does not contribute to the tax shield.

Evaluate

We know that in perfect capital markets, financing transactions have an NPV of zero—the interest and principal repayment have a present value of exactly the amount of the bonds: $2 billion. However, the interest tax deductibility makes this a positive-NPV transaction for the firm. Because the government effectively subsidizes the payment of interest, issuing these bonds has an NPV of $309 million.

The Interest Tax Shield with Permanent Debt

Many factors can affect the future tax savings from interest. Typically, the level of future interest payments varies due to:

Changes the firm makes in the amount of debt outstanding,
Changes in the interest rate on that debt,
Changes in the firm's marginal tax rate, and
The risk that the firm may default and fail to make an interest payment.

Rather than attempting to account for all possibilities here, we will consider the special case in which the firm issues debt and plans to keep the dollar amount of debt constant forever.

For example, the firm might issue a perpetual consol bond, making only interest payments but never repaying the principal. More realistically, suppose the firm issues short-term debt such as a five-year coupon bond. When the principal is due, the firm raises the money needed to pay it by issuing new debt. In this way, the firm never pays off the principal but simply refinances it whenever it comes due. In this situation, the debt is effectively permanent.

Many large firms have a policy of maintaining a certain amount of debt on their balance sheets. As old bonds and loans mature, they start new loans and issue new bonds. Note that we are considering the value of the interest tax shield with a *fixed* dollar amount of outstanding debt, rather than an amount that changes with the size of the firm.

As we learned in Chapter 6, if the debt is fairly priced, the Valuation Principle implies that the market value of the debt today must equal the present value of the future interest payments:[3]

$$\text{Market Value of Debt} = D = PV\,(\text{Future Interest Payments}) \tag{16.6}$$

If the firm's marginal tax rate (T_c) is constant, then we have the following general formula:

Value of the Interest Tax Shield of Permanent Debt

$$\begin{aligned} PV\,(\text{Interest Tax Shield}) &= PV(T_c \times \text{Future Interest Payments}) \\ &= T_c \times PV\,(\text{Future Interest Payments}) \\ &= T_c \times D \end{aligned} \tag{16.7}$$

This formula shows the magnitude of the interest tax shield. Given a 35% corporate tax rate, it implies that for every $1 in new permanent debt that the firm issues, the value of the firm increases by $0.35.

Leverage and the WACC with Taxes

There is another way we can incorporate the benefit of the firm's future interest tax shield. Recall from Chapter 13 that for companies without preferred stock (which is most companies) we defined the WACC with taxes as:

Weighted Average Cost of Capital with Taxes

$$r_{wacc} = r_E \frac{E}{E+D} + r_D(1 - T_C)\frac{D}{E+D} \tag{16.8}$$

In Eq. 16.8, we incorporate the benefit of the interest tax shield by adjusting the cost of debt to the firm. If the firm pays interest rate r_D on its debt, then because it receives a tax shield of $T_C \times r_D$, the effective after-tax cost of debt is reduced to $r_D(1 - T_C)$. Comparing Eq. 16.8 with Eq. 16.2 for the pretax WACC, we can see that corporate taxes lower the effective cost of debt financing, which translates into a reduction in the weighted average cost of capital. In fact, Eq. 16.8 implies:

$$r_{wacc} = \underbrace{r_E \frac{E}{E+D} + r_D \frac{D}{E+D}}_{\text{Pretax WACC}} \underbrace{- r_D T_C \frac{D}{E+D}}_{\substack{\text{Reduction Due} \\ \text{to Interest Tax Shield}}} \tag{16.9}$$

Thus, the reduction in the WACC increases with the amount of debt financing. The higher the firm's leverage, the more the firm exploits the tax advantage of debt and the lower its WACC. Figure 16.7 illustrates this decline in the WACC with leverage. The figure also shows the pretax WACC as shown in Figure 16.5.

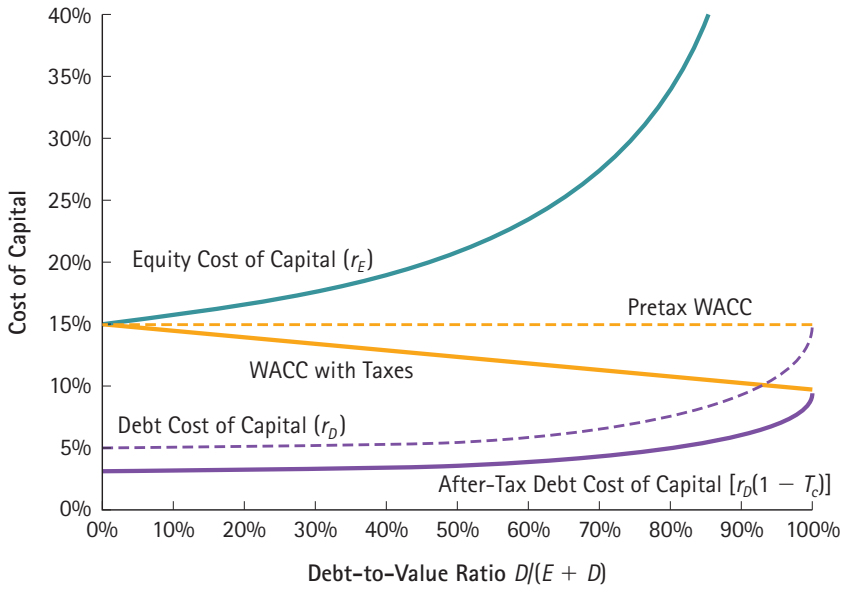
Debt and Taxes: The Bottom Line

In this section, we have seen that the deductibility of interest expenses for corporate taxes creates an advantage for debt financing. We can calculate the value of this benefit to the firm in two ways. First, we can forecast the firm's future interest tax shields and deter-

[3]Equation 16.6 is valid even if interest rates fluctuate and the debt is risky, as long as any new debt is also fairly priced. It requires only that the firm never repay the principal on the debt (it either refinances or defaults on the principal).

FIGURE 16.7
The WACC with and without Corporate Taxes

We compute the WACC as a function of leverage using Eq. 16.9. Whereas the pretax WACC remains constant, with taxes the WACC declines as the firm increases its reliance on debt financing and the benefit of the interest tax deduction grows. The figure assumes a marginal corporate income tax rate of 35%.



mine their present value. This approach is especially simple when the amount of debt is fixed permanently, in which case the value of the tax shield is equal to $T_C \times D$.

A second way to calculate the benefit of the interest tax shield is to incorporate it in the firm's cost of capital by using the WACC. Unlike the pretax WACC, the WACC with taxes declines with leverage because of the interest tax shield. If we use this lower discount rate to compute the present value of the firm's or an investment's free cash flow, the present value will be higher by an amount that reflects the benefit of the future interest tax shields. This approach is simplest to apply when the firm adjusts its debt to keep the fraction of debt financing (its debt-to-value ratio) constant over time.

To summarize, we can include the interest tax shield when assessing the value of a firm or an investment by *either* of these methods:

1. Discounting its free cash flow using the pretax WACC, and adding the present value of expected future interest tax shields, or
2. Discounting its free cash flow using the WACC (with taxes).

Using either of these methods, we will find the value of the firm will increase with leverage. Thus, unlike the setting with perfect capital markets, capital structure matters. But now we face a new puzzle: Given the tax benefit, why don't firms use debt financing almost exclusively?

Concept Check

5. How does the interest tax deduction affect firm value?
6. How does the firm's WACC change with leverage?

16.4 The Costs of Bankruptcy and Financial Distress

The last section presents us with an interesting question: If increasing debt increases the value of the firm, why not shift to nearly 100% debt? One part of the answer comes from bankruptcy costs. With more debt, there is a greater chance that the firm will be unable to make its required interest payments and will default on its debt obligations. A firm that has trouble meeting its debt obligations is in **financial distress**. Bankruptcy is a long and complicated process that imposes both direct and indirect costs on the firm and its investors that the assumption of perfect capital markets ignores.

financial distress When a firm has difficulty meeting its debt obligations.

Direct Costs of Bankruptcy

Each country has a bankruptcy code that details the process for dealing with a firm in default of its debt obligations (see the appendix to this chapter). The bankruptcy code is designed to provide an orderly process for settling a firm's debts. However, the process is still complex, time-consuming, and costly. When a corporation becomes financially distressed, outside professionals, such as legal and accounting experts, consultants, appraisers, auctioneers, and others with experience selling distressed assets, are generally hired. Investment bankers may also assist with a potential financial restructuring.

In addition to the money spent by the firm, the creditors may incur costs during the bankruptcy process. In the case of reorganization, creditors must often wait several years for a reorganization plan to be approved and to receive payment. To ensure that their rights and interests are respected, and to assist in valuing their claims in a proposed reorganization, creditors may seek separate legal representation and professional advice.

Studies typically report that the average direct costs of bankruptcy are approximately 3% to 4% of the pre-bankruptcy market value of total assets. The costs are likely to be higher for firms with more complicated business operations and for firms with larger numbers of creditors, because it may be more difficult to reach agreement among many creditors regarding the final disposition of the firm's assets. Because many aspects of the bankruptcy process are independent of the size of the firm, the costs are typically higher, in percentage terms, for smaller firms.

Bankruptcy Can Be Expensive

Outside experts that specialize in aiding distressed firms are costly. At the time Enron entered bankruptcy, it reportedly spent a record $30 million per month on legal and accounting fees, and the total cost ultimately exceeded $750 million. WorldCom paid its advisors $657 million as part of its reorganization to become MCI. Between 2003 and 2005, United Airlines paid a team of over 30 advisory firms an average of $8.6 million per month for legal and professional services related to its bankruptcy reorganization. Whether paid by the firm or its creditors, these direct costs of bankruptcy reduce the value of the assets that the firm's investors will ultimately receive. In the case of Enron, reorganization costs were expected to approach 10% of the value of the assets.

Source: Julie Johnsson, "UAL a Ch. 11 Fee Machine," *Crain's Chicago Business,* June 27, 2005.

Indirect Costs of Financial Distress

Aside from the direct legal and administrative costs of bankruptcy, many other *indirect* costs are associated with financial distress (whether or not the firm has formally filed for bankruptcy). Whereas these costs are difficult to measure accurately, they are often much larger than the direct costs of bankruptcy.

Indirect bankruptcy costs often occur because the firm may renege on both implicit and explicit commitments and contracts when in financial distress. For example, a

bankrupt software manufacturer need not fulfill an obligation to support one of its products. Knowing this, customers who rely on such support might choose to buy software from companies that have a low chance of bankruptcy; that is, companies with lower leverage. Importantly, many of these indirect costs may be incurred even if the firm is not yet in financial distress, but simply faces a significant possibility of bankruptcy in the future. Consider the following examples:

Loss of Customers. Because bankruptcy may enable firms to walk away from future commitments to their customers, those customers may be unwilling to purchase products whose value depends on future support or service from the firm.

Loss of Suppliers. Suppliers may be unwilling to provide a firm with inventory if they fear they will not be paid. For example, Swiss Air was forced to shut down in 2001 because financial concerns caused its suppliers to refuse to fuel its planes.

Cost to Employees. One important cost that often receives a great deal of press coverage is the cost of financial distress to employees. Most firms offer their employees explicit long-term employment contracts, or an implicit promise regarding job security. However, during bankruptcy these contracts and commitments are often ignored and significant numbers of employees may be laid off. In anticipation of this, employees will be less willing to work for firms with significant bankruptcy risk and so will demand a higher compensation to do so. Thus, hiring and retaining key employees may be costly for a firm with high leverage: Pacific Gas and Electric Corporation implemented a retention program costing over $80 million to retain 17 key employees while it was in bankruptcy in 2003.

Fire Sales of Assets. A company in distress may be forced to sell assets quickly to raise cash, possibly accepting a lower price than the assets are actually worth to the firm. This cost is likely to be large when creditors are more pessimistic than management regarding the value of the assets, and so creditors will try to force liquidation even at low prices.

In total, the indirect costs of financial distress may be substantial. A study of highly levered firms by Gregor Andrade and Steven Kaplan estimated a potential loss due to financial distress of 10% to 20% of firm value.[4] Importantly, many of these indirect costs may be incurred even if the firm is not yet in financial distress, but simply faces a significant possibility that it may occur in the future.

Concept Check

7. What are the direct costs of bankruptcy?
8. Why are the indirect costs of financial distress likely to be more important than the direct costs of bankruptcy?

16.5 Optimal Capital Structure: The Tradeoff Theory

tradeoff theory The total value of a levered firm equals the value of the firm without leverage plus the present value of the tax savings from debt, less the present value of financial distress costs.

We can now combine our knowledge of the benefits of leverage from the interest tax shield with the costs of financial distress to determine the amount of debt that a firm should issue to maximize its value. The analysis presented in this section is called the *tradeoff theory* because it weighs the benefits of debt that result from shielding cash flows from taxes against the costs of financial distress associated with leverage. This theory is also sometimes referred to as the *static tradeoff theory*.

According to the **tradeoff theory**, the total value of a levered firm equals the value of the firm without leverage plus the present value of the tax savings from debt, less the

[4]Andrade, G. and S. Kaplan, 1998, "How Costly Is Financial (Not Economic) Distress? Evidence from Highly Leveraged Transactions that Became Distressed," *Journal of Finance* 53: 1443–1493.

present value of financial distress costs:

$$V^L = V^U + PV(\textit{Interest Tax Shield}) - PV(\textit{Financial Distress Costs}) \quad (16.10)$$

Equation 16.10 shows that leverage has costs as well as benefits. Firms have an incentive to increase leverage to exploit the tax benefits of debt. But with too much debt, they are more likely to risk default and incur financial distress costs.

Differences Across Firms

Whereas we have seen in Section 16.3 how to calculate the benefits of the interest tax shield, calculating the precise present value of financial distress costs is very difficult if not impossible. Two key qualitative factors determine the present value of financial distress costs: (1) the probability of financial distress and (2) the magnitude of the direct and indirect costs related to financial distress that the firm will incur.

What determines each of these factors? The magnitude of the financial distress costs will depend on the relative importance of the sources of these costs and is likely to vary by industry. For example, technology firms are likely to incur high costs associated with financial distress, due to the potential for loss of customers and key personnel, as well as a lack of tangible assets that can be easily liquidated. In contrast, forest products companies are likely to have low costs of financial distress, as much of their value derives from tangible assets (timber land and mills) that can be sold if necessary. Not surprisingly, by reviewing Figure 16.1 we can see that these two industries have very different leverage policies: Technology companies have very low debt, whereas forest products companies tend to be highly levered. Similarly, in Figure 16.2 we saw that Barnes & Noble has a lot of debt whereas Amazon has virtually none. Barnes & Noble has stores and land that can be sold to meet the claims of lenders. Amazon has distribution facilities, but little else other than its book inventory that could be sold to meet the demands of creditors.

The probability of financial distress depends on the likelihood that a firm will be unable to meet its debt commitments and therefore default. This probability increases with the amount of a firm's liabilities (relative to its assets). It also increases with the volatility of a firm's cash flows and asset values. Thus, firms with steady, reliable cash flows such as utility companies are able to use high levels of debt and still have a very low probability of default. Firms whose value and cash flows are very volatile (for example, semiconductor firms) must have much lower levels of debt to avoid a significant risk of default.

Optimal Leverage

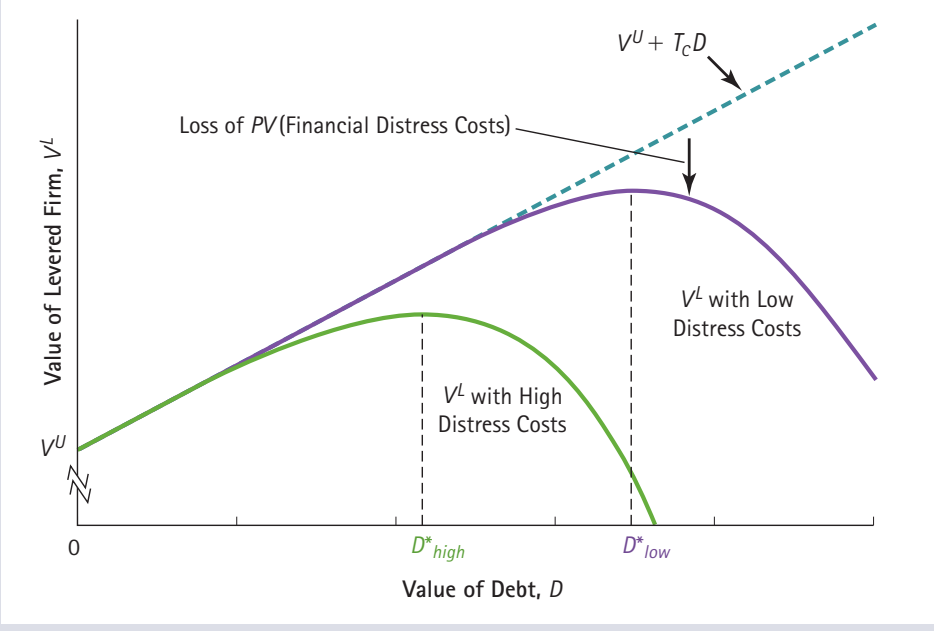
Figure 16.8 shows how the value of a levered firm, V^L, varies with the level of permanent debt, D, according to Eq. 16.10. With no debt, the value of the firm is V^U. For low levels of debt, the risk of default remains low and the main effect of an increase in leverage is an increase in the interest tax shield, which has present value $T_C D$ as shown in Eq. 16.7. If there were no costs of financial distress, the value would continue to increase at this rate until the interest on the debt exceeds the firm's earnings before interest and taxes, and the tax shield is exhausted.

The costs of financial distress reduce the value of the levered firm, V^L. The amount of the reduction increases with the probability of default, which in turn increases with the level of the debt D. The tradeoff theory states that firms should increase their leverage until it reaches the level D^* for which V^L is maximized. At this point, the tax savings that result from increasing leverage are just offset by the increased probability of incurring the costs of financial distress.

Figure 16.8 also illustrates the optimal debt choices for two types of firms. The optimal debt choice for a firm with low costs of financial distress is indicated by D^*_{low}, and the

FIGURE 16.8
Optimal Leverage with Taxes and Financial Distress Costs

As the level of debt increases, the tax benefits of debt increase until the interest expense exceeds the firm's EBIT. However, the probability of default, and hence the present value of financial distress costs, also increases. The optimal level of debt, D^*, occurs when these effects balance out and the value of the levered firm is maximized. D^* will be lower for firms with higher costs of financial distress.



optimal debt choice for a firm with high costs of financial distress is indicated by D^*_{high}. Not surprisingly, with higher costs of financial distress, it is optimal for the firm to choose lower leverage.

The tradeoff theory helps to resolve two important facts about leverage:

1. The presence of financial distress costs can explain why firms choose debt levels that are too low to fully exploit the interest tax shield.
2. Differences in the magnitude of financial distress costs and the volatility of cash flows can explain the differences in the use of leverage across industries.

Concept Check

9. According to the tradeoff theory, how should a financial manager determine the right capital structure for a firm?
10. Why would managers in one industry choose different capital structures than those in another industry?

16.6 Additional Consequences of Leverage: Agency Costs and Information

Taxes and financial distress costs are not the only capital market imperfections that arise in practice. In this section, we continue to relax the assumption of perfect capital markets to see other ways in which capital structure can affect a firm's value. We begin by discussing how leverage alters managers' incentives and changes their investment decisions. We then address the complexities of stakeholders in the firm having different information.

Critical Thinking

1. Absent tax effects, why can't we change the cost of capital of the firm by using more debt financing and less equity financing?
2. Explain what is wrong with the following argument: "If a firm issues debt that is risk-free because there is no possibility of default, the risk of the firm's equity does not change. Therefore, risk-free debt allows the firm to get the benefit of a low cost of capital of debt without raising its cost of capital of equity."
3. What are the channels through which financing choices can affect firm value?
4. How do taxes affect the choice of debt versus equity?
5. What is meant by "indirect costs of financial distress"?
6. Which type of firm is more likely to experience a loss of customers in the event of financial distress:
   a. Campbell Soup Company or Intuit, Inc. (a maker of accounting software)?
   b. Allstate Corporation (an insurance company) or Reebok International (a footwear and clothing firm)?
7. According to the tradeoff theory, how is capital structure determined?
8. For each pair below, which type of asset is more likely to be liquidated for close to its full market value in the event of financial distress:
   a. An office building or a brand name?
   b. Product inventory or raw materials?
   c. Patent rights or engineering "know-how"?
9. Which of the following industries have low optimal debt levels according to the tradeoff theory? Which have high optimal levels of debt?
   a. Tobacco firms
   b. Accounting firms
   c. Established restaurant chains
   d. Lumber companies
   e. Cell phone manufacturers
10. How can leverage alter the incentives of managers?

Problems

All problems in this chapter are available in MyFinanceLab.

Capital Structure in Perfect Capital Markets

For problems in this section assume no taxes or distress costs.

1. Consider a project with free cash flows in one year of $130,000 or $180,000, with each outcome being equally likely. The initial investment required for the project is $100,000, and the project's cost of capital is 20%. The risk-free interest rate is 10%.
   a. What is the NPV of this project?
   b. Suppose that to raise the funds for the initial investment, the project is sold to investors as an all-equity firm. The equity holders will receive the cash flows of the project in one year. How much money can be raised in this way—that is, what is the initial market value of the unlevered equity?
   c. Suppose the initial $100,000 is instead raised by borrowing at the risk-free interest rate. What are the cash flows of the levered equity, and what is its initial value according to MM?

2. You are an entrepreneur starting a biotechnology firm. If your research is successful, the technology can be sold for $30 million. If your research is unsuccessful, it will be worth nothing. To fund your research, you need to raise $2 million. Investors are willing to provide you with $2 million in initial capital in exchange for 50% of the unlevered equity in the firm.
   a. What is the total market value of the firm without leverage?
   b. Suppose you borrow $1 million. According to MM, what fraction of the firm's equity will you need to sell to raise the additional $1 million you need?
   c. What is the value of your share of the firm's equity in cases (a) and (b)?

3. Acort Industries owns assets that will have an 80% probability of having a market value of $50 million in one year. There is a 20% chance that the assets will be worth only $20 million. The current risk-free rate is 5%, and Acort's assets have a cost of capital of 10%.
   a. If Acort is unlevered, what is the current market value of its equity?
   b. Suppose instead that Acort has debt with a face value of $20 million due in one year. According to MM, what is the value of Acort's equity in this case?
   c. What is the expected return of Acort's equity without leverage? What is the expected return of Acort's equity with leverage?
   d. What is the lowest possible realized return of Acort's equity with and without leverage?

*4. Suppose there are no taxes. Firm ABC has no debt, and firm XYZ has debt of $5000 on which it pays interest of 10% each year. Both companies have identical projects that generate free cash flows of $800 or $1000 each year. After paying any interest on debt, both companies use all remaining free cash flows to pay dividends each year.

	ABC		XYZ	
FCF	Debt Payments	Equity Dividends	Debt Payments	Equity Dividends
$ 800				
$1000				

   a. Fill in the table above showing the payments debt and equity holders of each firm will receive given each of the two possible levels of free cash flows.
   b. Suppose you hold 10% of the equity of ABC. What is another portfolio you could hold that would provide the same cash flows?
   c. Suppose you hold 10% of the equity of XYZ. If you can borrow at 10%, what is an alternative strategy that would provide the same cash flows?

5. Hardmon Enterprises is currently an all-equity firm with an expected return of 12%. It is considering borrowing money to buy back some of its existing shares, thus increasing its leverage.
   a. Suppose Hardmon borrows to the point that its debt-equity ratio is 0.50. With this amount of debt, the debt cost of capital is 6%. What will the expected return of equity be after this transaction?
   b. Suppose instead Hardmon borrows to the point that its debt-equity ratio is 1.50. With this amount of debt, Hardmon's debt will be much riskier. As a result, the debt cost of capital will be 8%. What will the expected return of equity be in this case?
   c. A senior manager argues that it is in the best interest of the shareholders to choose the capital structure that leads to the highest expected return for the stock. How would you respond to this argument?

6. Suppose Microsoft has no debt and a WACC of 9.2%. The average debt-to-value ratio for the software industry is 5%. What would its cost of equity be if it took on the average amount of debt for its industry at a cost of debt of 6%?

Debt and Taxes

7. Pelamed Pharmaceuticals had EBIT of $325 million in 2010. In addition, Pelamed had interest expenses of $125 million and a corporate tax rate of 40%.
   a. What was Pelamed's 2010 net income?
   b. What was the total of Pelamed's 2010 net income and interest payments?
   c. If Pelamed had no interest expenses, what would its 2010 net income have been? How does it compare to your answer in part (b)?
   d. What was the amount of Pelamed's interest tax shield in 2010?

8. Grommit Engineering expects to have net income next year of $20.75 million and free cash flow of $22.15 million. Grommit's marginal corporate tax rate is 35%.
   a. If Grommit increases leverage so that its interest expense rises by $1 million, how will its net income change?
   b. For the same increase in interest expense, how will its free cash flow change?

9. Assume that Microsoft has a total market value of $300 billion and a marginal tax rate of 35%. If it permanently changes its leverage from no debt by taking on new debt in the amount of 13% of its current market value, what is the present value of the tax shield it will create?

10. Suppose the corporate tax rate is 40%. Consider a firm that earns $1000 before interest and taxes each year with no risk. The firm's capital expenditures equal its depreciation expenses each year, and it will have no changes to its net working capital. The risk-free interest rate is 5%.
    a. Suppose the firm has no debt and pays out its net income as a dividend each year. What is the value of the firm's equity?
    b. Suppose instead the firm makes interest payments of $500 per year. What is the value of equity? What is the value of debt?
    c. What is the difference between the total value of the firm with leverage and without leverage?
    d. To what percentage of the value of the debt is the difference in part (c) equal?



11. Your firm currently has $100 million in debt outstanding with a 10% interest rate. The terms of the loan require the firm to repay $25 million of the balance each year. Suppose that the marginal corporate tax rate is 40%, and that the interest tax shields have the same risk as the loan. What is the present value of the interest tax shields from this debt?

12. Arnell Industries has $10 million in permanent debt outstanding. The firm will pay interest only on this debt. Arnell's marginal tax rate is expected to be 35% for the foreseeable future.
    a. Suppose Arnell pays interest of 6% per year on its debt. What is its annual interest tax shield?
    b. What is the present value of the interest tax shield, assuming its risk is the same as the loan?
    c. Suppose instead that the interest rate on the debt is 5%. What is the present value of the interest tax shield in this case?

13. Rogot Instruments makes fine violins, violas, and cellos. It has $1 million in debt outstanding, equity valued at $2 million, and pays corporate income tax at a rate of 35%. Its cost of equity is 12% and its cost of debt is 7%.
a. What is Rogot's pretax WACC?
b. What is Rogot's (effective after-tax) WACC?

14. Rumolt Motors has 30 million shares outstanding with a price of $15 per share. In addition, Rumolt has issued bonds with a total current market value of $150 million. Suppose Rumolt's equity cost of capital is 10%, and its debt cost of capital is 5%.
a. What is Rumolt's pretax WACC?
b. If Rumolt's corporate tax rate is 35%, what is its after-tax WACC?

15. Summit Builders has a market debt-equity ratio of 0.65 and a corporate tax rate of 40%, and it pays 7% interest on its debt. By what amount does the interest tax shield from its debt lower Summit's WACC?

16. Milton Industries expects free cash flows of $5 million each year. Milton's corporate tax rate is 35%, and its unlevered cost of capital is 15%. The firm also has outstanding debt of $19.05 million, and it expects to maintain this level of debt permanently.
a. What is the value of Milton Industries without leverage?
b. What is the value of Milton Industries with leverage?

17. NatNah, a builder of acoustic accessories, has no debt and an equity cost of capital of 15%. NatNah decides to increase its leverage to maintain a market debt-to-value ratio of 0.5. Suppose its debt cost of capital is 9% and its corporate tax rate is 35%. If NatNah's pretax WACC remains constant, what will its (effective after-tax) WACC be with the increase in leverage?

***18.** Kurz Manufacturing is currently an all-equity firm with 20 million shares outstanding and a stock price of $7.50 per share. Although investors currently expect Kurz to remain an all-equity firm, Kurz plans to announce that it will borrow $50 million and use the funds to repurchase shares. Kurz will pay interest only on this debt, and it has no further plans to increase or decrease the amount of debt. Kurz is subject to a 40% corporate tax rate.
a. What is the market value of Kurz's existing assets before the announcement?
b. What is the market value of Kurz's assets (including any tax shields) just after the debt is issued, but before the shares are repurchased?
c. What is Kurz's share price just before the share repurchase? How many shares will Kurz repurchase?
d. What are Kurz's market value balance sheet and share price after the share repurchase?

19. Kohwe Corporation plans to issue equity to raise $50 million to finance a new investment. After making the investment, Kohwe expects to earn free cash flows of $10 million each year. Kohwe currently has 5 million shares outstanding, and it has no other assets or opportunities. Suppose the appropriate discount rate for Kohwe's future free cash flows is 8%, and the only capital market imperfections are corporate taxes and financial distress costs.
a. What is the NPV of Kohwe's investment?
b. What is Kohwe's share price today?

20. Suppose Kohwe borrows the $50 million instead. The firm will pay interest only on this loan each year, and it will maintain an outstanding balance of $50 million on the loan. Suppose that Kohwe's corporate tax rate is 40%, and expected free cash flows are still $10 million each year. What is Kohwe's share price today if the investment is financed with debt?

21. Now suppose that with leverage, Kohwe's expected free cash flows will decline to $9 million per year due to reduced sales and other financial distress costs. Assume that the appropriate discount rate for Kohwe's future free cash flows is still 8%. What is Kohwe's share price today given the financial distress costs of leverage?

Optimal Capital Structure: The Tradeoff Theory

22. Hawar International is a shipping firm with a current share price of $5.50 and 10 million shares outstanding. Suppose that Hawar announces plans to lower its corporate taxes by borrowing $20 million and repurchasing shares, that Hawar pays a corporate tax rate of 30%, and that shareholders expect the change in debt to be permanent.
    a. If the only imperfection is corporate taxes, what will the share price be after this announcement?
    b. Suppose the only imperfections are corporate taxes and financial distress costs. If the share price rises to $5.75 after this announcement, what is the PV of financial distress costs Hawar will incur as the result of this new debt?

23. Marpor Industries has no debt and expects to generate free cash flows of $16 million each year. Marpor believes that if it permanently increases its level of debt to $40 million, the risk of financial distress may cause it to lose some customers and receive less favorable terms from its suppliers. As a result, Marpor's expected free cash flows with debt will be only $15 million per year. Suppose Marpor's tax rate is 35%, the risk-free rate is 5%, the expected return of the market is 15%, and the beta of Marpor's free cash flows is 1.10 (with or without leverage).
    a. Estimate Marpor's value without leverage.
    b. Estimate Marpor's value with the new leverage.

Additional Consequences of Leverage: Agency Costs and Information

24. Dynron Corporation's primary business is natural gas transportation using its vast gas pipeline network. Dynron's assets currently have a market value of $150 million. The firm is exploring the possibility of raising $50 million by selling part of its pipeline network and investing the $50 million in a fiber-optic network to generate revenues by selling high-speed network bandwidth. Whereas this new investment is expected to increase profits, it will also substantially increase Dynron's risk. If Dynron is levered, would this investment be more or less attractive to equity holders than if Dynron had no debt?

25. Consider a firm whose only asset is a plot of vacant land, and whose only liability is debt of $15 million due in one year. If left vacant, the land will be worth $10 million in one year. Alternatively, the firm can develop the land at an upfront cost of $20 million. The developed land will be worth $35 million in one year. Suppose the risk-free interest rate is 10%, assume all cash flows are risk-free, and assume there are no taxes.
    a. If the firm chooses not to develop the land, what is the value of the firm's equity today? What is the value of the debt today?
    b. What is the NPV of developing the land?

Dutch auction A share repurchase method in which shareholders indicate how many shares they are willing to sell at each price. The firm then pays the lowest price at which it can buy back its desired number of shares.

The price is usually set at a substantial premium (10%–20% is typical) to the current market price. The offer often depends on shareholders tendering a sufficient number of shares. If shareholders do not tender enough shares, the firm may cancel the offer and no buyback occurs.

A related method is the **Dutch auction** share repurchase, in which the firm lists different prices at which it is prepared to buy shares, and shareholders in turn indicate how many shares they are willing to sell at each price. The firm then pays the lowest price at which it can buy back its desired number of shares.

targeted repurchase When a firm purchases shares directly from a specific shareholder; the purchase price is negotiated directly with the seller.

Targeted Repurchase. A firm may also purchase shares directly from a major shareholder in a **targeted repurchase**. In this case, the purchase price is negotiated directly with the seller. A targeted repurchase may occur if a major shareholder desires to sell a large number of shares but the market for the shares is not sufficiently liquid to sustain such a large sale without severely affecting the price. Under these circumstances, the shareholder may be willing to sell shares back to the firm at a discount to the current market price. Alternatively, targeted repurchases may be used if a major shareholder is threatening to take over the firm and remove its management. With **greenmail**, the firm may decide to eliminate the threat by buying out the shareholder—often at a large premium over the current market price.

Concept Check

1. How is a stock's ex-dividend date determined, and what is its significance?
2. What is an open market share repurchase?

17.2 Dividends Versus Share Repurchases in a Perfect Capital Market

greenmail When a firm avoids a threat of takeover and removal of its management by a major shareholder by buying out the shareholder, often at a large premium over the current market price.

If a corporation decides to pay cash to shareholders, it can do so through either dividend payments or share repurchases. How do firms choose between these alternatives? In this section, we show that in the perfect capital markets setting of Modigliani and Miller, the method of payment does not matter.

Consider the case of Genron Corporation, a hypothetical firm. Genron has $20 million in excess cash and no debt. The firm expects to generate additional free cash flows of $48 million per year in subsequent years. If Genron's unlevered cost of capital is 12%, then the enterprise value of its ongoing operations is:

$$\text{Enterprise Value} = PV(\text{Future FCF}) = \frac{\$48 \text{ million}}{12\%} = \$400 \text{ million}$$

Including the cash, Genron's total market value is $420 million.

Genron's board is meeting to decide how to pay out its $20 million in excess cash to shareholders. The board is considering three options:

1. Use the $20 million to pay a $2 cash dividend for each of Genron's 10 million outstanding shares.
2. Repurchase shares instead of paying a dividend.
3. Raise additional cash to pay an even larger dividend today and in the future.

Will the amount of the current dividend affect Genron's share price? Which policy would shareholders prefer?

To provide a baseline for our discussion of payout policy, we will analyze the consequences of each of these three alternative policies in the following sections and compare

them in a setting of perfect capital markets. We will also explore how market imperfections such as taxes and transaction costs affect payout policy.

Alternative Policy 1: Pay a Dividend with Excess Cash

Suppose the board opts for the first alternative and uses all excess cash to pay a dividend. With 10 million shares outstanding, Genron will be able to pay a $2 dividend immediately. Because the firm expects to generate future free cash flows of $48 million per year, it anticipates paying a dividend of $4.80 per share each year thereafter. The board declares the dividend and sets the record date as December 14, so that the ex-dividend date is December 12. To determine the impact of this decision, let's compute Genron's share price just before and after the stock goes ex-dividend.

Recall from Chapter 7 and the Valuation Principle that the fair price for the shares is the present value of the expected dividends given Genron's equity cost of capital. Because Genron has no debt, its equity cost of capital equals its unlevered cost of capital of 12%. Just before the ex-dividend date, the stock is said to trade **cum-dividend** ("with the dividend") because anyone who buys the stock will be entitled to the dividend. In this case:

cum dividend When a stock trades before the ex-dividend date, entitling anyone who buys the stock to the dividend.

$$P_{cum} = \text{Current Dividend} + PV(\text{Future Dividends}) = 2 + \frac{4.80}{0.12} = 2 + 40 = \$42$$

After the stock goes ex-dividend, new buyers will not receive the current dividend. At this point, the share price will reflect only the dividends in subsequent years:

$$P_{ex} = PV(\text{Future Dividends}) = \frac{4.80}{0.12} = \$40$$

The share price will drop on the ex-dividend date, December 12, from $42 to $40. The amount of the price drop is equal to the amount of the current dividend, $2. We can also determine this change in the share price using the market value balance sheet (values in millions of dollars):

	December 11 (Cum-Dividend)	December 12 (Ex-Dividend)
Cash	20	0
Other assets	400	400
Total market value	420	400
Shares (millions)	10	10
Share price	**$42**	**$40**

As the market value balance sheet shows, the share price falls when a dividend is paid because the reduction in cash decreases the market value of the firm's assets. Although the stock price falls, holders of Genron stock do not incur a loss overall. Before the dividend, their stock was worth $42. After the dividend, their stock is worth $40 and they hold $2 in cash from the dividend, for a total value of $42. Our analysis of both the stock price and the market value balance sheet shows that:

> *In a perfect capital market, when a dividend is paid, the share price drops by the amount of the dividend when the stock begins to trade ex-dividend.*

Alternative Policy 2: Share Repurchase (No Dividend)

Suppose that Genron does not pay a dividend this year, but instead uses the $20 million to repurchase its shares on the open market. How will the repurchase affect the share price?

With an initial share price of \$42, Genron will repurchase \$20 million ÷ \$42 per share = 0.476 million shares, leaving only 10 − 0.476 = 9.524 million shares outstanding. Once again, we can use Genron's market value balance sheet to analyze this transaction:

	December 11 (Before Repurchase)	December 12 (After Repurchase)
Cash	20	0
Other assets	400	400
Total market value of assets	420	400
Shares (millions)	10	9.524
Share price	**\$42**	**\$42**

In this case, the market value of Genron's assets falls when the company pays out cash, but the number of shares outstanding also falls from 10 million to 9.524 million. The two changes offset each other, so the share price remains the same at \$42.

Genron's Future Dividends. We can also see why the share price does not fall after the share repurchase by considering the effect on Genron's future dividends. In future years, Genron expects to have \$48 million in free cash flow, which can be used to pay a dividend of \$48 million ÷ 9.524 million shares = \$5.04 per share each year. Thus, with a share repurchase, Genron's share price today is:

$$P_{rep} = \frac{5.04}{0.12} = \$42$$

In other words, by not paying a dividend today and repurchasing shares instead, Genron is able to raise its dividends *per share* in the future. The increase in future dividends compensates shareholders for the dividend they give up today. This example illustrates the following general conclusion about share repurchases:

> *In perfect capital markets, an open market share repurchase has no effect on the stock price, and the stock price is the same as the cum-dividend price if a dividend were paid instead.*

Investor Preferences. Would an investor prefer that Genron issue a dividend or repurchase its stock? Both policies lead to the same *initial* share price of \$42. But is there a difference in shareholder value *after* the transaction? Consider an investor who currently holds 2000 shares of Genron stock. Assuming the investor does not trade the stock, the investor's holdings after a dividend or share repurchase are as follows:

Dividend	Repurchase
\$40 × 2,000 = \$80,000 stock	\$42 × 2,000 = \$84,000 stock
\$2 × 2,000 = \$4,000 cash	

In either case, the value of the investor's portfolio is \$84,000 immediately after the transaction. The only difference is the distribution between cash and stock holdings. Thus, it might seem the investor would prefer one approach or the other based on whether she needs the cash.

But if Genron repurchases shares and the investor wants cash, she can raise cash by selling shares. For example, she can sell 95 shares to raise 95 × \$42 per share = \$3990 in cash. She will then hold 1905 shares, or 1905 × \$42 = \$80,010 in stock. Thus, in the case of a share repurchase, by selling shares an investor can create a *homemade dividend.*

COMMON MISTAKE

Repurchases and the Supply of Shares

There is a misconception that when a firm repurchases its own shares, the price rises due to the decrease in the supply of shares outstanding. This intuition follows naturally from the standard supply and demand analysis taught in microeconomics. Why does that analysis not apply here?

When a firm repurchases its own shares, two things happen. First, the supply of shares is reduced. At the same time, however, the value of the firm's assets declines when it spends its cash to buy the shares. If the firm repurchases its shares at their market price, these two effects offset each other, leaving the share price unchanged.

This result is similar to the dilution fallacy discussed in Chapter 16: When a firm issues shares at their market price, the share price does not fall due to the increase in supply. The increase in supply is offset by the increase in the firm's assets that results from the cash it receives from the issuance.

Similarly, if Genron pays a dividend and the investor does not want the cash, she can use the $4000 proceeds of the dividend to purchase 100 additional shares at the ex-dividend share price of $40 per share. As a result, she will hold 2100 shares, worth $2100 \times \$40 = \$84{,}000$. In fact, many firms allow investors to register for a dividend reinvestment program, or *DRIP*, that automatically reinvests any dividends into new shares of the stock.

We summarize these two cases below:

Dividend + Buy 100 Shares	Repurchase + Sell 95 Shares
$\$40 \times 2{,}100 = \$84{,}000$ stock	$\$42 \times 1905 = \$80{,}010$ stock
	$\$42 \times 95 = \$3990$ cash

By selling shares or reinvesting dividends, the investor can create any combination of cash and stock desired. As a result, the investor is indifferent between the various payout methods the firm might employ:

In perfect capital markets, investors are indifferent between the firm distributing funds via dividends or share repurchases. By reinvesting dividends or selling shares, they can replicate either payout method on their own.

Alternative Policy 3: High Dividend (Equity Issue)

Let's look at a third possibility for Genron. Suppose the board wishes to pay an even larger dividend than $2 per share right now. Is that possible and, if so, will the higher dividend make shareholders better off?

Genron plans to pay $48 million in dividends starting next year. Suppose the firm wants to start paying that amount today. Because it has only $20 million in cash today, Genron needs an additional $28 million to pay the larger dividend now. It could raise cash by scaling back its investments. But if the investments have positive NPV, reducing them would lower the firm's value. An alternative way to raise more cash is to borrow money or sell new shares. Let's consider an equity issue. Given a current share price of $42, Genron could raise $28 million by selling $28 million $\div$ $42 per share $= 0.67$ million shares. Because this equity issue will increase Genron's total number of shares outstanding to 10.67 million, the amount of the dividend per share each year will be:

$$\frac{\$48 \text{ million}}{10.67 \text{ million shares}} = \$4.50 \text{ per share}$$

Under this new policy, Genron's cum-dividend share price is:

$$P_{cum} = 4.50 + \frac{4.50}{0.12} = 4.50 + 37.50 = \$42$$

As in the previous examples, the initial share value is unchanged by this policy, and increasing the dividend has no benefit to shareholders.

EXAMPLE 17.1
Homemade Dividends

Problem

Suppose Genron does not adopt the third alternative policy and instead pays a $2 dividend per share today. Show how an investor holding 2000 shares could create a homemade dividend of $4.50 per share × 2000 shares = $9000 per year on her own.

Solution

Plan

If Genron pays a $2 dividend, the investor receives $4000 in cash and holds the rest in stock. She can raise $5000 in additional cash by selling 125 shares at $40 per share just after the dividend is paid.

Execute

The investor creates her $9000 this year by collecting the $4000 dividend and then selling 125 shares at $40 per share. In future years, Genron will pay a dividend of $4.80 per share. Because she will own 2000 − 125 = 1875 shares, the investor will receive dividends of 1875 × $4.80 = $9000 per year from then on.

Evaluate

Again, the policy that the firm chooses is irrelevant—the investor can transact in the market to create a homemade dividend policy that suits her preferences.

Modigliani-Miller and Dividend Policy Irrelevance

In our analysis, we considered three possible dividend policies for the firm: (1) pay out all cash as a dividend, (2) pay no dividend and use the cash instead to repurchase shares, or (3) issue equity to finance a larger dividend. These policies are illustrated in Table 17.1.

Table 17.1 shows an important tradeoff: If Genron pays a higher *current* dividend per share, it will pay lower *future* dividends per share. For example, if the firm raises the current dividend by issuing equity, it will have more shares and therefore smaller free cash flows per share to pay dividends in the future. If the firm lowers the current dividend and repurchases its shares, it will have fewer shares in the future, so it will be able to pay a higher dividend per share. The net effect of this tradeoff is to leave the total present value of all future dividends, and hence the current share price, unchanged at $42.

The logic of this section matches that in our discussion of capital structure in Chapter 16. There, we explained that in perfect capital markets, buying and selling equity and debt are zero-NPV transactions that do not affect firm value. Moreover, any choice of leverage by a firm could be replicated by investors using homemade leverage. As a result, the firm's choice of capital structure is irrelevant. Here, we have established the same principle for a firm's choice of a dividend. Regardless of the amount of cash the firm has on hand, it can pay a smaller dividend (and use the remaining cash to repurchase shares) or a larger dividend (by selling equity to raise cash). Because buying or selling shares is a zero-NPV transaction, such transactions have no effect on the initial share price.

TABLE 17.1
Genron's Dividends per Share Each Year Under the Three Alternative Policies

		Dividend Paid ($ per share)			
	Initial Share Price	Year 0	Year 1	Year 2	. . .
Policy 1:	$42.00	2.00	4.80	4.80	. . .
Policy 2:	$42.00	0	5.04	5.04	. . .
Policy 3:	$42.00	4.50	4.50	4.50	. . .

Furthermore, shareholders can create a homemade dividend of any size by buying or selling shares themselves.

MM Dividend Irrelevance In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price.

Modigliani and Miller developed this idea in another influential paper published in 1961.[4] As with their result on capital structure, it went against the conventional wisdom that dividend policy could change a firm's value and make its shareholders better off even if there were no market imperfections. We state here their important proposition:

MM Dividend Irrelevance: *In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price.*

COMMON MISTAKE

The Bird in the Hand Fallacy

"A bird in the hand is worth two in the bush."

The bird in the hand hypothesis states that firms choosing to pay higher current dividends will enjoy higher stock prices because shareholders prefer current dividends to future ones (with the same present value). According to this view, alternative policy 3 would lead to the highest share price for Genron.

This view is a misconception. Modigliani and Miller showed that with perfect capital markets, shareholders can generate an equivalent homemade dividend at any time by selling shares. Thus, the dividend choice of the firm should not matter.

Dividend Policy with Perfect Capital Markets

The examples in this section illustrate the idea that by using share repurchases or equity issues a firm can easily alter its dividend payments. Because these transactions do not alter the value of the firm, neither does dividend policy.

This result may at first seem to contradict the idea that the price of a share should equal the present value of its future dividends. As our examples have shown, however, a firm's choice of dividend today affects the dividends it can afford to pay in the future in an offsetting fashion. Thus, although dividends *do* determine share prices, a firm's choice of dividend policy does not.

As Modigliani and Miller made clear, the value of a firm ultimately derives from its underlying free cash flows. A firm's free cash flows determine the level of payouts that it can make to its investors. In a perfect capital market, whether these payouts are made through dividends or share repurchases does not matter. Of course, in reality capital markets are not perfect. As with capital structure, it is the imperfections in capital markets that should determine the firm's payout policy. The main imperfection we will discuss is taxes: Repurchases and dividends have different tax implications and investors also face different tax rates. In addition, when managers have information investors do not, payout policy decisions can help signal that information. We start with taxes.

Concept Check

3. Explain the misconception that when a firm repurchases its own shares, the price rises due to the decrease in the supply of shares outstanding.
4. In a perfect capital market, how important is the firm's decision to pay dividends versus repurchase shares?

[4]See F. Modigliani and M. Miller, "Dividend Policy, Growth, and the Valuation of Shares," *Journal of Business* 34 (4) (1961): 411–433. See also J. B. Williams, *The Theory of Investment Value* (Cambridge, MA: Harvard University Press, 1938).

17.3 The Tax Disadvantage of Dividends

As with capital structure, taxes are an important market imperfection that influences a firm's decision to pay dividends or repurchase shares.

Taxes on Dividends and Capital Gains

Shareholders typically must pay taxes on the dividends they receive. They must also pay capital gains taxes when they sell their shares. Table 17.2 shows the history of U.S. tax rates from 1971–2010 applied to dividends and long-term capital gains for investors in the highest tax bracket.

TABLE 17.2

Long-Term Capital Gains Versus Dividend Tax Rates in the United States, 1971–2010

Year	Capital Gains	Dividends
1971–1978	35%	70%
1979–1981	28%	70%
1982–1986	20%	50%
1987	28%	39%
1988–1990	28%	28%
1991–1992	28%	31%
1993–1996	28%	40%
1997–2000	20%	40%
2001–2002	20%	39%
2003–2010*	15%	15%

*The current tax rates are set to expire in 2010 unless they are extended by Congress. The tax rates shown are for financial assets held for one year. For assets held less than one year, capital gains are taxed at the ordinary income tax rate (currently 35% for the highest bracket); the same is true for dividends if the assets are held for less than 61 days. Because the capital gains tax is not paid until the asset is sold, for assets held for longer than one year the *effective* capital gains tax rate is equal to the present value of the rate shown, when discounted by the after-tax risk-free interest rate for the additional number of years the asset is held.

Do taxes affect investors' preferences for dividends versus share repurchases? When a firm pays a dividend, shareholders are taxed according to the dividend tax rate. If the firm repurchases shares instead, and shareholders sell shares to create a homemade dividend, the homemade dividend will be taxed according to the capital gains tax rate. If dividends are taxed at a higher rate than capital gains, which has been true until the most recent change to the tax code, shareholders will prefer share repurchases to dividends. Recent changes to the tax code have equalized the tax rates on dividends and capital gains. It is probably not coincidental that Microsoft starting paying dividends shortly after these changes. Nonetheless, because long-term investors can defer the capital gains tax until they sell, *there is still a tax advantage for share repurchases over dividends.*

Not all countries tax dividends at a higher rate than capital gains. Figure 17.4 shows the dividend and capital gains tax rates for different countries. In Chile, for example, capital gains are taxed at a 45% rate, while dividends are taxed at 35%. A similar tax preference for dividends exists in Australia, Denmark, Finland, and Brazil.

Optimal Dividend Policy with Taxes

When the tax rate on dividends exceeds the tax rate on capital gains, shareholders will pay lower taxes if a firm uses share repurchases for all payouts rather than dividends. This tax savings will increase the value of a firm that uses share repurchases rather than dividends.

7. Natsam Corporation has $250 million of excess cash. The firm has no debt and 500 million shares outstanding with a current market price of $15 per share. Natsam's board has decided to pay out this cash as a one-time dividend.
   a. What is the ex-dividend price of a share in a perfect capital market?
   b. If the board instead decided to use the cash to do a one-time share repurchase, in a perfect capital market, what is the price of the shares once the repurchase is complete?
   c. In a perfect capital market, which policy in part (a) or (b) makes investors in the firm better off?

8. Suppose the board of Natsam Corporation decided to do the share repurchase in Problem 7(b), but you as an investor would have preferred to receive a dividend payment. How can you leave yourself in the same position as if the board had elected to make the dividend payment instead?

The Tax Disadvantage of Dividends

9. The HNH Corporation will pay a constant dividend of $2 per share, per year, in perpetuity. Assume all investors pay a 20% tax on dividends and that there is no capital gains tax. The cost of capital for investing in HNH stock is 12%.
   a. What is the price of a share of HNH stock?
   b. Assume that management makes a surprise announcement that HNH will no longer pay dividends but will use the cash to repurchase stock instead. What is the price of a share of HNH stock now?

10. You purchased CSH stock for $40 and it is now selling for $50. The company has announced that it plans a $10 special dividend.
    a. Assuming 2010 tax rates, if you sell the stock or wait and receive the dividend, will you have different after-tax income?
    b. If the capital gains tax rate is 20% and the dividend tax rate is 40%, what is the difference between the two options in part (a)?

Payout Versus Retention of Cash

11. Assume perfect capital markets. Kay Industries currently has $100 million invested in short-term Treasury securities paying 7%, and it pays out the interest payments on these securities as a dividend. The board is considering selling the Treasury securities and paying out the proceeds as a one-time dividend payment.
    a. If the board went ahead with this plan, what would happen to the value of Kay stock upon the announcement of a change in policy?
    b. What would happen to the value of Kay stock on the ex-dividend date of the one-time dividend?
    c. Given these price reactions, will this decision benefit investors?

12. Redo Problem 11, but assume that Kay must pay a corporate tax rate of 35%, and that investors pay no taxes.

13. Redo Problem 11, but assume that investors pay a 15% tax on dividends but no capital gains taxes, and that Kay does not pay corporate taxes.

Signaling with Payout Policy

Use the following information to answer Problems 14 through 18:

AMC Corporation currently has an enterprise value of $400 million and $100 million in excess cash. The firm has 10 million shares outstanding and no debt. Suppose AMC uses

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Financial Markets And Corporate Strategy{Mark Grinblatt, She

- Hamada model and the adjusted cost of capital: Ch. 13 (p. 464–465)
- Hamada model and the adjusted cost of capital: Ch. 13 (p. 482–482)
- MM theorem, undoing capital structure, risky debt: Ch. 14, §14.1–14.3 (p. 501–509)
- Bankruptcy costs: Ch. 16, §16.1 (p. 559–561)

EXHIBIT 13.1 Balance Sheet for a Firm with Leverage When Debt Interest Is Corporate Tax Deductible

Assets		Liabilities and Equity	
Debt Tax Shield *(TX)*	T_cD	Debt	D
Unlevered Assets *(UA)*	$D + E - T_cD$	Equity	E

presents the two sides of the balance sheet of a firm for which there is a corporate tax deduction for debt interest payments, but no personal taxes. Exhibit 13.1 illustrates that typically, the assets of the firm contain two components, one associated directly with the firm's operations and the other an indirect asset associated with a financing subsidy. The former component, the **unlevered assets (*UA*)**, is defined as the present value of the unlevered cash flows; the other component, the **debt tax shield (*TX*)**, is the present value of the financing subsidy (that is, the present value of the debt-interest deduction for all corporate profits taxes: federal, state, and city). The more debt the firm has, the bigger this tax shield. Note that the two sides of the balance sheet must add up to the same number—that is, balance—implying that the value of the unlevered assets can be viewed as the sum of the debt and equity, $D + E$, less the value of the debt tax shield.[5]

Viewing the assets of the firm with value A as a portfolio of unlevered assets with value UA and debt tax shields with value TX implies that the beta (or expected return) of the assets is the portfolio-weighted average of the betas (or expected returns) of the unlevered assets and debt tax shields; that is

$$\beta_A = \left(\frac{UA}{D+E}\right)\beta_{UA} + \left(\frac{TX}{D+E}\right)\beta_{TX} \tag{13.1a}$$

and

$$\bar{r}_A = \left(\frac{UA}{D+E}\right)\bar{r}_{UA} + \left(\frac{TX}{D+E}\right)\bar{r}_{TX} \tag{13.1b}$$

where

β_{UA} = beta risk of the unlevered assets
β_{TX} = beta risk of the debt tax shield
$\bar{r}_{UA}$ = expected return of the unlevered assets
$\bar{r}_{TX}$ = expected return of the debt tax shield

Static Perpetual Risk-Free Debt. Let T_c denote the effective corporate tax rate. Exhibit 13.1 has the firm's debt tax shield expressed as

$$TX = T_cD \tag{13.2}$$

and thus

$$UA = D + E - T_cD \tag{13.3}$$

The values for TX and UA both in Exhibit 13.1 and in equations (13.2) and (13.3) are developed in a model by Hamada (1972). If the firm issues default-free perpetual debt (that is, debt that never matures) with aggregate face amount of D and interest payments

[5]More generally, TX can be viewed as the present value of *any* debt financing subsidy.

equal to the risk-free rate, each interest payment of Dr_f saves T_cDr_f in taxes in the year it is paid. The present value of the tax savings from the interest payments (see Chapter 9 for the formula for the present value of a perpetuity) is

$$T_cD = \frac{T_cDr_f}{r_f}$$

The same present value is also achieved when the risk-free rate changes over time and the firm rolls over one-period risk-free debt.

The Hamada model assumes that the tax shield is riskless and thus, each period's tax deduction arising from an interest payment should be discounted back to date 0 at the risk-free rate. This implies that the beta of the debt tax shield in the Hamada model is zero. Using this observation, and substituting equations (13.2) and (13.3) into equation (13.1a) yields

$$\beta_A = \left[\frac{D + E - T_cD}{D + E}\right]\beta_{UA} \tag{13.4}$$

In the typical case where the beta of the unlevered assets of the firm, β_{UA}, is positive, equation (13.4) states that the beta of the combination of the unlevered assets and the debt tax shield assets, β_A, must decline with an increase in leverage to reflect the addition of the risk-free tax savings. To see this, note that the portfolio weight on β_{UA}, $\frac{D + E - T_cD}{D + E}$, which equals $1 - \frac{T_cD}{D + E}$, declines as the leverage ratio $\frac{D}{D + E}$ increases, reflecting the fact that risk-free debt tax shields constitute a larger proportion of the firm's assets as leverage increases.

The key assumptions that lead to this result are: (1) The debt is perpetual; in other words, it is either a perpetuity or consists of rolled over short-term debt positions. (2) The debt is default-free and pays the risk-free rate. (3) The face value of the debt and the tax rate do not change over time. The third assumption distinguishes the Hamada model from other models that will be discussed later in this chapter.

Equity Betas and Asset Betas. Equity betas are also affected by taxes. Chapter 11 indicated that (assuming risk-free debt) the equity beta is

$$\beta_E = \left(1 + \frac{D}{E}\right)\beta_A \tag{13.5}$$

Substituting the right-hand side of equation (13.4) for β_A in equation (13.5) gives

$$\beta_E = \left(1 + \frac{D}{E}\right)\left[\frac{D + E - T_cD}{D + E}\right]\beta_{UA}$$

With a little simplification, this equation, which is based on the Hamada model, becomes

$$\beta_E = \left[1 + (1 - T_c)\frac{D}{E}\right]\beta_{UA} \tag{13.6}$$

Assuming that β_{UA} does not change with leverage,[6] equation (13.6) states that, for a given debt increase, the beta of the firm's equity increases less the larger is the corporate tax rate, and increases the most when there are no taxes. Reversing this equation also allows us to identify β_{UA} from β_E, which is often a necessary step when using the discounted cash flow method for valuation when debt tax shields exist.

[6]Chapters 16–19 discuss situations where this may not be the case.

Recall, now, that earlier in this chapter we learned that the expected return of assets is a portfolio-weighted average of the expected returns of the unlevered assets and the debt tax shield. As leverage increases, the portfolio weight on the debt tax shield increases. It would be extremely aberrational to have a debt tax shield with higher beta risk than the unlevered assets. Because of this, it is safe to conclude that the expected return on assets does not increase as leverage increases.

Equation (13.11)[13] thus implies the following result:

Result 13.4 When debt interest is tax deductible, the WACC will decline as the firm's leverage ratio, D/E, increases.

The Adjusted Cost of Capital Formula. In the Hamada model, with static perpetual debt, both debt and its tax shield are risk-free, implying $\bar{r}_D = r_f, \beta_{TX} = 0$, and $TX = T_c D$. In this case, the portfolio-weighted average of the expected returns of the unlevered assets and the debt tax shield, given in equation (13.1b), simplifies to

$$\bar{r}_A = \left(1 - \frac{T_c D}{D + E}\right)\bar{r}_{UA} + \frac{T_c D}{D + E}\bar{r}_D$$

If we substitute this equation into equation (13.11), equation (13.11) reduces to Modigliani and Miller's (1963) **adjusted cost of capital formula**, which gives the firm's WACC as a function of its debt to value ratio:

$$\text{WACC} = \bar{r}_{UA}\left[1 - T_c \frac{D}{D + E}\right]$$

An application of this formula is provided in Example 13.12.

Example 13.12: The Effect of Leverage on the WACC with Corporate Taxes

Example 13.8 found that United Technologies (UT), with liabilities consisting of 20 percent debt and 80 percent equity, had a WACC of 13.2 percent when the corporate tax rate was 34 percent. In a financial restructuring designed to raise to 40 percent the proportion of UT financed with debt, UT issues debt and buys back its equity with the proceeds. Compute the firm's new WACC given the assumptions of the Hamada model.

Answer: To calculate the firm's new cost of capital, first estimate UT's unlevered cost of capital. From the adjusted cost of capital formula, a WACC of 13.2 percent with 20 percent debt financing corresponds to an $\bar{r}_{UA}$ that satisfies

$$.132 = \bar{r}_{UA}[1 - (.34)(.20)]$$

Therefore, $\bar{r}_{UA} = .1416$. Plugging this value into the adjusted cost of capital formula at a 40 percent $\frac{D}{D + E}$ ratio leaves a WACC satisfying

$$\text{WACC} = .1416[1 - (.34)(.4)] = .122$$

Therefore, the new WACC is 12.2 percent.

Graphing the WACC, Cost of Debt, and Cost of Equity with Corporate Taxes. Exhibit 13.5 graphs United Technologies' WACC, cost of equity capital, and cost of

[13] Equation (13.11) applies to risky debt as long as one adjusts T_c to account for nonuse of the tax shields, as discussed earlier in the chapter.

what will the WACC become? How would your WACC answer change if the government raises the tax rate from zero to 30 percent?

13.15. The Akron Company consists of $50 million in perpetual riskless debt and $50 million in equity. The current market value of its assets is $100 million and the beta of its equity return is 1.2. Assume the risk-free rate is 8 percent, the expected return of the market portfolio is 13 percent per year, and the CAPM is true. Compute the expected return of Akron's equity and its WACC assuming a 40 percent corporate tax rate.

13.16. Akron, from the last example, is considering an exchange offer where half of Akron's outstanding debt ($25 million) is retired. The purchase of this debt would be financed by issuing $25 million in equity to the debt holders of Akron. Assuming debt policy that is consistent with the Hamada model, what will Akron's new WACC be after the exchange offer?

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Chapter 13 discussed how taxes and a firm's financial structure can interact to affect its real investment decisions. This chapter and the one that follows it take one step backward and examine how taxes affect the proportions of debt and equity used to finance a firm. To understand how taxes affect a firm's financing choices, one must first understand what financing choices would be like without taxes. Hence, the springboard for discussion of this topic is the no-tax capital structure irrelevance theorem offered by Franco Modigliani and Merton Miller.

The **Modigliani-Miller Theorem,** which was largely responsible for both authors winning Nobel Prizes in economics, states that if the capital structure decision has no effect on the total cash flows that a firm can distribute to its debt and equity holders, the decision also will have no effect—in the absence of transaction costs—on the total value of the firm's debt and equity. This means that a manager who is contemplating whether it is cheaper to finance the firm primarily with junk bonds (that is, very high-yield, high-risk debt) or with equity and perhaps a small amount of high-quality debt should stop worrying: Neither financing decision is superior to the other!

The premise of the Modigliani-Miller Theorem, that capital structure has no effect on cash flows, is not true in the real world. Because the interest on debt is tax deductible, the after-tax cash flows of firms increase when they include more debt in their capital structures, leading firms to favor debt over equity financing. However, the capital structure choice becomes more complicated when one considers personal as well as corporate taxes. Personal taxes tend to favor the use of equity in a firm's capital structure since a large portion of the returns on stock are taxed at the capital gains rate, which is generally more favorable than the ordinary tax rate that applies to interest income.

14.1 The Modigliani-Miller Theorem

The first step in understanding the firm's capital structure choice is the Modigliani-Miller Theorem. Much of the rest of this chapter—indeed, much of the rest of this text—will build results based on this theorem.

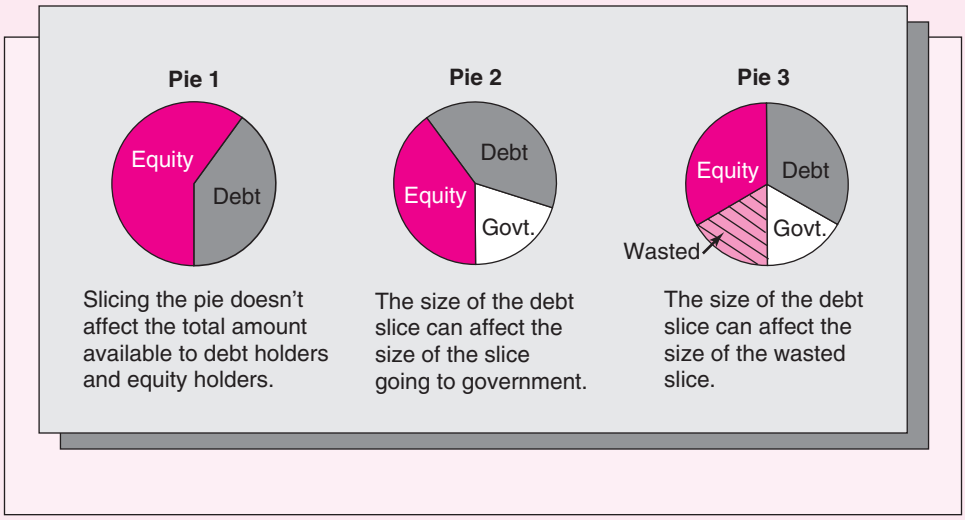
Slicing the Cash Flows of the Firm

Exhibit 14.1 illustrates the total cash flows generated by a firm as a pie chart and the various claims on those cash flows as slices of the pie. Pie 1 illustrates the case in which all of the cash flows accrue to debt holders and equity holders; the way that the pie is sliced does not affect the total cash flows available. This is the assumption of the Modigliani-Miller Theorem.

Pies 2 and 3 illustrate how a firm's capital structure affects the total cash flows to its debt and equity holders. Pie 2 has a piece removed for tax payments to the government. Again, how the pie is sliced does not affect its the total size. However, the owners of the firm are not interested in the total size of the pie. They are interested only in those cash flows they can sell to security holders—that is, the debt and equity portions. Hence, the owners would like to minimize the size of the slice going to the government, which they cannot sell. When more of the pie is allocated to the debt holders, less is allocated to the government, implying that the combined debt and equity slices increase.

Finally, pie 3 includes an additional piece that reflects the possibility that part of the pie is wasted. In other words, a slice of the pie is lost because of inefficiency, lost

EXHIBIT 14.1 Slicing the Cash Flows of the Firm



opportunities, or avoidable costs. Chapters 16 through 19 describe various reasons why the size of this wasted slice may be determined partly by how a firm is financed.

This section, and the two sections that follow it, consider the case illustrated by pie 1, where the total *future* cash flows the firm generates for its debt and equity holders are unaffected by the debt-equity mix. As we will show, if the total cash flows are unaffected by the debt-equity mix, the total *value* of a firm's debt and equity also is unaffected by their mix.

The remainder of this chapter focuses on Pie 2. Here, again, the total cash flows generated by the firm's assets are unaffected by the debt-equity mix. However, in contrast to Pie 1, the corporate profits tax, studied in Section 14.4, and the personal income and capital gains taxes, studied in Section 14.5, claim a portion of these cash flows and the size of these government claims depends on the debt-equity mix.

Proof of the Modigliani-Miller Theorem

Modigliani and Miller (1958) proved the irrelevance theorem by showing that if two firms are identical except for their capital structures, an opportunity to earn arbitrage profits exists if the total values of the two firms are not the same. To illustrate this, assume that two firms exist for one year, produce identical pretax cash flows ($\tilde{X}$) at the end of that year, and then liquidate. However, the firms are financed differently: company U is unleveraged (that is, it has no debt) and company L is leveraged (that is, it has some debt in its capital structure).

Exhibit 14.2 presents the cash flows of companies U and L, the split of the cash flows between debt and equity, and the present values of the cash flows. Since company U has no debt, its uncertain cash flow $\tilde{X}$ is split only among the firm's equity holders. Thus, the current value of company U, or V_U, is the same as the value of its equity. Company L, in contrast, has a debt obligation in one year of $(1 + r_D)D$ dollars. If, for simplicity, we assume that the debt is riskless, and r_D is equal to the risk-

EXHIBIT 14.2 Liability Cash Flows and Their Market Values for Two Firms with Different Capital Structures

	Company U		Company L	
	Future Cash Flow	*Current Value*	*Future Cash Flow*	*Current Value*
Debt	0	0	$(1 + r_D)D$	D
Equity	$\tilde{X}$	V_U	$\tilde{X} - (1 + r_D)D$	E_L
Total	$\tilde{X}$	V_U	$\tilde{X}$	$V_L = D + E_L$

less rate, company L's debt holders will receive $(1 + r_D)D$ at the end of the year and its equity holders will receive the remaining $\tilde{X} - (1 + r_D)D$ dollars. The current value of company L, or V_L, is the current value of its outstanding debt D plus its equity E_L.

According to the Modigliani-Miller Theorem, V_U must equal $D + E_L$, which can be seen by applying the tracking approach to valuation introduced in Part II. Since company U's equity is perfectly tracked in the future by company L's debt plus equity, the value of company U's equity must equal the combined value of company L's debt plus equity. If these values are unequal, an opportunity to earn arbitrage profits exists.

Earning Arbitrage Profits When the Modigliani-Miller Theorem Fails to Hold. To illustrate this arbitrage opportunity, first consider the case where the value of company U is greater than the value of company L ($V_U > D + E_L$). Suppose, for example, that company U has an equity value of \$100 million and company L has an equity value of \$60 million and a debt value of \$30 million. If this were the case, a shrewd investor could profit by buying, for example, 10 percent of the outstanding shares (equity) of company L (costing \$6 million) and 10 percent (\$3 million) of its outstanding debt, and selling short 10 percent (or \$10 million) of the shares (equity) of company U.

Assuming that the investor receives the proceeds of the short sale, this transaction yields a net cash inflow of \$1 million [\$10 million − (\$6 million + \$3 million)]. However, when cash flows are realized at year-end, the investor will receive

$$.1[\tilde{X} - (1 + r_D)D] + .1(1 + r_D)D$$

in cash from company L's stock and bonds and must pay out $.1\tilde{X}$ to cover the short sale of company U's stock. Thus, the net combined future cash flow from these transactions is

$$.1[\tilde{X} - (1 + r_D)D] + .1(1 + r_D)D - .1\tilde{X}$$

which is zero, regardless of the future value that $\tilde{X}$ realizes. In other words, the transaction generates cash for the investor at the beginning of the period, but requires nothing from the investor at the end of the period. Similar opportunities for arbitrage exist if company U has a lower value than company L, that is, $V_U < D + E_L$. Since we assume that such arbitrage opportunities cannot exist, the total value of the two firms must be the same regardless of how they are financed.

Example 14.1 illustrates this type of arbitrage opportunity in a simple case where the firm's cash flows can take on one of only two possible values.

Example 14.1: An Arbitrage Opportunity If the Modigliani-Miller Theorem Is False

Assume:

- Company U is financed totally with equity and is worth $100 million.
- An otherwise identical company L is financed with $40 million in equity plus $50 million (market value) in riskless debt that offers a 10 percent interest rate. Thus, the bonds pay $55 million (= $5 million in interest plus $50 million in principal) at the end of the year.
- If the economy is weak, cash flows for each company will be $80 million; if the economy is strong, cash flows for each company will be $200 million.

Show how a shrewd investor, able to purchase or short up to $10 million in any security, would profit from this violation of the Modigliani-Miller Theorem.

Answer: The shrewd investor can profit by purchasing 10 percent of company L's equity ($4 million) and 10 percent of its outstanding debt ($5 million) while selling short 10 percent of the shares of company U. The investor would then realize an immediate cash inflow of $1 million (= .10 ($100 million − $40 million − $50 million). However, as the following table shows, the investor would have no net obligations at the end of the year, so the initial $1 million inflow can be considered a risk-free profit.

	Cash Flow to Investor (in $ millions) at		
	Beginning of Year	End of Year	
		Weak Economy	*Strong Economy*
Short sale of U equity	$10	($8.0)	($20.0)
Purchase of L equity	(4)	2.5	14.5
Purchase of L debt	(5)	5.5	5.5
Net cash inflow	$ 1	$0.0	$ 0.0

The year-end equity figures for company L are, by definition, the total cash flows for company L less the principal and interest payments to debt holders.

Stating the Modigliani-Miller Theorem Explicitly. Example 14.1 illustrates how a shrewd investor can realize arbitrage profits when the Modigliani-Miller Theorem is violated. Result 14.1 states explicitly the assumptions and the implications of this theorem.

Result 14.1 *(The Modigliani-Miller Theorem.)* Assume: (1) a firm's total cash flows to its debt and equity holders are not affected by how it is financed, (2) there are no transaction costs, and (3) no arbitrage opportunities exist in the economy. Then the total market value of the firm, which is the same as the sum of the market values of the items on the right-hand side of the balance sheet (that is, its debt and equity), is not affected by how it is financed.

Assumptions of the Modigliani-Miller Theorem

Result 14.1 indicates that the capital structure irrelevance theorem holds only under some restrictive assumptions. The Modigliani-Miller theorem is, however, important because it provides a framework that allows managers to focus on those factors that are important determinants of the optimal capital structure choice. Examining the

different assumptions of the theorem provides important insights into how the capital structure decision affects firm values.

The Key Assumption. The first assumption of the Modigliani-Miller Theorem—that the sum of all future cash flows distributed to the firm's debt and equity investors is unaffected by capital structure—is really the key, and it will be the focus of much of the remainder of this text. In reality, capital structure can affect a firm's cash flows for a number of reasons. This chapter focuses on how capital structure can affect a firm's total cash flows by altering its tax liabilities.[2]

The Importance of Transaction Costs. The second assumption, no transaction costs, was used throughout Parts II and III of the text. However, transaction costs play a special role here that requires some additional discussion. In the absence of transaction costs, the cash flows to investors from any new security that a firm issues can be tracked by other securities that already exist in the market.[3] This means that the issuing firm is not really offering the investing public a pattern of returns that they could not otherwise obtain. However, transaction costs limit the availability of return patterns that can be obtained in the market and sometimes provide firms with an opportunity to improve their value by issuing special securities that investors desire.

For example, in the early 1990s in Hong Kong, there was a strong demand for warrants because they provide investors with highly leveraged positions with limited downside risk. In the absence of transaction costs, warrants are not special securities because they can be tracked with a dynamic strategy involving the underlying stock and the risk-free asset (see Chapter 8). However, if transaction costs are high, tracking a warrant with a dynamic strategy may be impossible. As a result, Hong Kong firms were able to take advantage of the demand for option-like payoffs by issuing warrants at prices exceeding the theoretical Black/Scholes prices discussed in Chapter 8. For similar reasons, a firm might be able, in some instances, to obtain relatively attractive financing with a convertible bond.

The Absence of Arbitrage. The final Modigliani-Miller assumption, no arbitrage, is made throughout the text. Analyzing any sort of decision that affects a corporation's value requires some framework for determining valuation. The Modigliani-Miller Theorem is consistent with all pricing models that satisfy the most basic assumption in asset pricing: Equilibrium prices cannot provide opportunities for riskless arbitrage profits.

14.2 How an Individual Investor Can "Undo" a Firm's Capital Structure Choice

The proof of the Modigliani-Miller Theorem described in the last section implicitly assumes that firms with identical cash flows but different capital structures exist. An alternative interpretation of the Modigliani-Miller Theorem is that, under its assumptions, a firm's shareholders are indifferent to a change in its capital structure. This is illustrated in Example 14.2.

[2]Chapters 16–19 examine how the financing mix affects a firm's investment choices and the efficiency of its operations.

[3]This tracking does not have to be perfect. For example, if the CAPM holds, the Modigliani-Miller Theorem still applies. The key is that investors cannot be made better off by the issuance of a new security.

Example 14.2: Undoing Elco's Capital Structure Change

Elco is considering a change in its capital structure. Before the change, the firm's stock sells for \$100 per share, and the firm has 1,000 shares outstanding. The firm also is financed with riskless zero-coupon debt maturing in one year, and having a current market value of $D = \$10{,}000$.

Stanley Kowalski currently owns 100 shares of Elco stock (10 percent of its equity). In the absence of a capital structure change, Stanley's payoff next year will equal

$$.1[\tilde{X} - (1 + r_D)\$10{,}000] = .1\tilde{X} - (1 + r_D)\$1{,}000$$

Here X is the firm's cash flows and r_D is the risk-free interest rate to be paid to debt holders, so that the firm's future debt service obligation is $(1 + r_D)\$10{,}000$. The firm plans to repurchase 500 shares of its outstanding stock for \$50,000 and will finance the equity repurchase by issuing \$50,000 in risk-free debt.

First, show what Stanley's payoff will be if he does nothing to counteract the firm's actions. Then show how he, as an individual investor, can alter his personal portfolio to undo the effect of any change in the firm's leverage, so that he attains the same cash flows he would have received without the leverage change.

Answer: If Stanley chooses not to alter his portfolio, his share of Elco's cash flow next period will be

$$.2[\tilde{X} - (1 + r_D)\$60{,}000]$$

because he would own 20 percent (= 100/500) of the firm's outstanding shares if half the shares were repurchased. He would then have a riskier investment than he held previously, which he may or may not prefer.

However, if Stanley sells 50 shares of his stock, using the proceeds to buy \$5,000 in bonds, he will once again own 10 percent (= 50/500) of the firm's shares and will realize a return equal to

$$.1[\tilde{X} - (1 + r_D)\$60{,}000] + (1 + r_D)\$5{,}000 = .1\tilde{X} - (1 + r_D)\$1{,}000$$

Example 14.2 illustrates how shareholders can undo the effect of a change in a firm's capital structure by making offsetting changes to their own portfolio. The shareholder can achieve the same cash flow pattern and continue to control the same percent of the firm's shares. Thus, without transaction costs, the shareholder is indifferent to changes in the firm's capital structure.

14.3 How Risky Debt Affects the Modigliani-Miller Theorem

Up to this point, we have assumed that all debt is riskless; that is, there is no bankruptcy. This section discusses situations in which firms can go bankrupt, thus making their debt risky.

The Modigliani-Miller Theorem with Costless Bankruptcy

The assumptions of the Modigliani-Miller Theorem permit bankruptcy, but no bankruptcy costs. In other words, the theorem assumes that if a firm is unable to meet its debt obligations and goes bankrupt, the ownership and control of the firm's assets move costlessly from the equity holders to the debt holders.[4]

[4]Of course, this assumption is unrealistic. Lawyers are very much involved in the bankruptcy process and they are expensive. The various costs of bankruptcy are discussed in detail in Chapter 16.

To understand what we mean by costless bankruptcy, return to Example 14.2, in which the payoff to Stanley Kowalski, who owns 10 percent of Elco's shares, is 10 percent of the cash flow less Elco's debt obligations:

$$.1[\tilde{X} - (1 + r_{DS})\$10{,}000]$$

where r_{DS} is the promised interest rate to be paid to the original or senior debt holders.

If the debt in Example 14.2 had been risky, the cash flow described in the above expression would be the payoff to Stanley *only if this quantity is positive*. Because stockholders enjoy limited liability, they receive nothing if the debt obligation exceeds the cash flows for the period; in this case, debt holders receive the entire cash flow $\tilde{X}$ of the firm, assuming no bankruptcy costs. The debt is risky because in these instances, the entire cash flow can still be considerably less than the promised debt payment.

Leverage Increases and Wealth Transfers

When bankruptcy can occur, we must be more explicit about what we mean when we say that the capital structure decision is irrelevant. The possibility of bankruptcy implies that capital structure changes can result in transfers of wealth between the firm's equity holders and its debt holders. Hence, a change in a firm's debt-equity mix can affect its share price even if the change does not affect the sum of the firm's debt and equity values.

The potential for wealth transfers from existing debt holders to the firm's equity holders depends on whether the new debt must be subordinated. In bankruptcy proceedings, *subordinated* or *junior debt* (see Chapter 2) has a claim to the assets of the firm only after the debt with higher priority has been fully paid off. As we will show, if a new debt issue is not subordinated to the old debt, the new debt can generate a transfer of wealth from the existing debt holders to the equity holders. However, if the debt is subordinated, shareholders will continue to be indifferent to a capital structure change that does not affect the assets of the firm.

The Effect of Leverage Changes When New Debt Is Subordinated to Existing Debt. To understand the effect of a capital structure change when debt is risky, consider again the possibility that Elco issues $50,000 in new debt and uses the proceeds to repurchase 500 shares. Assume initially that the new debt is junior to existing debt ($10,000 face amount) in the event of bankruptcy, and, as a consequence, its coupon rate, r_{DJ}, is greater than r_{DS}, the promised rate on senior debt.

Exhibit 14.3 illustrates that Mr. Kowalski can, in effect, undo the effects of additional risky debt in Elco's capital structure by buying $5,000 of the new debt and selling 50 shares. This leaves him with 50 shares (100 shares less the 50 sold) and $5,000 of the new junior debt.

Note that in each scenario of Exhibit 14.3 the cash flow is identical to what Mr. Kowalski would have received as a shareholder, given the firm's original capital structure. In scenarios A and B—where the firm has sufficient cash to pay off its obligations to senior debt holders—the investor receives the same cash flow he or she would have received as an equity holder, given the firm's original capital structure, that is, $.1[\tilde{X} - (1 + r_{DS})\$10{,}000]$. In scenario C, the state in which the firm generates insufficient cash flows to pay off its senior debt, an equity holder would have received nothing even in the absence of a new debt issue. Now the investor still receives nothing as an investor holding both equity and junior debt.

EXHIBIT 14.3 Undoing the Effects of Additional Risky Debt When New Debt Is Junior to Old Debt

	Cash Flows		
Investment	*Scenario A* Cash flow exceeds all debt obligations: $\tilde{X} > (1 + r_{DS})\$10{,}000 + (1 + r_{DJ})\$50{,}000$	*Scenario B* Cash flow exceeds sr. but not jr. debt obligation $(1 + r_{DJ})\$50{,}000 > \tilde{X} - (1 + r_{DS})\$10{,}000 > 0$	*Scenario C* Senior debt obligation exceeds cash flow: $\tilde{X} < (1 + r_{DS})\$10{,}000$
50 shares of stock	$.1[\tilde{X} - (1 + r_{DS})\$10{,}000] - .1[(1 + r_{DJ})\$50{,}000]$	0	0
\$5,000 of new debt	$.1(1 + r_{DJ})\$50{,}000$	$.1[\tilde{X} - (1 + r_{DS})\$10{,}000]$	0
Total	$.1[\tilde{X} - (1 + r_{DS})\$10{,}000]$	$.1[\tilde{X} - (1 + r_{DS})\$10{,}000]$	0

Note: r_{DS} represents the promised interest rate on the senior debt and r_{DJ} is the promised interest rate on the junior debt.

Result 14.2 summarizes the main points about how capital structure changes affect debt and equity holders when new debt is subordinated to existing debt.

Result 14.2 Assume (1) a firm's total cash flows to its debt and equity holders are unaffected by how it is financed; (2) there are no transaction costs; and (3) no arbitrage opportunities exist in the economy. Then, if a firm's existing debt holders have a senior claim in the event of bankruptcy, both the firm's stock price per share and the value of its existing senior debt claims are unaffected by changes in the firm's capital structure.

What Occurs When New Debt Is Not Subordinated to Old Debt? When firms issue junior debt, its promised yield must exceed r_{DS}, the promised yield on the existing senior debt, to compensate for the greater default risk born by the junior debt holders. If a new debt issue is not junior, but rather has the same priority as existing debt, both new and existing debt holders proportionately share the assets of the bankrupt firm. The new debt holders greater claim to bankrupt assets makes the existing debt holders worse off, but it allows the new debt issue to attract investors with a lower promised yield than if it were junior to the existing debt.

This cost to the existing debt holders benefits the stockholders. To understand this, consider again the case analyzed in Exhibit 14.3, except now assume that the new debt is *not* subordinated. Assume again that the investor undoes the capital structure increase by selling 50 shares of stock and buying \$5,000 of new debt. In scenario A, the investor still receives the same cash flow that would have been achieved under the original capital structure. Under scenarios B and C where the firm is bankrupt, however, the investor receives greater cash flows than Exhibit 14.3 specifies because the new debt the investor holds now receives a proportional share of the bankruptcy proceeds.

Result 14.3 If a firm's existing debt holders do not have a senior claim in the event of bankruptcy, a new debt issue can decrease the value of existing debt. Under the assumptions listed in Result 14.1, however, the loss to the old debt holders would be offset by a gain to the equity holders, leaving the total value of the firm unaltered by this type of capital structure change.

During the 1980s, a number of firms increased their debt ratios substantially, most dramatically in leveraged buyout transactions that took public companies private. In a number of these transactions, the transfer of wealth from the original debt holders to

the equity holders was quite large. In the RJR Nabisco leveraged buyout, for example, some existing bonds lost about 20 percent of their value.[5]

14.4 How Corporate Taxes Affect the Capital Structure Choice

Financial managers spend a great deal of time making decisions about their firms' capital structures. In addition, stock prices react dramatically when firms make major changes in their capital structures.[6] This suggests that it probably would be unwise to stick with the conclusion that the capital structure decision is irrelevant.

The apparent relevance of the capital structure decision suggests that some of the assumptions underlying the Modigliani-Miller Theorem are unrealistic and that relaxing them may have important implications for the firm's capital structure choice. The most obviously unrealistic assumption is that of no taxes. Taxes have a major effect on the cash flows of firms and, as a result, strongly influence their capital structure decisions. Indeed, a recent study of 392 CFOs by Graham and Harvey (2001) found that 45% surveyed agreed that tax considerations played either an important or very important role in their capital structure choices. The preceding two sections indicated that, in the absence of taxes, a firm's value does not depend on its capital structure. It follows that, in the absence of other market frictions, minimizing the amount paid in taxes maximizes the cash flows to the firm's equity holders and debt holders, thereby maximizing the firm's total value.

In the U.S., the existence of corporate taxes favors debt financing. This is because, in the U.S., interest is a tax-deductible corporate expense. However, since dividends are viewed as distributions of profits rather than expenses of doing business, they are not tax deductible.[7]

Howard Hughes's perceptive analysis of the tax benefits of debt financing, described in the chapter's opening vignette, predated the academic discussion of debt and taxes by several years. This section explores the types of tax benefits he mentioned in more detail, specifically considering situations in which there are (1) corporate taxes, (2) tax-deductible interest expenses, and (3) no personal taxes. In the next section and in Chapter 15 we will explore the effects of personal taxes.

How Debt Affects After-Tax Cash Flows

If the corporate tax rate is T_c, a firm with pretax cash flows of $\tilde{X}$, which we will assume for simplicity is EBIT, and interest payments of $r_D D$ has taxable income of $\tilde{X} - r_D D$ and pays a corporate tax of $(\tilde{X} - r_D D)T_c$. Since interest expense is tax deductible, a firm can reduce its tax liabilities and thus increase the amount it distributes to its security holders by issuing additional debt. Therefore, in the absence of personal taxes, transaction costs, and bankruptcy costs, and holding the pretax cash flow generated by the firm constant, the value-maximizing capital structure includes enough debt to eliminate the firm's tax liabilities.

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[5]Studies by Asquith and Wizman (1990) and Warga and Welch (1993) indicate that bondholders lost money in a number of other leveraged buyouts as well.

[6]Stock price reactions to corporate events like capital structure changes are discussed in Chapter 19.

[7]A firm can carry back the net losses in its current year as far as two years. When there is not enough income in the previous two years to allow the loss carryback, the firm can carry forward those losses for up to 20 years to offset future taxable profits. Hence, the corporate tax advantage of debt financing applies even to firms that are temporarily unprofitable, although the debt tax shield is used less efficiently if it must be carried forward.

total value of the firm unaltered by this type of capital structure change.

Result 14.4: Assume that the pretax cash flows of the firm are unaffected by a change in a firm's capital structure, and that there are no transaction costs or opportunities for arbitrage. With corporate taxes at the rate T_c, but no personal taxes, the value of a levered firm with static risk-free perpetual debt is the value of an otherwise equivalent unlevered firm plus the product of the corporate tax rate and the market value of the firm's debt; that is

$$V_L = V_U + T_c D$$

Result 14.5: Assume that the pretax cash flows of the firm are unaffected by a change in a firm's capital structure, and that there are no transaction costs or opportunities for arbitrage. With corporate taxes but no personal taxes, a firm's optimal capital structure will include enough debt to completely eliminate the firm's tax liabilities.

Result 14.6: Assume that the pretax cash flows of the firm are unaffected by a change in a firm's capital structure, and that there are no transaction costs of opportunities for arbitrage. If investors all have personal tax rates on debt and equity income of T_D and T_E, respectively, and if the corporate tax rate is T_c, then the value of a levered firm exceeds the value of an otherwise equivalent unlevered firm by $T_g D$; that is

$$V_L = V_U + T_g D,$$

where

$$T_g = 1 - \left[\frac{(1 - T_c)(1 - T_E)}{1 - T_D}\right]$$

Result 14.7: Assume there is a tax gain from leverage, but the taxable earnings of firms are low relative to their present values.

- With riskless future cash flows, firms will want to use debt financing up to the point where they eliminate their entire corporate tax liabilities, but they will not want to borrow beyond that point.
- With uncertainty, firms will pick the debt ratio that weighs the benefits associated with the debt tax shield when it can be used against the higher cost of debt in cases where the debt tax shield cannot be used.
- Firms with more nondebt tax shields are likely to use less debt financing.

Result 14.8: For low tax bracket investors, it is often cheaper to lease an asset than to buy it.

Key Terms

accelerated depreciation 524
economic depreciation 523
effective marginal tax rates 518
financial lease (capital lease) 523
lessee-lessor 523
Modigliani-Miller Theorem 501
nondebt tax shields 515
operating lease 523
true lease 525

Exercises

14.1. Suppose $r_D = 12\%$, $\bar{r}_E = 10\%$, $T_c = 33\%$, $T_D = 20\%$.

a. What is the marginal tax rate on stock income T_E which would make an investor indifferent in terms of after-tax returns between holding stock or bonds? Assume all betas are zero.

b. What is the probability that a firm will not utilize its tax shield if, on the margin, the firm is indifferent between issuing a little more debt or equity?

14.2. Consider a single period binomial setting where the riskless interest rate is zero, and there are no taxes. A firm consists of a machine that will produce cash flows of $210 if the economy is good and $80 if the economy is bad. The good and bad states occur with equal risk-neutral probability. Initially, the firm has 100 shares outstanding and debt with a face value of $50 due at the end of the period. What is the share price of the firm?

14.3. Suppose the firm in exercise 14.2 unexpectedly announces that it will issue additional debt, with the same seniority as existing debt and a face value of $50. The firm will use the entire proceeds to repurchase some of the outstanding shares.
a. What is the market price of the new debt?
b. Just after the announcement, what will the price of a share jump to?
c. Show how a shareholder with 20 percent of the shares outstanding is better off as a result of this transaction when he or she undoes the leverage change.
d. Show how the Modigliani-Miller Theorem still holds.

14.4. Assume that the real riskless interest rate is zero and the corporate tax rate is 33 percent. IGWT Industries can borrow at the riskless interest rate. It will have an inflation-adjusted EBIT next year of $200 million. It would like to borrow $50 million today. Its only deductions will be interest payments (if any).
a. What are its interest payments, taxable income, tax payments, and income left for shareholders in a no-inflation environment?
b. Suppose there is inflation of 10 percent per year, but the real interest rate stays at zero. This means that investors now will require a sure payment of $1.10 next year for each $1.00 loaned today. Repeat part *a,* assuming that EBIT is affected by inflation.
c. In which environment is the inflation-adjusted income left for shareholders higher? Why?

14.5. As owner of 10 percent of ABC Industries, you have control of its capital structure decision. The current corporate tax rate is 34 percent and your personal tax rate is 31 percent. Assume that the returns to stockholders accrue as nontaxable capital gains. ABC currently has no debt and can finance the repurchase of 10 percent of its outstanding shares by borrowing $100 million at the risk free rate of 10 percent. The AAA (tax-free) municipal bond rate is 8 percent. If you hold your 10 percent of the firm constant and buy the municipal bonds, what is your annual after-tax gain from this transaction?

14.6. Explain how inflation affects the capital structure decision. Does inflation affect the capital structure choice differently for different firms?

14.7. Assume the corporate tax rate is 50 percent, AAA corporate bonds are trading at a yield of 9 percent, and municipal bonds are trading at a yield of 6 percent. How can the shareholders of an AAA-rated firm gain by increasing the leverage of their firm without increasing the leverage of their personal portfolio? Assume the probability of bankruptcy is zero.

14.8. During the early 1990s, most new airplanes were leased by the airlines. This was not true during the early and mid-1980s. Explain why.

14.9. Restaurant chains like McDonald's sometimes franchise their restaurants and sometimes own them outright. The franchised restaurants are usually owned by individuals who hold them in subchapter S corporations which pass income through directly to the owners. There is no corporate tax on this income, but the owner must pay personal taxes on the income.
a. From the perspective of the owner of the franchise, is there a tax advantage to debt financing?
b. Which organizational form is better from the perspective of tax minimization: corporate ownership of the individual restaurants or franchises?

14.10. Real estate investment trusts (REITs) are companies set up to manage investment properties like office buildings and apartment houses. REITs are not subject to corporate taxes and are required to pass through 95 percent of their income to their shareholders who are taxed at the personal level. How do we expect taxes to affect the capital structure choice of REITs?

14.11. X-Tex Industries has large depreciation tax deductions and can thus eliminate all of its taxable income with a relatively small amount of debt. In contrast, Unique Scientific Equipment Corporation is generating a substantial amount of taxable income. Despite the tax advantage of debt, Unique uses only a modest amount of debt financing because the nature of its products would make financial distress very costly. Suppose the rate of inflation increased from 3 percent to 6 percent, increasing borrowing rates from 6 percent to 9 percent. How would this affect the optimal capital structures of these two firms?

14.12. Jeff started an Internet company, Finstrat.com, which, unlike others in the industry, generated taxable earnings almost immediately. Jeff owns 10 percent of the shares, and the rest of the shares are held by tax-exempt institutions. The firm needs to raise $100 million in new capital. Jeff would like to see the firm issue equity and would be willing to purchase $10 million of the new equity to keep his ownership stake constant. However, the institutions would like to see the firm raise the capital through debt. Explain how part of this disagreement might be related to taxes.

however, firms generally will want to limit the amount of debt in their capital structure even when there are substantial tax benefits to debt financing.

Most of the key issues in this chapter are developed with a set of relatively simple numerical examples. Unless specified otherwise, valuation in these examples assumes risk neutrality, no direct bankruptcy costs, no taxes, and a risk-free rate of zero.

16.1 Bankruptcy

The U.S. Bankruptcy Code

Firms in the U.S. that are unable to make required payments to their creditors can file for either a **Chapter 7** bankruptcy,[1] which leads to the liquidation of the firm's assets in the settlement of the creditors' claims, or a **Chapter 11** bankruptcy, which allows the firm to restructure its debt and equity claims and continue to operate.

Chapter 7 Bankruptcy. In a Chapter 7 bankruptcy, the bankruptcy court selects a trustee from outside the company who liquidates the assets of the firm and distributes the proceeds to the debt holders. Any proceeds from the liquidation that remain after settling the debt holders' claims are then distributed to the company's shareholders. Under Chapter 7, these proceeds are divided among the claim holders according to the **absolute priority rule**, which states that debt holders must be paid in full before equity holders receive any proceeds of the bankruptcy. The rule also states that secured debt holders must be paid before unsecured debt holders and that the more senior of the unsecured debt holders must be paid in full before the more junior debt holders.

Chapter 11 Bankruptcy. Chapter 11 bankruptcies are much more complicated than Chapter 7 bankruptcies. Under Chapter 11, the claims of debt and equity holders cannot be settled with cash realized from the liquidation of assets. Rather, the debt and equity holders receive new financial claims in exchange for their existing claims. For example, debt holders often end up with equity in the newly reorganized firm that emerges from a Chapter 11 bankruptcy.

Almost all bankruptcies of large corporations start out as Chapter 11 bankruptcies and become Chapter 7 bankruptcies only when the various claimants fail to agree on a **reorganization plan**, which specifies how the new financial claims are to be distributed among the claim holders. After filing for a Chapter 11 bankruptcy, the corporation has 120 days to submit a reorganization plan, which creditors can either accept or reject.[2] If creditors reject the offer or the corporation fails to submit a reorganization plan, the judge overseeing the bankruptcy case can give the corporation an extension during which it must come up with an acceptable plan or ask the creditors to come up with their own reorganization plan. In most cases, at least one extension is granted.

To determine the acceptability of a plan, each class of **impaired creditors**—the creditors who will not be paid in full—as well as the equity holders must vote on it. For a specific class of creditors (for example, convertible bondholders) to accept the plan, a simple majority of the claimants and those who hold two-thirds of the dollar amount of the claims must vote favorably. A class of stockholders exhibits approval for the plan when at least two-thirds of the firm's shares vote to accept it.

[1] In reality, it is often the creditors who file for Chapter 7 bankruptcy.

[2] Judges frequently extend this 120-day period.

If all classes of claimants accept the plan, the court will approve it. However, it may not be necessary to have all classes of claimants approve the plan. The court may confirm the plan over the objections of the dissenting classes if it judges the plan to be nondiscriminatory, fair, and equitable. A plan that is forced on dissenting claimants is known as a **cramdown**. A necessary condition for a cramdown is that the dissenters receive at least what they would receive under a Chapter 7 liquidation.

The absolute priority rule is often violated in a Chapter 11 settlement.[3] Junior debt holders and equity holders often receive a share of the bankruptcy proceeds even when senior debt holders are not paid in full. Senior debt holders allow these deviations from the absolute priority rule to facilitate a timely settlement of the bankruptcy and to avoid future lawsuits. For example, junior debt holders often sue the bankrupt firm's bank for taking actions that may have caused the firm to go into bankruptcy. In an attempt to settle those lawsuits, the banks often allow these more junior claimants to receive a portion of the bankruptcy settlement.

The Direct Costs of Bankruptcy

If the bankruptcy process is relatively costless, the possibility of bankruptcy does not affect a firm's capital structure decision. Recall from Chapter 14 of this text that the Modigliani-Miller Theorem does not require that there be no bankruptcy, only that the bankruptcy be costless.

There are, however, a number of costs directly related to bankruptcy, including the time management spends dealing with creditors and the additional time creditors spend with the managers of a bankrupt firm. Legal expenses, court costs, and advisory fees also are direct costs of bankruptcy. For example, when Macy's department store went bankrupt in 1992, it spent an estimated $100 million on "lawyers, accountants, investment bankers, and other highly paid advisors."[4] The $100 million represented approximately 2 to 3 percent of the firm's estimated value at the time. Macy's emerged from Chapter 11 bankruptcy in mid-1994 as a division of Federated Department Stores. The two and one-half years that Macy's spent in bankruptcy proceedings was slightly more than the average time spent in bankruptcy.[5]

Estimates of the Direct Costs of Bankruptcy. The 2 to 3 percent of assets that Macy's lost owing to direct bankruptcy costs is fairly typical for large U.S. corporations. Warner (1977) estimated that direct costs of bankruptcy for large railroads, although substantial in magnitude, are quite small relative to the size of the firms, averaging only 5.3 percent of their (already diminished) debt and equity values just before the bankruptcy filings. A study of bankruptcies of large firms by Weiss (1990) determined that direct costs averaged about 3.1 percent of the total value of the debt and equity of the bankrupt firms.

Because most of the direct costs of bankruptcy are the same for both small and large firms, the bankruptcy costs of small firms, as a proportion of the value of their assets, are much larger. For small firms, these costs may be fairly large, perhaps 20–25 percent of a firm's value.[6] However, even these high percentages suggest that the direct costs of bankruptcy add very little to a firm's borrowing costs except in cases where

[3]See Franks and Torous (1989).
[4]*The Economist,* May 21, 1994.
[5]Hotchkiss (1995) reported that on average firms in her sample spent 19.5 months in bankruptcy.
[6]See Ang, Chua, and McConnell (1982) and Altman (1984).

bankruptcy is quite likely. This is because these costs (1) are percentages of an already diminished firm value and (2) must be multiplied by the probability of bankruptcy, and then discounted, to determine their impact on borrowing costs.

Who Bears the Bankruptcy Costs? Under the absolute priority rule, most of a firm's value in the event of bankruptcy is transferred to its debt holders. Since the direct costs of bankruptcy diminish the value of the firm, most direct bankruptcy costs are thus ultimately borne by the firm's debt holders.

Why should managers, who represent the interests of the firm's equity holders, be concerned about the costs borne by debt holders? While equity holders are not directly concerned with costs that may be imposed on their firm's lenders, they are concerned about the rates at which the firm can borrow. Since lenders realize they will be bearing costs in the event of bankruptcy, they will demand a *default premium* on the interest rate they charge the firm. The **default premium**, the difference between the promised yield to the bond's lenders and the yield on a bond with no default, reflects the probability and cost of the firm's bankruptcy. In paying this default premium, shareholders are, in effect, paying the expected bankruptcy costs whenever they issue risky debt. Therefore, they must consider that at least a portion of the premium (that due to the cost of bankruptcy) reflects a cost that may offset other advantages associated with debt financing.

Example 16.1: How Bankruptcy Affects Borrowing Costs

Westlake Corporation would like to borrow $1 million for one year. There is a 90 percent chance that the loan will be repaid in full and a 10 percent chance that the firm will be bankrupt at the end of the year. In the event that the firm is bankrupt, its assets can be sold for $600,000. However, the legal costs of seizing Westlake's assets will cost the bank $100,000. How much will a bank charge for the loan if it prices loans to companies like Westlake to earn, on average, 10 percent? How is the interest rate charged on the loan affected by the potential bankruptcy costs?

Answer: To realize its 10 percent expected return, the bank needs to receive, on average, $1.1 million at the end of the year. If Westlake is bankrupt, the bank will receive only $500,000. Hence, to receive $1.1 million, on average, it must receive $1.167 million if the firm is not bankrupt (since $.1 \times \$.5 \text{ million} + .9 \times \$1.167 \text{ million} = \$1.1 \text{ million}$). In other words, the bank must charge 16.7 percent on the loan. If there were no costs associated with bankruptcy, the bank would recover $600,000 instead of $500,000 in the event of bankruptcy, and it would then need to charge only 15.6 percent on the loan.

The general principle to remember from Example 16.1 is that lenders are likely to anticipate future events and thus require compensation for the costs they expect to bear.

Result 16.1 Debt holders charge an interest premium that reflects the expected costs they must bear in the event of default. Therefore, equity holders indirectly bear the expected costs of bankruptcy and must consider these costs when choosing their optimal capital structures.

16.2 Debt Holder–Equity Holder Conflicts: An Indirect Bankruptcy Cost

If direct bankruptcy costs were the only costs associated with debt financing, it would be difficult to justify the relatively low debt ratios of firms like IBM that have long histories of generating taxable earnings and could presumably lower their costs of capital by increasing their leverage. The expected present value of the direct costs of

Result 16.9: Firms with the potential to select high-risk projects may be unable to obtain debt financing at any borrowing rate when risk-free interest rates are high.

Result 16.10: Since debt holders have priority in the event of liquidation, they have a stronger interest in liquidating the assets of a distressed firm than the firm's equity holders, who profit from the possible upside benefits that may be realized if the firm continues to operate. As a result, a firm's financial structure partially determines the conditions under which it liquidates.

Result 16.11: In Chapter 11 bankruptcy, firms are able to obtain debtor-in-possession (DIP) financing. To some extent, this provision of the bankruptcy code mitigates the debt overhang/underinvestment problem. However, the provision also may allow some firms to continue operating when they would be better off liquidating.

Result 16.12: The adverse effects of debt financing on a firm's unlevered cash flows arising from debt holder–equity holder conflicts may be mitigated by using

- protective covenants,
- bank debt and privately placed debt,
- short-term debt instead of long-term debt,
- convertible bonds,
- project financing, and
- properly designed management compensation contracts.

However, many of these remedies have downsides to them as well.

Key Terms

absolute priority rule 559
asset substitution problem 563
Chapter 7 559
Chapter 11 559
cramdown 560
debtor-in-possession (DIP) financing 580
debt overhang problem 563
default premium 561
direct bankruptcy costs 558
financial distress costs 562
financially distressed firms 558
free-rider problem 565
going concern value 576
hold up problem 584
impaired creditors 559
indirect bankruptcy costs 558
liquidation costs 576
liquidation value 576
nonfinancial stakeholders 558
non-recourse debt 586
project financing 586
reluctance to liquidate problem 563
reorganization plan 559
shortsighted investment problem 563
technical default 581
underinvestment problem 563

Exercises

16.1. A firm has $100 million in cash on hand and a debt obligation of $100 million due in the next period. With this cash, it can take on one of two projects—A or B—which cost $100 million each. Assume that the firm cannot raise any additional outside funds. If the economy is favorable, project A will pay $120 million and project B will pay $101 million. If the economy is unfavorable, project A will pay $60 million and project B will pay $101 million. Assume that investors are risk neutral, there are no taxes or direct costs of bankruptcy, the riskless interest rate is zero, and the probability of each state is .5.
a. What is the *NPV* of each project?
b. Which project will equity holders want the managers to take? Why?

16.2. Nigel decides he can make zippers at night for one period and will have cash flows next period of $210 if the economy is favorable, and $66 if the economy is unfavorable. One-third of these proceeds must be paid out in taxes if the firm is all equity financed; however, because of the tax advantage of debt, Nigel saves $0.05 in taxes for every $1.00 of debt financing that he uses. Assume investors are risk neutral, the riskless rate is 10 percent per period, and the probability of each state is .5. Also assume that if Nigel's firm goes bankrupt and debt holders take over, the legal fees and other bankruptcy costs total $20.

a. If Nigel organizes his firm as all equity, what will it be worth?
b. Suppose Nigel's firm sold a zero-coupon bond worth $44 at maturity next period. How much would the firm receive for the debt?
c. With the debt level above, how much would the equity be worth?
d. How much would the firm be worth?
e. Would the firm be worth more if it had a debt obligation of $70 next period?

16.3. A firm has a senior bond obligation of $20 due this period and $100 next period. It also has a subordinated loan of $40 owed to Jack and Jill and due next period. It has no projects to provide cash flows this period. Therefore, if the firm cannot get a loan of $20, it must liquidate. The firm has a current liquidation value of $120. If the firm does not liquidate, it can take one of two projects with no additional investment. If it takes project A, it will receive cash flows of $135 next period, for sure. If the firm takes project B, it will receive either cash flows of $161 or $69 with equal probability. Assume risk neutrality, a zero interest rate, no direct bankruptcy costs, and no taxes.
a. What has a higher *PV*: liquidating, project A, or project B?
b. Should Jack and Jill agree to loan the firm the $20 it needs to stay operating if they receive a (subordinated) bond with a face value of $20.50?
c. If the firm does receive the loan from Jack and Jill, which project will the managers choose if they act in the interest of the equity holders?

16.4. Hiroko Fashion Corporation (HFC) can pursue either project Dress or project Cosmetic, with possible payoffs at year-end as follows:

	Bad Economy (prob. = 30%) (in $ millions)	*Good Economy (prob. = 70%) (in $ millions)*
Project Dress	$2	$9
Project Cosmetic	7	6

Each project costs $6 million at the beginning of the year. Assume there are no taxes, there are no direct bankruptcy costs, all investors are risk neutral, and the risk-free interest rate is zero.
a. Which project should HFC pursue if it is all equity financed? Why?
b. If HFC has a $5 million bond obligation at the end of the year, which project would its equity holders want to pursue? Why?

16.5. Sigma Design, a computer interface start-up firm with no tangible assets, has invested $50,000 in R&D. The success of the R&D effort as well as the state of the economy will be observed in one year. If the R&D is successful (*prob.* = 90%), Sigma requires a $53,000 investment to start manufacturing. If the economy is favorable (*prob.* = 90%), the project is worth $153,000, and if it is unfavorable, the project will have a value of $61,000. Demonstrate how the value of Sigma is affected by whether or not it was originally financed with debt or equity. Assume no taxes, no direct bankruptcy costs, all investors are risk neutral, and the risk-free interest rate is zero.

16.6. In Japan, financial institutions hold significant equity interests in the borrowing firms. How does this affect the costs of financial distress and bankruptcy?

16.7. Describe the relation between the zero-beta expected return on common stock and the zero-beta expected return on corporate bonds in an economy where stock returns are taxed more favorably than bond returns, interest payments are tax deductible, and bankruptcy costs are important determinants of a firm's capital structure choice.

16.8. ABC Corp., which currently has no assets, is considering two projects that each cost $100. Project A pays off $120 next year in the good state of the economy and $90 in the bad state of the economy. Project B pays off $140 next year in the good state of the economy and $60 in the bad state of the economy. If the two states are equally likely, there are no taxes or direct bankruptcy costs, the risk-free rate of interest is zero, and investors are all risk neutral, which project would equity holders prefer if the firm is 100 percent equity financed? Which project would equity holders prefer if the firm has an $85 bond obligation due next year?

16.9. Suppose you are hired as a consultant for Tailways, Inc., just after a recapitalization that increased the firm's debt-to-assets ratio to 80 percent. The firm has the opportunity to take on a risk-free project yielding 10 percent, which you must analyze. You note that the risk-free rate is 8 percent and apply what you learned in Chapter 11 about taking positive net present value projects; that is, accept those projects that generate expected returns that exceed the appropriate risk-adjusted discount rate of the project. You recommend that Tailways take the project.

Unfortunately, your client is not impressed with your recommendation. Because Tailways is highly leveraged and is in risk of default, its borrowing

Higgins

Jennifer L. Koski_ Robert C. Higgins_ Todd Mitton - Analysis

- Asset beta and adjusted present value: Ch. 8, §8.A.3 (p. 321–325)

price data for a single division, because stock prices are based on the prospects of the firm as a whole. But multidivision firms can estimate an equity beta for a single division based on the asset betas of other companies focused in that particular industry.

As an example of the mechanics of unlevering and relevering, let's suppose that Polaris, concerned about the accuracy of its directly estimated equity beta, decides to estimate its equity beta based on industry information. Table 8A.1 illustrates an estimate of Polaris's industry asset beta based on numbers for six competitors of Polaris, as identified in Chapter 2. To avoid giving undue weight to smaller firms, I weighted the asset betas by relative market value of equity in calculating the industry figure. The resulting industry asset beta is 1.65, which matches our estimate of Polaris's own asset beta. The exact match is unusual; typically the industry asset beta would be in the ballpark of the single-firm asset beta, but not exactly the same. The last step is to relever the industry beta to reflect Polaris's unique capital structure, which can be accomplished using a simple rearrangement of the asset beta formula above. In this case, because the industry asset beta is the same as the single-firm asset beta, relevering the beta yields an estimated equity beta that is the same as Polaris's original equity beta [($7,334/$5,898) × 1.65 = 2.05]. Such a result should give Polaris management increased confidence in the accuracy of their directly estimated equity beta.

Asset Beta and Adjusted Present Value

In the standard WACC-based approach to investment appraisal described in the chapter, we ask the weighted-average cost of capital to do double duty: to adjust for the risk of the cash flows being discounted, and to capture the tax-shield advantages of the debt financing used by the firm. We reflect these tax-shield advantages by using the after-tax cost of debt in the weighted-average

TABLE 8A.1 **Estimate of Industry Asset Beta for Polaris Inc.**

Source: Compustat and Yahoo! Finance.

Company	Equity Beta	Equity/Firm Value	Asset Beta	Market Value Equity	% of Total Market Value	Weighted-Asset Beta
Brunswick	1.88	85%	1.60	$5,937	31%	0.49
Fox Factory Holding	1.80	91%	1.65	$4,419	23%	0.38
Malibu Boats	1.89	93%	1.76	$1,286	7%	0.12
Marine Products	1.22	100%	1.22	$ 492	3%	0.03
Thor Industries	2.37	75%	1.77	$5,133	27%	0.47
Winnebago	2.00	79%	1.57	$2,017	10%	0.16
				Industry asset beta		**1.65**

calculation. In most instances, this creates no problem; however, difficulties can arise when the firm's capital structure is changing over time, or when the project's debt capacity differs from that implicit in the WACC.

In these situations it becomes advantageous to use an APV approach, or what is sometimes called "valuation by parts.'' First, abstract entirely from anything to do with debt financing by estimating the project's NPV assuming all-equity financing. Then, capture the tax-shield effects of debt financing, and any other "side effects," in separate add-on terms. If the sum of these separate present value terms is positive, the opportunity is financially attractive, and vice versa. In symbols,

$$APV = NPV_{\text{all-equity financing}} + PV_{\text{interest tax shields}} + PV_{\text{any other side effects}}$$

At its root, APV is nothing more than a formalization of the idea that when evaluating investment opportunities, the whole should equal the sum of the parts.

Asset beta and APV are ideal partners because asset beta enables us to estimate the appropriate discount rate for valuing investments that are all-equity financed. A moment's review of the WACC equation in the chapter will convince you that in the absence of debt financing, WACC collapses to the cost of equity. The discount rate for evaluating all-equity financed investments is therefore represented by Equation 8.2 in the chapter, with β_A replacing β_E.

$$K_A = i_g + \beta_A \times Rp$$

where i_g is a government bond rate, β_A is the investment's asset beta, and Rp is the risk premium, usually approximated by the excess return on common stocks over government bonds.

To illustrate the combined use of APV and asset beta, consider the investment opportunity under review by Delaney Pumps. Delaney Pumps manufactures and distributes an extensive line of agricultural irrigation systems. In recent years, computerized control systems used to automate irrigation and to conserve water have become increasingly important in selling high-end systems. Delaney management is actively considering investing $160 million to develop a state-of-the-art, computerized controller that promises to leapfrog competition. Development work would be contracted to a software development company on a cost-plus basis. Revenue would come from a new product line featuring the controller and from license fees from selected competitors who elected to include the controller in their products. Projected cash flows for the investment appear in Table 8A.2. The projections extend for only four

TABLE 8A.2 **Adjusted Present Value Analysis of Automated Irrigation Controller ($ millions)***

	Year				
	0	**1**	**2**	**3**	**4**
Earnings before interest and taxes		$50.0	$150.0	$80.0	$30.0
Expected free cash flow	(160.0)	30.0	120.0	60.0	70.0
Interest expense		5.0	15.0	8.0	3.0
Interest tax shield @ 30% tax rate		1.5	4.5	2.4	0.9
Asset beta	2.41				
NPV all-equity	27.7				
PV tax shields	6.5				
APV	**$ 34.3**				

*Totals may not add due to rounding.

years because management anticipates that other, more advanced controllers will be available by this time.

Two challenges confronted Delaney management as they began their deliberations. Because the digital controller appeared much riskier than the company's usual capital expenditures, managers were uncomfortable using the company's 10 percent weighted-average cost of capital as the hurdle rate. In addition, Delaney had traditionally financed its business with the goal of maintaining a target times-interest-earned ratio of about 3 to 1. But because this project consisted almost entirely of intangible computer code and because its cash flows were quite uncertain, Delaney's treasurer thought it prudent to target a higher interest coverage of 10 to 1 on this project.

To address these challenges, the treasurer decided to do an APV analysis. Reasoning that the digital controller would probably be an average-risk investment for software companies, she identified five smaller, publicly traded firms specializing in business automation software. She then unlevered the equity betas of these firms and calculated an industry average asset beta equal to 2.41, confirming her intuition that business automation software is indeed a risky business. Combining this asset beta with a 1.7 percent riskless borrowing rate and a 6.4 percent historical risk premium in the earlier equation, she calculated a hurdle rate for unlevered, business automation software investments equal to 17.1 percent ($1.7\% + 2.41 \times 6.4\%$). Using this rate to discount the expected free cash flows in Table 8A.2, she found the project's NPV assuming all-equity financing to be $27.7 million.

The investment's principal side effect was the interest-tax shields it would generate over time. At a target times-interest-earned ratio of 10 to 1 and a 30 percent tax rate, the annual interest expense appearing in the table equals one-tenth of projected EBIT, while the corresponding tax shield is 30 percent of this amount. The discount rate used to calculate the present value of these tax shields should, of course, reflect the risk of the cash flows being discounted. Some executives argue that because interest tax shields are debt-like in terms of risk, they should be discounted at a corporate debt rate. Others maintain that while individual debt contracts may generate predictable cash flows, the total debt a business carries varies with its size and cash flows, in which case a discount rate more like K_A is appropriate. Here, because the tax shields are tied mechanically to operating income, K_A is the proper rate. Discounting at this rate, the tax shields are worth \$6.5 million, so the investment's APV is an attractive \$34.3 million.

$$APV = NPV_{\text{all-equity financing}} + PV_{\text{interest tax shields}}$$
$$\$34.3 \text{ million} = \$27.7 \text{ million} + \$6.5 \text{ million}$$

Note carefully in this analysis that the treasurer's tax-shield calculations had nothing to do with the way Delaney intended to finance the investment and everything to do with how much debt the treasurer believed the project could prudently support. For tactical reasons, companies routinely finance some investments entirely with debt and others entirely with retained profits, but this information is irrelevant to judging an investment's debt capacity and its consequent claim to interest tax shields. To think otherwise would be to commit a variation of the "marginal cost of capital fallacy."

This example deals with a straightforward investment possessing one simple side effect, but I hope it hints at the power of the technique. APV's divide-and-conquer perspective makes it possible to break even very complex problems into a series of tractable, smaller problems, and to solve the complex problem by stringing together solutions to the smaller ones. We can thus analyze a cross-border investment involving several currencies and subsidized financing as the sum of separate NPV calculations for cash flows in each currency translated into the home currency at prevailing exchange rates, plus a separate term capturing the value of the subsidized finance. And we can even apply a separate, customized hurdle rate to each cash flow stream. In a complicated world, APV and its cousin, asset beta, are indeed welcome additions to our tool kit.

SUMMARY

1. An investment's total risk
   - Refers to the range of possible returns.
   - Can be estimated for traded assets as the standard deviation of returns.
   - Can be avoided to some extent by diversifying.
2. Systematic risk
   - Is the part of total risk that cannot be avoided by diversifying.
   - Equals about half of total risk, on average, for stocks.
   - Is the only part of total risk that should affect asset prices and returns.
   - Is positively related to the return demanded by risk-averse investors.
   - Can be estimated as the product of total risk and the correlation coefficient between an asset's returns and those on a well-diversified portfolio.
3. The cost of capital
   - Is a risk-adjusted discount rate.
   - Equals the value-weighted average of the opportunity costs incurred by owners and creditors.
   - Is the return a firm must earn on existing assets to at least maintain stock price.
   - Is relevant for private firms and not-for-profits as well as public firms.
   - Is the appropriate hurdle rate for evaluating carbon copy investments.
   - Can be an appropriate hurdle rate for evaluating non–carbon copy investments when it is the cost of capital of other firms for which the investment *is* a carbon copy.
4. The cost of equity capital
   - Is the opportunity cost incurred by owners.
   - Is the most challenging variable to estimate when measuring a firm's cost of capital.
   - Is best approximated as the sum of an interest rate on a government bond plus a risk premium.
   - Increases with financial leverage.
5. Beta
   - Measures an asset's relative systematic risk.
   - Can be estimated by regressing an asset's periodic realized returns on those of a well-diversified portfolio.
   - When multiplied by the realized excess return on stocks relative to bonds, yields a suitable risk premium for estimating the cost of equity.
   - Increases with financial leverage.

d. Estimate the present value of the interest tax shields on the acquisition debt discounted at K_A.
e. What is the highest price the investors can reasonably justify paying for the target company?
f. What does your estimated maximum acquisition price in question (e) assume about the costs of financial distress?

23. You are valuing a project for Gila Corp. using the APV method. You already found the net present value of free cash flows from the project (discounted at the appropriate cost of equity) to be $500,000. The only important side effect of financing is the present value of interest tax shields. The project will be partially financed by a constant $1 million in debt over the life of the project, which is three years. Gila's tax rate is 40 percent, and the current interest rate applicable for the project's debt is 8 percent (assume this rate is also appropriate for discounting the interest tax shields). Assuming tax shields are realized at the end of each year, what is the APV of the project?

24. Sinaloa Appliance, Inc., a private firm that manufactures home appliances, has hired you to estimate the company's beta. You have obtained the following equity betas for publicly traded firms that also manufacture home appliances.

		($ millions)	
Firm	**Beta**	**Debt**	**Market Value of Equity**
iRobot	0.99	$0	$2,740
Middleby's	2.10	732	7,310
National Presto	0.12	0	739
Newell Brands	1.16	11,893	25,770
Whirlpool	1.69	4,470	13,560

a. Estimate an asset beta for Sinaloa Appliance.
b. What concerns, if any, would you have about using the betas of these firms to estimate Sinaloa Appliance's asset beta?

Ross

ROSS, WESTERFIELD, JORDAN_ LAMB - Fundamentos de Administraç

- Alavancagem caseira; Proposições I e II de M&M: Cap. 16, §16.2–16.3 (p. 542–551)
- M&M com impostos sobre o lucro da pessoa jurídica: Cap. 16, §16.4 (p. 552–556)

TABELA 16.1 Valores possíveis para a empresa: sem dívida *versus* dívida mais dividendos

		Dívida mais dividendos		
	Sem dívida	I	II	III
Dívida	$ 0	$ 500	$ 500	$500
Valor do capital próprio	1.000	750	500	250
Valor da empresa	$1.000	$1.250	$1.000	$750

TABELA 16.2 Possíveis resultados para os acionistas: dívida mais dividendos

	Dívida mais dividendos		
	I	II	III
Redução no valor do capital próprio	−$250	−$500	−$750
Dividendos	500	500	500
Efeito líquido	+$250	$ 0	−$250

A principal observação a ser feita aqui é que a variação no valor da empresa é igual ao efeito líquido sobre os acionistas. Portanto, os administradores financeiros podem tentar encontrar a estrutura de capital que maximize o valor da empresa. Em outras palavras, a regra do VPL se aplica às decisões sobre estrutura de capital, e a variação no valor da empresa em geral é o VPL de uma reestruturação. Assim, a JC deve fazer um empréstimo de $500 se esperar o Cenário I.

A pergunta crucial para determinar a estrutura de capital de uma empresa é: qual o cenário que tem probabilidade de ocorrer?

Estrutura de capital e custo de capital

No Capítulo 14, discutimos o conceito do custo médio ponderado de capital (CMPC) da empresa. Talvez você se lembre de que o CMPC nos mostra que o custo de capital total da empresa é uma média ponderada dos custos de diversos componentes da estrutura de capital. Quando descrevemos o CMPC, usamos uma estrutura de capital da empresa que nos havia sido dada. Assim, uma questão importante que queremos explorar neste capítulo é o que acontece ao custo de capital quando variamos o montante do financiamento por dívidas ou o índice dívida/capital próprio.

Um dos principais motivos para o estudo do CMPC é que o valor da empresa é maximizado quando o CMPC é minimizado. Lembre-se de que o CMPC é a taxa de desconto apropriada para os fluxos de caixa totais da empresa. Como os valores e as taxas de desconto se movem em direções opostas, a minimização do CMPC maximizará o valor dos fluxos de caixa da empresa.

Assim, queremos selecionar a estrutura de capital da empresa, para que o CMPC seja minimizado. Por esse motivo, diremos que uma estrutura de capital é melhor do que outra se ela resultar em um CMPC menor. Além disso, dizemos que determinado índice dívida/capital próprio representa a *estrutura ótima de capital* se resultar no menor CMPC possível. Essa estrutura ótima de capital também é chamada de estrutura-*meta* de capital da empresa.

Questões conceituais

16.1a Por que os administradores financeiros deveriam escolher a estrutura de capital que maximiza o valor da empresa?

16.1b Qual é a relação entre o CMPC e o valor da empresa?

16.1c O que é uma estrutura ótima de capital?

16.2 O efeito da alavancagem financeira

A seção anterior descreveu o motivo pelo qual a estrutura de capital que produz o mais alto valor para a empresa (ou o mais baixo custo de capital) é a mais benéfica para os acionistas.

Nesta seção, examinaremos o impacto da alavancagem financeira sobre os resultados para os acionistas. Como você deve lembrar, a *alavancagem financeira* se refere a quanto uma empresa se apoia na dívida. Quanto maior o financiamento por meio de dívidas que uma empresa usar em sua estrutura de capital, maior será a alavancagem financeira que ela usa.

Como descrevemos, a alavancagem financeira pode alterar drasticamente os resultados para os acionistas da empresa. Curiosamente, porém, a alavancagem financeira pode não afetar o custo de capital total. Se isso for verdadeiro, a estrutura de capital de uma empresa não será importante, pois as alterações na estrutura de capital não afetarão o valor da empresa. Retornaremos a essa questão mais tarde.

Fundamentos da alavancagem financeira

Começamos ilustrando como a alavancagem financeira funciona. Por enquanto, ignoramos o impacto dos impostos sobre lucros. Da mesma maneira, para facilitar a apresentação, descrevemos o impacto da alavancagem sobre os lucros por ação (LPA) e sobre a rentabilidade sobre o patrimônio líquido (ROE). Obviamente, esses são números contábeis e, como tais, não são nossa principal preocupação. O uso dos fluxos de caixa em vez desses números contábeis levaria exatamente às mesmas conclusões, mas isso exigiria um pouco mais de trabalho. Discutiremos o impacto sobre os valores de mercado em uma seção posterior.

Alavancagem financeira, LPA e ROE: um exemplo No momento, a Transmadruga S/A não tem dívidas em sua estrutura de capital. A diretora financeira, a Sra. Augusta, está pensando em uma reestruturação que envolva a emissão de dívida. Com o valor recebido, pretende recomprar parte das ações em circulação. A Tabela 16.3 apresenta as estruturas de capital atual e proposta. Como mostrado, os ativos da empresa têm um valor de mercado de **$8 milhões**, e há **400.000** ações em circulação. Como a Transmadruga é uma empresa financiada somente por capital próprio, o preço por ação é de $20.

A emissão proposta de dívida levantaria $4 milhões, e a taxa de juros seria de 10%. Como a ação é vendida por $20, os $4 milhões da nova dívida seriam usados para comprar $4 milhões/20 = 200.000 ações, deixando 200.000 ações no mercado.

Após a reestruturação, a Transmadruga teria uma estrutura de capital com 50% da dívida, de modo que o índice dívida/capital próprio seria 1.

Por enquanto, assumimos que o preço da ação permanecerá em $20. Para investigar o impacto da reestruturação proposta, a Sra. Augusta preparou o Tabela 16.4, que compara a estrutura de capital atual da empresa com a estrutura de capital proposta em três cenários. Os cenários refletem diferentes hipóteses sobre o lucro antes de juros e impostos (Lajir) da empresa. Dentro do cenário esperado, o Lajir é de **$1 milhão**. No cenário de recessão, o Lajir cai para **$500.000**. No cenário de expansão, ele sobe para **$1,5 milhão**.

Para ilustrar alguns dos cálculos dos números do Tabela 16.4, considere o caso de expansão. O Lajir é de $1,5 milhão. Sem dívida (a estrutura de capital atual) e sem impostos sobre lucros, o lucro líquido também é de $1,5 milhão. Neste caso, existem 400.000 ações no valor total de $8 milhões. Portanto, o LPA é de $1,5 milhão/400.000 = $3,75. Do mesmo modo, como o ROE é o lucro líquido dividido pelo patrimônio total, o ROE é $1,5 milhão/8 milhões = 18,75%.[2]

TABELA 16.3 Estruturas de capital atual e proposta para a Transmadruga S/A

	Atual	Proposta
Ativos	$8.000.000	$8.000.000
Dívida	$ 0	$4.000.000
Capital próprio	$8.000.000	$4.000.000
Índice dívida/capital próprio	0	1
Preço da ação	$ 20	$ 20
Ações em circulação	400.000	200.000
Taxa de juros	10%	10%

[2] O ROE é discutido de forma mais detalhada no Capítulo 3.

TABELA 16.4 Cenários para a estrutura de capital da Transmadruga S/A

	Recessão	Esperado	Expansão
Estrutura de capital atual: sem dívida			
Lajir	$500.000	$1.000.000	$1.500.000
Juros	0	0	0
Lucro líquido	$500.000	$1.000.000	$1.500.000
ROE	6,25%	12,50%	18,75%
LPA	$ 1,25	$ 2,50	$ 3,75
Estrutura de capital proposta: dívida = $4 milhões			
Lajir	$500.000	$1.000.000	$1.500.000
Juros	400.000	400.000	400.000
Lucro líquido	$100.000	$ 600.000	$1.100.000
ROE	2,50%	15,00%	27,50%
LPA	$ 0,50	$ 3,00	$ 5,50

Com $4 milhões de dívida (a estrutura de capital proposta), a situação é um pouco diferente. Como a taxa de juros é de 10%, os juros são de $400.000. Com Lajir de $1,5 milhão, juros de $400.000 e sem impostos, o lucro líquido é de $1,1 milhão. Agora, existem apenas 200.000 ações no valor total de $4 milhões. Portanto, o LPA é $1,1 milhão/200.000 = $5,50, em contraste com os $3,75 que calculamos no cenário anterior. Além disso, o ROE é $1,1 milhão/4 milhões = 27,5%. Isso está bem acima dos 18,75% que calculamos para a estrutura de capital atual.

LPA *versus* Lajir O impacto da alavancagem é evidente quando examinamos o efeito da reestruturação sobre o LPA e o ROE. Em particular, a variabilidade do LPA e do ROE é muito maior na estrutura de capital proposta. Isso ilustra como a alavancagem financeira atua para ampliar ganhos e perdas para os acionistas.

Na Figura 16.1, vemos mais de perto o efeito da reestruturação proposta. Essa figura mostra o LPA em relação ao Lajir para as estruturas de capital atual e proposta. A primeira linha, intitulada "Sem dívida", representa o caso em que não há alavancagem. Essa linha começa na

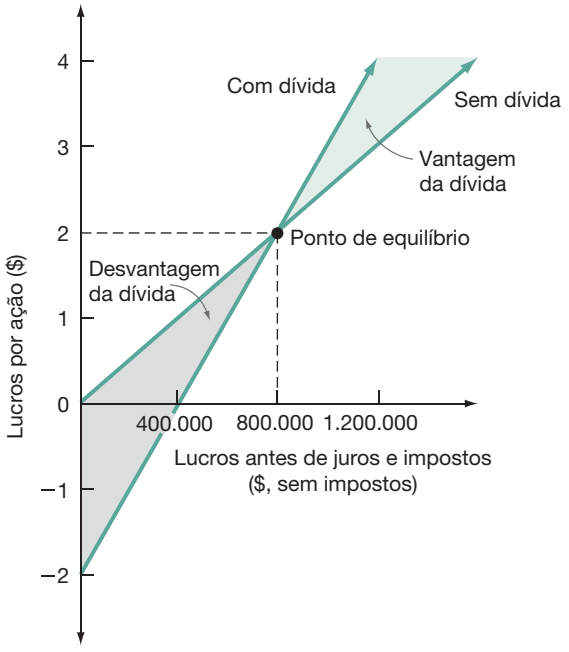


FIGURA 16.1 Alavancagem financeira: LPA e Lajir da Transmadruga S/A.

origem, indicando que o LPA seria zero se o Lajir fosse zero. Desse ponto, cada aumento de $400.000 no Lajir aumenta o LPA em $1 (porque existem 400.000 ações em circulação).

A segunda linha representa a estrutura de capital proposta. Aqui, o LPA é negativo se o Lajir for zero. Isso acontece porque $400.000 de juros devem ser pagos independentemente dos lucros da empresa. Como existem 200.000 ações nesse caso, o LPA é de − $2, como mostrado. Da mesma maneira, se o Lajir fosse de $400.000, o LPA seria exatamente zero.

O importante a ser observado na Figura 16.1 é que a inclinação da linha no segundo caso é maior. Na verdade, para cada aumento de $400.000 no Lajir, o LPA sobe $2, de modo que a linha tem o dobro da inclinação. Isso mostra que o LPA tem o dobro de sensibilidade a mudanças do que o Lajir, por causa da alavancagem financeira usada.

Outra observação a ser feita na Figura 16.1 é que as linhas se cruzam. No ponto de cruzamento, o LPA é exatamente igual para as duas estruturas de capital. Para encontrar esse ponto, lembre que o LPA é igual ao Lajir/400.000 no caso sem dívida. No caso com dívida, o LPA é (Lajir − $400.000)/200.000. Se definirmos um igual ao outro, o Lajir será de:

$$\begin{aligned} \text{Lajir}/400.000 &= (\text{Lajir} - \$400.000)/200.000 \\ \text{Lajir} &= 2 \times (\text{Lajir} - \$400.000) \\ &= \$800.000 \end{aligned}$$

Quando o Lajir é $800.000, o LPA é $2 em cada uma das duas estruturas de capital. Isso é chamado de ponto de equilíbrio na Figura 16.1. Também poderíamos chamar de ponto de indiferença. Se o Lajir estiver acima desse nível, a alavancagem será benéfica. Caso contrário, não será benéfica.

Existe outra maneira mais intuitiva de saber por que o ponto de equilíbrio é $800.000. Observe que, se a empresa não tiver dívida e seu Lajir for $ 800.000, seu lucro líquido também será $800.000. Nesse caso, o ROE é de 10%. Essa é exatamente a mesma taxa de juros sobre a dívida. Portanto, com esse Lajir a empresa ganha um rendimento suficiente apenas para pagar os juros.

EXEMPLO 16.1 Lajir do ponto de equilíbrio

A Companhia MPD optou por uma reestruturação de capital. Atualmente, a empresa não usa dívida para seu financiamento. Porém, após a reestruturação, a dívida será de $1 milhão. A taxa de juros sobre a dívida será de 9%. No momento, a MPD tem 200.000 ações em circulação, e o preço por ação é $20. Se a reestruturação deve aumentar o LPA, qual é o nível mínimo de Lajir esperado pela administração da MPD? Ignore os impostos sobre lucros em sua resposta.

Para responder, calculamos o Lajir do ponto de equilíbrio. Em qualquer Lajir acima dele, o aumento da alavancagem financeira aumentará o LPA, e isso nos dirá o nível mínimo do Lajir. Na estrutura de capital antiga, o LPA é simplesmente o Lajir/200.000. Na nova estrutura de capital, as despesas com juros serão de $1 milhão × 0,09 = $90.000. Além disso, com a receita de $1 milhão, a MPD recomprará $1 milhão/20 = 50.000 ações, deixando 150.000 em circulação. O LPA será de (Lajir − $90.000)/150.000.

Agora que sabemos como calcular o LPA em ambos os cenários, nós os definimos iguais entre si e calculamos o Lajir do ponto de equilíbrio:

$$\begin{aligned} \text{Lajir}/200.000 &= (\text{Lajir} - \$90.000)/150.000 \\ \text{Lajir} &= 4/3 \times (\text{Lajir} - \$90.000) \\ &= \$360.000 \end{aligned}$$

Comprove que, em ambos os casos, o LPA é $1,80 quando o Lajir é $360.000. Aparentemente, a administração da MPD está convencida de que o LPA excederá os $1,80.

Empréstimos na empresa e alavancagem caseira

Com base nas Tabelas 16.3 e 16.4 e na Figura 16.1, a Sra. Augusta chega às seguintes conclusões:

1. O efeito da alavancagem financeira depende do Lajir da empresa. Quando o Lajir é relativamente alto, a alavancagem é benéfica.

2. Dentro do cenário esperado, a alavancagem aumenta os retornos aos acionistas, medidos pelo ROE e pelo LPA.
3. Os acionistas são expostos a mais risco na estrutura de capital proposta porque, nesse caso, o LPA e o ROE são muito mais sensíveis às alterações no Lajir.
4. Devido ao impacto que a alavancagem financeira tem no retorno esperado para os acionistas e no grau de risco da ação, a estrutura de capital é uma consideração importante.

As três primeiras conclusões estão claramente corretas. A última conclusão está necessariamente certa? Surpreendentemente, a resposta é não. Como discutiremos a seguir, o motivo é que os acionistas podem ajustar o montante da alavancagem financeira, tomando emprestado e concedendo empréstimos por conta própria. Esse uso de empréstimos pessoais para alterar o grau de alavancagem financeira é chamado de **alavancagem caseira**.

alavancagem caseira
o uso de empréstimo pessoal para mudar o montante geral da alavancagem financeira à qual o indivíduo está exposto.

Mostraremos agora que o fato de a Transmadruga adotar ou não a estrutura de capital proposta não faz diferença, pois qualquer acionista que prefira a estrutura de capital proposta pode simplesmente criá-la usando a alavancagem caseira. Para começar, a primeira parte da Tabela 16.5 mostra o que acontecerá a um investidor que comprar $2.000 em ações da Transmadruga se a estrutura de capital proposta for adotada. Esse investidor compra 100 ações. Na Tabela 16.4, vemos que o LPA será $0,50, $3 ou $5,50, de modo que os lucros totais de 100 ações serão $50, $300 ou $550 na estrutura de capital proposta.

Agora, suponhamos que a Transmadruga não adote a estrutura de capital proposta. Nesse caso, o LPA será $1,25, $2,50 ou $3,75. A segunda parte da Tabela 16.5 demonstra como um acionista que prefere os resultados da estrutura proposta pode criá-los usando o empréstimo pessoal. Para isso, o acionista toma emprestados $2.000 a 10% por conta própria. Nosso investidor usa esse montante, juntamente com os $2.000 originais, para comprar 200 ações. Como mostrado, os resultados líquidos são exatamente iguais àqueles da estrutura de capital proposta.

Como sabemos que o acionista teria de tomar emprestados $2.000 para criar os resultados corretos? Estamos tentando replicar a estrutura de capital proposta da Transmadruga no nível pessoal. A estrutura de capital proposta resulta em um índice dívida/capital próprio de 1. Para replicar essa estrutura no nível pessoal, o acionista deve tomar emprestado o suficiente para criar esse mesmo índice dívida/capital próprio. Como o acionista tem $2.000 investidos no capital próprio, o empréstimo de outros $2.000 criará um índice dívida/capital próprio de 1.

Esse exemplo demonstra que os investidores sempre podem aumentar a alavancagem financeira por conta própria para criar um padrão diferente de resultados. Assim, não fará diferença se a Transmadruga escolher ou não a estrutura de capital proposta.

TABELA 16.5 Estrutura de capital proposta *versus* estrutura de capital original com alavancagem caseira

Estrutura de capital proposta			
	Recessão	**Esperado**	**Expansão**
LPA	$ 0,50	$ 3,00	$ 5,50
Lucros de 100 ações	50,00	300,00	550,00
Custo líquido = 100 ações × $20 = $2.000			
Estrutura de capital original e alavancagem caseira			
LPA	$ 1,25	$ 2,50	$ 3,75
Lucros de 200 ações	250,00	500,00	750,00
Menos: juros sobre $2.000 a 10%	200,00	200,00	200,00
Lucros líquidos	$ 50,00	$ 300,00	$ 550,00
Custo líquido = 200 ações × $20 − Montante do empréstimo = $4.000 − 2.000 − $2.000			

EXEMPLO 16.2 Desalavancagem de ações

No exemplo da Transmadruga, suponhamos que a administração adote a estrutura de capital proposta. Suponhamos também que um investidor que tenha 100 ações preferisse a estrutura de capital original. Mostre como esse investidor poderia "desalavancar" a ação para recriar os resultados originais.

Para criar a alavancagem, os investidores tomam empréstimos por conta própria. Para desfazer a alavancagem, os investidores devem emprestar dinheiro. No caso da Transmadruga, a empresa tomou emprestado um montante igual à metade de seu valor. O investidor pode desalavancar a ação emprestando dinheiro na mesma proporção. Nesse caso, o investidor vende 50 ações pelo total de $1.000 e, em seguida, empresta os $1.000 (realiza uma aplicação financeira) a 10%. Os resultados são calculados no próximo quadro:

	Recessão	Esperado	Expansão
LPA (estrutura proposta)	$ 0,50	$ 3,00	$ 5,50
Lucros de 50 ações	25,00	150,00	275,00
Mais: juros sobre $1.000	100,00	100,00	100,00
Resultado total	$125,00	$250,00	$375,00

Esses são exatamente os resultados que o investidor teria de acordo com a estrutura de capital original.

Questões conceituais

16.2a Qual é o impacto da alavancagem financeira sobre os acionistas?

16.2b O que é a alavancagem caseira?

16.2c Por que a estrutura de capital da Transmadruga é irrelevante?

16.3 Estrutura de capital e o custo do capital próprio

Vimos que nada haveria de especial na tomada de empréstimos pelas empresas, pois os investidores poderiam tomar emprestado ou emprestar por conta própria. Como resultado, seja qual for a estrutura de capital escolhida pela Transmadruga, o preço das suas ações seria o mesmo. Portanto, a estrutura de capital da empresa não é relevante, pelo menos não dentro do quadro simples que examinamos.

Nosso exemplo da Transmadruga se baseia no famoso argumento proposto por dois ganhadores do prêmio Nobel, Franco Modigliani e Merton Miller, que daqui em diante serão chamados de M&M.

Por meio da Transmadruga S/A, ilustramos um caso especial da **Proposição I de M&M**. Essa proposição afirma que o modo como uma empresa escolhe organizar suas finanças é completamente irrelevante.

Proposição I de M&M
a afirmação de que o valor da empresa é independente da sua estrutura de capital.

Proposição I de M&M: o modelo *pizza*

Uma forma de ilustrar a Proposição I de M&M é imaginar duas empresas que são idênticas no lado esquerdo do balanço patrimonial. Seus ativos e operações são exatamente iguais. Os lados direitos são diferentes porque as duas empresas financiam suas operações de maneira diferente. Nesse caso, podemos ver a questão da estrutura de capital num modelo de *pizza*. A Figura 16.2 mostra duas maneiras possíveis de cortar a *pizza*, entre a fatia do capital próprio, E, e a fatia da dívida, D: 40%-60% e 60%-40%. Entretanto, o tamanho da *pizza* da Figura 16.2 é igual para ambas as empresas, já que o valor dos ativos é o mesmo. Isso é exatamente o que diz a Proposição I de M&M: o tamanho da *pizza* não depende de em quantas fatias ela é cortada.

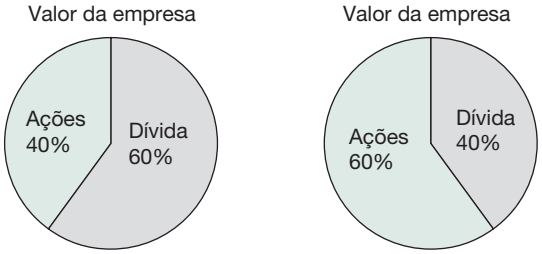


FIGURA 16.2 Dois modelos de *pizza* para a estrutura de capital.

O custo do capital próprio e a alavancagem financeira: Proposição II de M&M

Embora a mudança na estrutura de capital da empresa não altere o valor *total* dela, isso causa variações importantes na dívida e no capital próprio da empresa. Agora, examinaremos o que acontece a uma empresa financiada com dívida e capital próprio quando o índice dívida/capital próprio é alterado. Para simplificar nossa análise, continuaremos ignorando os impostos sobre lucros.

Com base em nossa discussão do Capítulo 14, se ignorarmos os impostos, o CMPC será:

$$\text{CMPC} = (E/V) \times R_E + (D/V) \times R_D$$

onde $V = E + D$. Além disso, vimos que um modo de interpretar o CMPC é entendê-lo como o retorno exigido sobre os ativos totais da empresa. Para lembrarmos disso, usaremos o símbolo R_A para representar o CMPC e escreveremos:

$$R_A = (E/V) \times R_E + (D/V) \times R_D$$

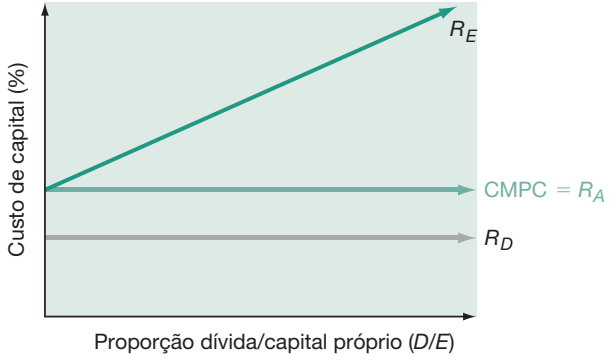
Se reorganizarmos essa expressão para calcular o custo do capital próprio, veremos que:

$$R_E = R_A + (R_A - R_D) \times (D/E) \qquad [16.1]$$

Proposição II de M&M
a afirmação de que o custo do capital próprio de uma empresa é uma função linear positiva da sua estrutura de capital.

Essa é a famosa **Proposição II de M&M**. Ela nos diz que o custo do capital próprio depende de três fatores: da taxa de retorno exigida sobre os ativos da empresa, R_A; do custo da dívida da empresa, R_D; e do índice dívida/capital próprio da empresa, D/E.

A Figura 16.3 resume nossa discussão até aqui por meio de gráficos mostrando a evolução do custo do capital próprio, R_E, e do índice dívida/capital próprio. Como mostrado, a Proposição II de M&M indica que o custo do capital próprio (R_E) é dado por uma linha reta com uma



$R_E = R_A + (R_A - R_D) \times (D/E)$ de acordo com a Proposição II de M&M

$R_A = \text{CMPC} = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D$

onde $V = D + E$

FIGURA 16.3 O custo do capital próprio e o CMPC: proposições I e II de M&M sem impostos.

inclinação de $(R_A - R_D)$. O intercepto y corresponde a uma empresa com índice dívida/capital próprio zero; nesse caso, $R_A = R_E$. A Figura 16.3 mostra que, à medida que a empresa aumenta seu índice dívida/capital próprio, o aumento na alavancagem eleva o risco do capital próprio e, portanto, o retorno exigido ou o custo do capital próprio (R_E).

Note, pela Figura 16.3, que o CMPC não depende da proporção dívida/capital próprio. Ele é o mesmo independentemente de qual for essa proporção. Essa é outra maneira de expressar a Proposição I de M&M: o custo de capital total da empresa não é afetado por sua estrutura de capital. Como mostrado, o fato do custo da dívida ser inferior ao custo do capital próprio é exatamente contrabalançado pelo aumento no custo do capital próprio decorrente da tomada de empréstimos. Em outras palavras, a variação nos pesos da estrutura de capital (E/V e D/V) é perfeitamente contrabalançada pela variação no custo do capital próprio (R_E), de modo que o CMPC permanece igual.

EXEMPLO 16.3 Custo do capital próprio

A Ricardo S/A tem um CMPC (ignorando os impostos) de 12%. A empresa pode tomar emprestado a 8%. Assumindo que a Ricardo tenha uma estrutura-meta de capital de 80% de capital próprio e 20% de dívida, qual é o custo do capital próprio? Qual é o custo do capital próprio se a estrutura-meta de capital for 50% de capital próprio? Calcule o CMPC usando suas respostas para confirmar se ele é o mesmo.

De acordo com a Proposição II de M&M, o custo do capital próprio (R_E) é:

$$R_E = R_A + (R_A - R_D) \times (D/E)$$

No primeiro caso, a proporção dívida/capital próprio é 0,2/0,8 = 0,25. Portanto, o custo do capital próprio é:

$$\begin{aligned} R_E &= 12\% + (12\% - 8\%) \times 0{,}25 \\ &= 13\% \end{aligned}$$

No segundo caso, a proporção dívida/capital próprio é 1, de modo que o custo do capital próprio é 16%.

Agora, podemos calcular o CMPC assumindo que a porcentagem de capital próprio no financiamento é 80%, o custo do capital próprio é 13% e a alíquota tributária é zero:

$$\begin{aligned} \text{CMPC} &= (E/V) \times R_E + (D/V) \times R_D \\ &= 0{,}80 \times 13\% + 0{,}20 \times 8\% \\ &= 12\% \end{aligned}$$

No segundo caso, a porcentagem de capital próprio no financiamento é 50%, e o custo do capital próprio é 16%. O CMPC é:

$$\begin{aligned} \text{CMPC} &= (E/V) \times R_E + (D/V) \times R_D \\ &= 0{,}50 \times 16\% + 0{,}50 \times 8\% \\ &= 12\% \end{aligned}$$

Como tínhamos calculado, o CMPC é 12% em ambos os casos.

Risco do negócio e risco financeiro

A Proposição II de M&M mostra que o custo do capital próprio da empresa pode ser dividido em dois componentes. O primeiro componente, R_A, é o retorno exigido sobre os ativos totais da empresa e depende da natureza das suas atividades operacionais. O risco inerente às operações de uma empresa é chamado de **risco do negócio** para o capital próprio na empresa. Voltando ao Capítulo 13, observe que o risco do negócio depende do risco sistemático dos ativos da empresa. Quanto maior for o risco do negócio de uma empresa, maior será o R_A e, com todos os demais fatores iguais, maior será seu custo de capital próprio.

risco do negócio
o risco para o capital próprio proveniente da natureza das atividades operacionais da empresa.

O segundo componente do custo do capital próprio, $(R_A - R_D) \times (D/E)$, é determinado pela estrutura financeira da empresa. Para uma empresa financiada somente por capital próprio, esse

EM SUAS PRÓPRIAS PALAVRAS...

Merton H. Miller sobre a estrutura de capital: M&M 30 anos depois

Percebi com clareza a dificuldade de resumir rapidamente a contribuição desses artigos após Franco Modigliani receber o Prêmio Nobel de Economia, em parte – obviamente, apenas em parte – por seu trabalho na área de finanças. Imediatamente, as equipes de repórteres e as câmeras da televisão local de Chicago desceram sobre mim. Eles diziam: "Sabemos que você trabalhou com Modigliani há alguns anos no desenvolvimento dos teoremas de M&M e queremos saber se você poderia explicá-los rapidamente para nossos telespectadores". "Quão rápido?", perguntei. "Ah, uns 10 segundos", foi a resposta.

Dez segundos para explicar o trabalho de uma vida inteira! Dez segundos para descrever dois artigos pensados com tanto cuidado, cada um com mais de 30 páginas impressas e 60 notas de rodapé! Ao ver o olhar de desespero em meu rosto, disseram: "Não é preciso entrar em detalhes. Basta nos dar os pontos em termos simples e práticos."

O ponto principal do artigo sobre custo de capital foi, pelo menos a princípio, fácil de explicar. Ele dizia que, no mundo ideal de um economista, o valor total de mercado de todos os títulos emitidos por uma empresa seria regido pela rentabilidade e pelo risco de seu ativo real básico e não dependeria de como a combinação de títulos emitidos para financiá-lo seria dividida entre instrumentos de dívida e de capital patrimonial. Alguns gestores financeiros podem muito bem achar que conseguiriam ampliar o valor total aumentando a proporção dos instrumentos de dívida, já que os retornos dos instrumentos de dívida, dado seu risco mais baixo, são consideravelmente mais baixos do que aqueles exigidos do capital próprio. Mas, nas condições ideais assumidas, o risco agregado para os acionistas pela emissão de mais dívida aumentará os retornos exigidos do capital próprio o suficiente para contrabalançar o ganho aparente resultante do uso de dívida de baixo custo.

Tal resumo não apenas teria sido muito longo, como também dependia de termos e conceitos resumidos, ricos em conotações para os economistas, que raramente são entendidos pelo público em geral. Em vez disso, pensei em uma analogia que nós mesmos usamos no artigo original. "Pense na empresa", eu disse, "como um pote gigante de leite integral. O fazendeiro pode vender o leite integral como está. Ou ele pode separar a nata e vendê-la por um preço consideravelmente mais alto do que o preço do leite integral. (A venda da nata é como uma empresa que vende títulos de dívida de baixo retorno e alto preço.) Mas, obviamente, o fazendeiro ficaria com o leite desnatado, de baixo teor de gordura, que seria vendido por muito menos do que o leite integral. O leite desnatado corresponde ao capital próprio alavancado. A proposição de M&M diz que, se não houver custos de separação (e, naturalmente, nenhum programa governamental de apoio à produção de laticínios), a soma da nata mais o leite desnatado teria o mesmo preço do leite integral".

O pessoal da televisão conversou em particular por algum tempo. Eles me informaram que continuava longo, complicado e acadêmico demais. "Você tem algo mais simples?", eles perguntaram. Pensei em outra maneira de apresentar a proposição de M&M que enfatizasse o papel dos títulos como dispositivos de "divisão" dos resultados de uma empresa entre o grupo de seus fornecedores de capital. "Pense na empresa", eu disse, "como uma enorme *pizza* dividida em quatro partes iguais. Se você cortar cada pedaço pela metade terá oitavos, a proposição de M&M diz que você terá mais pedaços, mas não terá mais *pizza*".

Novamente, eles cochicharam um pouco. Dessa vez, eles desligaram as luzes. Guardaram o equipamento. Agradeceram minha cooperação e disseram que voltariam a me procurar. Mas sabia que, de algum modo, havia perdido a chance de iniciar uma nova carreira como apresentador de sabedoria econômica para telespectadores em um simples bordão de 10 segundos. Alguns têm o talento para isso, e outros simplesmente não o têm.

O falecido Merton H. Miller ficou famoso por seu trabalho inovador com Franco Modigliani sobre a estrutura de capital das empresas, o custo de capital e a política de dividendos. Ele recebeu o Prêmio Nobel de Economia por suas contribuições pouco depois desse artigo ter sido preparado.

componente é zero. À medida que a empresa começa a depender de dívidas para seu financiamento, o retorno exigido sobre o capital próprio aumenta. Isso acontece porque o financiamento por dívidas aumenta os riscos para os acionistas. Esse risco extra, que surge do uso de dívidas para o financiamento dos ativos, é chamado de **risco financeiro** para capital próprio na empresa.

risco financeiro
o risco para o capital próprio proveniente da política financeira (ou seja, da estrutura de capital) da empresa.

Portanto, o risco sistemático total do capital próprio da empresa tem dois elementos: o risco do negócio e o risco financeiro. O primeiro elemento (o risco do negócio) depende dos ativos e das operações da empresa e não é afetado pela estrutura de capital. Dado um nível de risco de negócio de uma empresa (e o correspondente custo de sua dívida), o segundo elemento (o risco financeiro) é determinado completamente pela sua política financeira. Como ilustramos, o custo do capital próprio da empresa aumenta quando a empresa aumenta o uso da alavancagem financeira, pois o risco financeiro para o capital próprio aumenta enquanto o risco do negócio permanece igual.

Questões conceituais

16.3a O que a Proposição I de M&M afirma?

16.3b Quais são os três fatores determinantes do custo do capital próprio de uma empresa?

16.3c O risco sistemático total para o capital próprio de uma empresa tem dois elementos. Quais são eles?

16.4 Proposições I e II de M&M com impostos sobre o lucro da pessoa jurídica

A dívida tem duas características distintivas às quais não demos a devida atenção. Em primeiro lugar, como já mencionamos em várias situações, os juros pagos sobre a dívida são dedutíveis do lucro sobre o qual incidem os impostos. Isso é bom para a empresa, e pode ser somado como um benefício agregado pelo financiamento por dívidas. Em segundo lugar, o não cumprimento das obrigações pode resultar em falência. Isso não é bom para a empresa, e pode ser somado como um custo agregado pelo financiamento por dívidas. Como não consideramos explicitamente essas duas características da dívida, nos damos conta de que podemos obter uma resposta diferente para a estrutura de capital se fizermos essas considerações. Assim, consideraremos os impostos sobre os lucros da pessoa jurídica nesta seção e a falência na próxima.

Podemos começar considerando o que acontece às Proposições I e II de M&M quando levamos em conta o efeito dos impostos para a pessoa jurídica. Para tanto, examinaremos duas empresas: a empresa U (não alavancada) e a empresa L (alavancada). Essas duas empresas são idênticas no lado esquerdo do balanço patrimonial, ou seja, seus ativos e suas operações são iguais.

Assumimos que o Lajir das duas empresas será de $1.000 para sempre. A diferença entre as empresas é que a empresa L emitiu $1.000 em títulos perpétuos sobre os quais ela paga 8% de juros ao ano. Dessa forma, os juros são de $0,08 \times \$1.000 = \$80$ para sempre. Além disso, suponha que a sua alíquota de impostos da pessoa jurídica seja de 30%.

Agora, para as duas empresas, U e L, podemos calcular:

	Empresa U	Empresa L
Lajir	$1.000	$1.000
Juros	0	80
Lucro tributável	$1.000	$ 920
Impostos sobre lucros (30%)	300	276
Lucro líquido	$ 700	$ 644

O benefício fiscal dos juros

Para simplificar, assumiremos que a depreciação seja zero. Também assumiremos que os gastos de capital sejam zero e que não haja variações na necessidade de capital de giro. Nesse caso, o fluxo de caixa dos ativos é simplesmente igual ao Lajir – Impostos. Portanto, para as empresas U e L, temos:

Fluxo de caixa dos ativos	Empresa U	Empresa L
Lajir	$1.000	$1.000
– Impostos sobre lucros	300	276
Total	$ 700	$ 724

De imediato, vemos que a estrutura de capital agora tem algum efeito, pois os fluxos de caixa de U e L não são iguais, embora as duas empresas tenham ativos idênticos.

Para saber o que está acontecendo, podemos calcular os fluxos de caixa para os acionistas e para os credores:

Fluxo de caixa	Empresa U	Empresa L
Para os acionistas	$700	$644
Para os credores	0	80
Total	$700	$724

Observamos que o fluxo de caixa total de L é $24 maior. Isso acontece porque os impostos de L (que são um fluxo de saída de caixa) são $24 menores. O fato de os juros serem dedutíveis

para fins tributários gerou uma economia fiscal igual ao pagamento dos juros ($80) multiplicado pela alíquota de impostos para a pessoa jurídica (30%): $80 × 0,30 = $24. Chamamos essa economia de impostos de **benefício fiscal dos juros**.

benefício fiscal dos juros
a economia fiscal obtida por uma empresa por meio das despesas com juros.

Impostos e Proposição I de M&M

Como a dívida é perpétua, o mesmo benefício fiscal de $24 será gerado indefinidamente. O fluxo de caixa após impostos para L será igual aos $700 que U ganha mais o benefício fiscal de $24. Como o fluxo de caixa de L sempre será $24 maior, a empresa L vale mais do que a empresa U, e a diferença é o valor dessa perpetuidade de $24.

Como o benefício fiscal é gerado pelo pagamento de juros, ele tem o mesmo risco da dívida e, portanto, 8% (o custo da dívida) é a taxa de desconto apropriada. Assim, o valor do benefício fiscal é:

$$\text{VP} = \frac{\$24}{0,08} = \frac{0,30 \times \$1.000 \times 0,08}{0,08} = 0,30 \times \$1.000 = \$300$$

Como ilustra nosso exemplo, o valor presente do benefício fiscal dos juros pode ser escrito como:

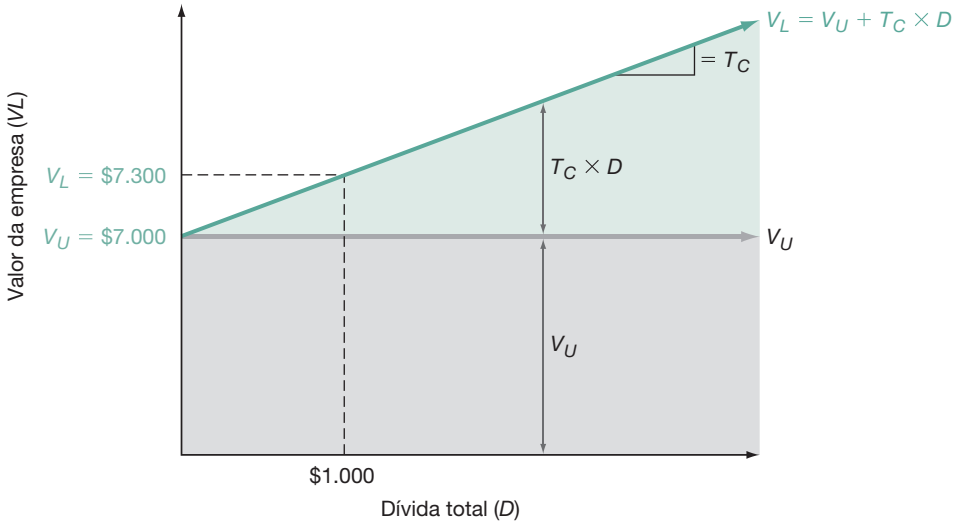
$$\begin{aligned}\text{Valor presente do benefício fiscal dos juros} &= (T_C \times D \times R_D)/R_D \\ &= T_C \times D\end{aligned} \qquad [16.2]$$

Agora, chegamos a outro resultado famoso, a Proposição I de M&M com impostos da pessoa jurídica. Vimos que o valor da empresa L (V_L) excede o valor da empresa U (V_U) pelo valor presente do benefício fiscal dos juros ($T_C \times D$). Portanto, a Proposição I de M&M com impostos afirma que:

$$V_L = V_U + T_C \times D \qquad [16.3]$$

Nesse caso, o efeito da dívida é ilustrado na Figura 16.4. Indicamos o valor da empresa alavancada (V_L) em relação ao montante da dívida (D). A Proposição I de M&M com impostos da pessoa jurídica indica que a relação é dada por uma linha reta com uma inclinação de T_C e um intercepto y de V_U.

Na Figura 16.4, também traçamos uma linha horizontal que representa V_U. Como indicado, a distância entre as duas linhas é $T_C \times D$, o valor presente do benefício fiscal.



O valor da empresa aumenta à medida que a dívida total aumenta por causa do benefício fiscal dos juros. Esse é o fundamento da Proposição I de M&M com impostos sobre lucros.

FIGURA 16.4 Proposição I de M&M com impostos.

Suponhamos que o custo de capital da empresa U seja de 10%. Chamaremos isso de **custo não alavancado de capital** e usaremos o símbolo R_U para representá-lo. Podemos imaginar o R_U como o custo de capital que uma empresa teria se não tivesse dívida. O fluxo de caixa da empresa U é \$700 por ano indefinidamente e, como U não tem dívida, a taxa de desconto apropriada é $R_U = 10\%$. O valor da empresa não alavancada (V_U) é simplesmente:

custo não alavancado de capital
o custo de capital de uma empresa que não tem dívida.

$$\begin{aligned} V_U &= \frac{\text{Lajir} \times (1 - T_C)}{R_U} \\ &= \frac{\$700}{0{,}10} \\ &= \$7.000 \end{aligned}$$

O valor da empresa alavancada (V_L) é:

$$\begin{aligned} V_L &= V_U + T_C \times D \\ &= \$7.000 + 0{,}30 \times 1.000 \\ &= \$7.300 \end{aligned}$$

Como indica a Figura 16.4, o valor da empresa sobe em \$0,30 para cada \$1 de dívida. Em outras palavras, o VPL por real de dívida é de \$0,30. É difícil imaginar por que uma empresa não tomaria emprestado o máximo absoluto nessas circunstâncias.

O resultado da análise desta seção é a compreensão de que, após incluirmos os impostos sobre os lucros da pessoa jurídica, a estrutura de capital definitivamente importa. Entretanto, de imediato chegamos à conclusão ilógica de que a estrutura ótima de capital é a com 100% de dívida.

Impostos, o CMPC e a Proposição II

Também podemos concluir que a melhor estrutura de capital é 100% de dívida verificando o custo médio ponderado de capital (CMPC). No Capítulo 14, vimos que, após considerarmos o efeito dos impostos, o CMPC é:

$$\text{CMPC} = (E/V) \times R_E + (D/V) \times R_D \times (1 - T_C)$$

Para calcular esse CMPC, precisamos saber qual é o custo do capital próprio. A Proposição II de M&M com impostos da pessoa jurídica afirma que o custo do capital próprio é:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C) \qquad [16.4]$$

Para ilustrar, relembre que vimos anteriormente que a empresa L vale \$7.300 no total. Como a dívida vale \$1.000, o capital próprio deve valer \$7.300 − 1.000 = \$6.300. Para a empresa L, o custo do capital próprio é:

$$\begin{aligned} R_E &= 0{,}10 + (0{,}10 - 0{,}08) \times (\$1.000/6.300) \times (1 - 0{,}30) \\ &= 10{,}22\% \end{aligned}$$

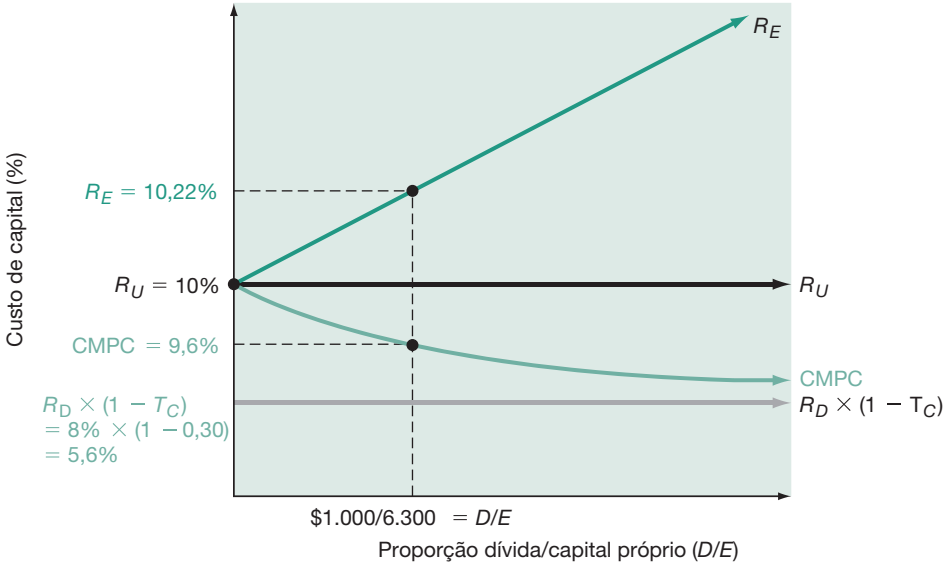
O custo médio ponderado de capital é:

$$\begin{aligned} \text{CMPC} &= (\$6.300/7.300) \times 10{,}22\% + (1.000/7.300) \times 8\% \times (1 - 0{,}30) \\ &= 9{,}6\% \end{aligned}$$

Sem dívida, o CMPC é mais de 10% e, com dívida, ele é 9,6%. Portanto, a empresa fica melhor com dívida.

Conclusão

A Figura 16.5 resume nossa discussão acerca da relação entre o custo do capital próprio, o custo da dívida após impostos e o CMPC. Para referência, incluímos o R_U, o custo não alavancado de capital. Na Figura 16.5, temos o índice dívida/capital próprio no eixo horizontal. Observe como



A Proposição I de M&M com impostos implica em o CMPC de uma empresa diminuir à medida que a empresa recorre, de forma mais expressiva, ao financiamento por dívidas:

$$\text{CMPC} = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D \times (1 - T_C)$$

A Proposição II de M&M com impostos implica em o custo do capital próprio (R_E) de uma empresa diminuir à medida que a empresa recorre, de forma mais expressiva, ao financiamento por dívidas:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$

FIGURA 16.5 O custo do capital próprio e o CMPC: Proposição II de M&M com impostos.

o CMPC diminui à medida que o índice dívida/capital próprio aumenta. Novamente, isso ilustra que, quanto mais dívida a empresa usar, mais baixo será seu CMPC. O Tabela 16.6 resume os principais resultados de nossa análise das proposições de M&M para referência futura.

TABELA 16.6 Resumo de Modigliani e Miller

I. O caso sem impostos sobre lucros da pessoa jurídica

A. Proposição I de M&M: o valor da empresa alavancada (V_L) é igual ao valor da empresa não alavancada (V_U):

$$V_L = V_U$$

Implicações da Proposição I:

1. A estrutura de capital de uma empresa não é relevante.
2. O custo médio ponderado de capital (CMPC) é o mesmo independentemente da combinação entre dívida e capital próprio usada para financiar a empresa.

B. Proposição II de M&M: o custo do capital próprio (R_E) é:

$$R_E = R_A + (R_A - R_D) \times (D/E)$$

onde R_A é o CMPC, R_D é o custo da dívida e D/E é o índice dívida/capital próprio.

Implicações da Proposição II:

1. O custo do capital próprio aumenta à medida que a empresa aumenta o uso de financiamento por dívidas.
2. O risco para o capital próprio depende diretamente de dois fatores: o nível de risco das operações da empresa (risco do negócio) e o grau de alavancagem financeira (risco financeiro). O risco do negócio determina o R_A, e o risco financeiro é determinado por D/E.

II. O caso com impostos sobre lucros da pessoa jurídica

A. Proposição I com impostos: o valor da empresa alavancada (V_L) é igual ao valor da empresa não alavancada (V_U) mais o valor presente do benefício fiscal dos juros:

$$V_L = V_U + T_C \times D$$

onde T_C é a alíquota de impostos da pessoa jurídica, e D é o montante da dívida.

Implicações da Proposição I:

1. O financiamento por dívidas é altamente vantajoso e, ao extremo, a estrutura ótima de capital de uma empresa é 100% de dívida.
2. O custo médio ponderado de capital (CMPC) de uma empresa diminui à medida que a empresa recorre, de forma mais expressiva, ao financiamento por dívidas.

B. Proposição II com impostos: o custo do capital próprio (R_E) é:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$

onde R_U é o custo não alavancado de capital, isto é, o custo de capital da empresa caso não haja dívida. Ao contrário do caso com a Proposição I, as implicações gerais da Proposição II são as mesmas, com ou sem impostos.

EXEMPLO 16.4 O custo do capital próprio e o valor da empresa

Este é um exemplo abrangente que ilustra a maior parte dos pontos que discutimos até agora. Você recebe as seguintes informações da Formato S/A:

$$\begin{aligned} \text{Lajir} &= \$151{,}52 \\ T_C &= 0{,}34 \\ D &= \$500 \\ R_U &= 0{,}20 \end{aligned}$$

O custo do capital da dívida é de 10%. Qual é o valor do capital próprio da Formato? Qual é o custo do capital próprio da Formato? Qual é o CMPC?

Essa é mais fácil do que parece. Lembre-se de que todos os fluxos de caixa são perpetuidades. O valor da empresa, se ela não tiver dívida, V_U, será:

$$\begin{aligned} V_U &= \frac{\text{Lajir} - \text{Impostos}}{R_U} = \frac{\text{Lajir} \times (1 - T_C)}{R_U} \\ &= \frac{\$100}{0{,}20} \\ &= \$500 \end{aligned}$$

Com base na Proposição I de M&M com impostos, sabemos que o valor da empresa com dívida é:

$$\begin{aligned} V_L &= V_U + T_C \times D \\ &= \$500 + 0{,}34 \times 500 \\ &= \$670 \end{aligned}$$

Como a empresa vale \$670 no total e a dívida vale \$500, o capital próprio vale \$170:

$$\begin{aligned} E &= V_L - D \\ &= \$670 - 500 \\ &= \$170 \end{aligned}$$

Com base na Proposição II de M&M com impostos, o custo do capital próprio é:

$$\begin{aligned} R_E &= R_U + (R_U - R_D) \times (D/E) \times (1 - T_C) \\ &= 0{,}20 + (0{,}20 - 0{,}10) \times (\$500/170) \times (1 - 0{,}34) \\ &= 39{,}4\% \end{aligned}$$

(*continua*)

(*continuação*)

Por fim, o CMPC é:

$$\text{CMPC} = (\$170/670) \times 39{,}4\% + (500/670) \times 10\% \times (1 - 0{,}34)$$
$$= 14{,}92\%$$

Observe que isso é substancialmente menor do que o custo do capital para a empresa sem dívidas ($R_U = 20\%$), de modo que o financiamento com dívidas é altamente vantajoso.

Questões conceituais

16.4a Qual é a relação entre o valor de uma empresa não alavancada e o valor de uma empresa alavancada depois de considerarmos o efeito dos impostos da pessoa jurídica?

16.4b Se considerarmos apenas o efeito dos impostos, qual é a estrutura ótima de capital?

16.4c Discuta sobre qual das empresas brasileiras a seguir têm o benefício fiscal da dívida: a tributada pelo lucro real, a tributada pelo lucro presumido ou a tributada no regime simples?

16.5 Custos de falência

Um fator limitante que afeta o montante de dívidas que uma empresa poderia usar surge na forma de custos de falência. À medida que o índice dívida/capital próprio aumenta, a probabilidade de a empresa não conseguir pagar o prometido a seus credores também aumenta. Quando isso acontece, a propriedade dos ativos da empresa é transferida dos acionistas para os credores ao final do processo.

Em princípio, uma companhia[†] estaria quebrada quando o valor do ativo da empresa for igual ao valor da sua dívida. Quando isso ocorre, o valor do patrimônio líquido é igual a zero, e os acionistas perderiam o controle da empresa para os credores. Nesse caso, os credores têm ativos cujo valor é exatamente igual ao que é devido pelo financiamento por dívida. Em uma situação de mercados perfeitos, não há custos associados a essa transferência de propriedade, e os credores nada perderiam.

Obviamente, essa visão idealizada do processo de falência não é o que acontece no mundo real. Ironicamente, é caro entrar em processo de falência. Como discutimos, os custos associados à recuperação de uma empresa quebrada podem, por fim, eliminar os ganhos decorrentes da economia de impostos obtida com a alavancagem.

Custos diretos de falência

Quando o valor dos ativos de uma empresa é igual ao valor de sua dívida, a companhia está economicamente falida no sentido de que o patrimônio líquido dos sócios não tem valor. Entretanto, a passagem formal dos ativos para os credores é um processo legal, não um processo econômico.

A lei de falências prevê três tratamentos para as empresas que não conseguem cumprir com suas obrigações: a recuperação judicial, a recuperação extrajudicial e a falência. Neste capítulo, os procedimentos conduzidos sob a lei de falências serão designados, em geral, como processos de falência.

Existem custos legais e administrativos nos processos de falência, e já houve quem dissesse que as falências são para os advogados aquilo que o sangue é para os tubarões.

[†] N. de R. T.: A distinção entre *companhia* e *empresa* é fundamental aqui. Pela lei societária, *companhia* é a sociedade, o conjunto dos sócios ou acionistas, e *empresa* é o objeto da companhia (ver o Artigo 2º da Lei 6.406/76). Quem quebra é a sociedade, a empresa poderia continuar funcionando, sob a gestão agora dos credores.

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13. Cálculo do CMPC [OA1] A Imperatrix S/A não tem dívidas, mas pode tomar empréstimos a 8,2%. No momento, o CMPC da empresa é 11%, e a alíquota tributária é 34%.

a. Qual é o custo do capital próprio da empresa?

b. Se a empresa converter sua estrutura de capital para 25% de dívida, qual será o custo do capital próprio?

c. Se a empresa converter sua estrutura de capital para 50% de dívida, qual será o custo do capital próprio?

d. Qual é o CMPC da empresa na parte (b)? E na parte (c)?

14. M&M e impostos [OA2] A Frederico & Co. espera um Lajir de $92.000 por ano para sempre. A empresa pode tomar empréstimos a 9%. Atualmente, a Frederico não tem dívida, e o custo do capital próprio é 15%. Se a alíquota tributária for 35%, qual será o valor da empresa? Qual será o valor se a empresa tomar empréstimos de $60.000 e usar a receita para recomprar ações?

15. M&M e impostos [OA2] No Problema 14, qual é o custo do capital próprio após a recompra? Qual é o CMPC? Quais são as implicações da decisão sobre a estrutura de capital da empresa?

INTERMEDIÁRIO
(Questões 16-18)

16. M&M [OA2] A Manufatura Ferramental tem um Lajir esperado de $64.000 em perpetuidade e uma alíquota tributária de 34%. A empresa tem $95.000 em dívida em circulação a uma taxa de juros de 8,5%, e seu custo não alavancado de capital é de 15%. Qual é o valor da empresa de acordo com a Proposição I de M&M com impostos? A empresa deve mudar seu índice dívida/capital próprio se o objetivo for maximizar o valor da empresa? Explique.

17. Valor da empresa [OA2] A Até Dormir S/A espera um Lajir de $14.000 por ano para sempre. No momento, a empresa não tem dívidas, e seu custo do capital próprio é de 16%. Ela pode tomar empréstimos a 9%. Se a alíquota de impostos for de 34%, qual será o valor da empresa? Qual será o valor se a empresa converter sua estrutura de capital para 50% de dívida? E para 100% de dívida?

18. Alavancagem caseira [OA1] As companhias Weber e Cavaleiro são idênticas em todos os aspectos, exceto que a Weber não é alavancada. As informações financeiras das duas empresas são mostradas no quadro a seguir. Todos os grupos de lucros são perpetuidades, e nenhuma das empresas paga impostos. As duas empresas distribuem imediatamente todos os lucros disponíveis aos acionistas.

	Weber	Cavaleiro
Lucro operacional projetado	$ 400.000	$ 400.000
Juros sobre a dívida ao fim do ano	—	$ 72.000
Valor de mercado do capital próprio	$2.500.000	$1.632.000
Valor de mercado da dívida	—	$1.200.000

a. Um investidor que pode tomar emprestado a 6% ao ano deseja comprar 5% do capital da Cavaleiro. Ele poderia aumentar seu retorno comprando 5% do capital da Weber se tomar emprestado de tal forma que os custos líquidos das estratégias sejam iguais?

b. Dadas as duas estratégias de investimentos de (a), qual será a estratégia escolhida pelos investidores? Quando esse processo terminará?

DESAFIO
(Questões 19-22)

19. Custo médio ponderado de capital [OA1] Em um mundo com impostos apenas sobre lucros da pessoa jurídica, mostre que o CMPC pode ser escrito como CMPC $= R_U \times [1 - T_C(D/V)]$.

20. Custo do capital próprio e alavancagem [OA1] Supondo um mundo com impostos apenas sobre lucros da pessoa jurídica, mostre que o custo do capital próprio (R_E) é dado neste capítulo pela Proposição II de M&M com impostos.